

# Relation Algebra

## §1. Notation

Traditionally Adjunction  $F \dashv G$  is defined as  $F(x) \leq y$  iff  $x \leq G(y)$ , and you derive laws like this:

$$\begin{aligned} F(x) \leq F(x) &\implies x \leq G(F(x)) \implies F(x) \leq F(G(F(x))) \\ G(y) \leq G(y) &\implies F(G(y)) \leq y \implies G(F(G(y))) \leq G(y) \end{aligned}$$

By define  $X : * \mapsto x$ ,  $Y : * \mapsto y$  and use diagram order, application can be replaced by composition:  $XF \leq Y$  iff  $X \leq YG$

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• write  $F$  as  $/$ ,  $G$  as  $\backslash$

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• render  $a \leq b$  as  $\boxed{\begin{array}{c} a \\ b \end{array}}$

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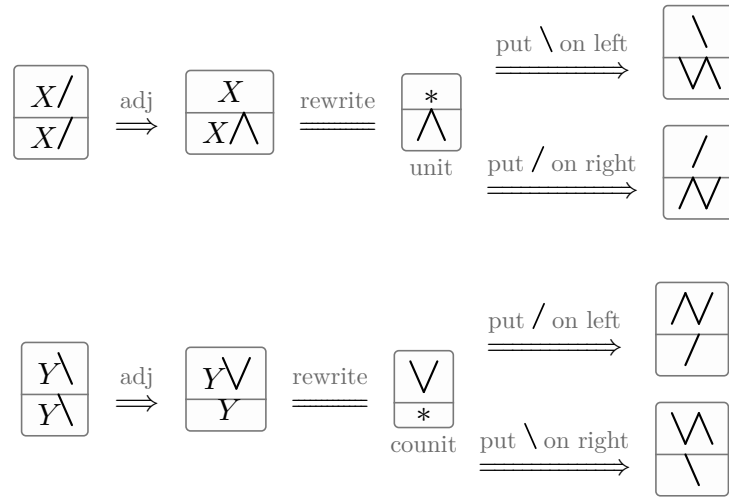
• **rewrite**  $X \leq XF$  as  $1 \leq F$

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(1a)

$$\boxed{\frac{X/}{Y}} = \boxed{\frac{X}{Y\backslash}}$$

adj



now we have  $\wedge = /$  and  $\vee = \backslash$ , and you just discovered string diagrams.

## §1.1. Adjunctions

(1.1a)

| $F \dashv G$                                           | $F$ 's type                                                           | monad $FG$                        | $1 \vDash FG$                                                   | $GF \vDash 1$                                                   | $F$ preserves $u$                                        | $G$ preserves $n$                                             | $FGF=F$                                                     | $GFG=G$                                                         |
|--------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------|
| $\triangleleft \dashv \triangleleft$                   | $A \rightarrow A \otimes A$                                           | $\triangleleft \triangleleft = 1$ | $1 \vDash \triangleleft \triangleleft$                          | $\triangleleft \triangleleft \vDash 1$                          | $(RuS) \triangleleft = R \triangleleft uS \triangleleft$ | $(RnS) \triangleright = R \triangleright nS \triangleright$   | $\triangleleft \triangleleft \triangleleft = \triangleleft$ | $\triangleright \triangleleft \triangleright = \triangleright$  |
| $\multimap \dashv \multimap$                           | $A \rightarrow I$                                                     | $\multimap \multimap = T$         | $1 \vDash \multimap \multimap$                                  | $\multimap \multimap \vDash 1$                                  | $(RuS) \multimap = R \multimap uS \multimap$             | $(RnS) \multimap = R \multimap nS \multimap$                  | $\multimap \multimap \multimap = \multimap$                 | $\multimap \multimap \multimap = \multimap$                     |
| $\circ \dashv \circ$                                   | $(A \rightarrow B) \rightarrow (B \rightarrow A)$                     | $\circ \circ = 1$                 | $R = R \circ \circ$                                             | $R = R \circ \circ$                                             | $(RuS) \circ = R \circ uS \circ$                         | $(RnS) \circ = R \circ nS \circ$                              | $R \circ \circ \circ = R \circ$                             | $R \circ \circ \circ = R \circ$                                 |
| $\multimap \dashv \triangleleft \dashv \multimap$      | $(X \otimes A \rightarrow Y) \rightarrow (X \rightarrow Y \otimes A)$ | $1$                               | $(\multimap \triangleleft \otimes 1) (1 \otimes \multimap) = 1$ | $(1 \otimes \multimap) (\triangleright \otimes 1) = 1$          | $=$                                                      | $=$                                                           | $=$                                                         | $=$                                                             |
| $\triangleright \dashv \multimap \dashv \triangleleft$ | $(X \rightarrow Y \otimes A) \rightarrow (X \otimes A \rightarrow Y)$ | $1$                               | $(1 \otimes \multimap) (\triangleright \otimes 1) = 1$          | $(\multimap \triangleleft \otimes 1) (1 \otimes \multimap) = 1$ | $=$                                                      | $=$                                                           | $=$                                                         | $=$                                                             |
| $\Delta \dashv n$                                      | $(A \rightarrow B) \rightarrow (A \rightarrow B)^2$                   | $R \mapsto RnR = R$               | $R \vDash RnR$                                                  | $RnS \vDash R$                                                  | $\Delta(RuS) = \Delta Ru \Delta S$                       | $(RnT) n (SnU) = (RnS) n (TnU)$                               | $RnR = R$                                                   | $RnR = R$                                                       |
| $u \dashv \Delta$                                      | $(A \rightarrow B)^2 \rightarrow (A \rightarrow B)$                   | $(R, S) \mapsto (RuS, RuS)$       | $R \vDash RuS$                                                  | $RuR \vDash R$                                                  | $(RuT) u (SuU) = (RuS) u (TuU)$                          | $\Delta(RnS) = \Delta Rn \Delta S$                            | $RuR = R$                                                   | $RuR = R$                                                       |
| $\perp \dashv !$                                       | $\{*\} \rightarrow (A \rightarrow B)$                                 | —                                 | —                                                               | $\perp \vDash R$                                                | —                                                        | —                                                             | —                                                           | —                                                               |
| $\cdot S \dashv /S$                                    | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $S/S$                             | $R \vDash (RS)/S$                                               | $(T/S)S \vDash T$                                               | $(RuT)S = RSuTS$                                         | $(RnT)/S = R/SnT/S$                                           | $((RS)/S)S = RS$                                            | $((T/S)S)/S = T/S$                                              |
| $S \cdot \dashv S \setminus$                           | $(B \rightarrow C) \rightarrow (A \rightarrow C)$                     | $S \setminus S$                   | $R \vDash S \setminus (SR)$                                     | $S(S \setminus T) \vDash T$                                     | $S(RuT) = SRuST$                                         | $S \setminus (RnT) = S \setminus RnS \setminus T$             | $S(S \setminus (SR)) = SR$                                  | $S \setminus (S(S \setminus T)) = S \setminus T$                |
| $Rn \dashv R \Rightarrow$                              | $(A \rightarrow B) \rightarrow (A \rightarrow B)$                     | $X \mapsto R \Rightarrow (XnR)$   | $X \vDash R \Rightarrow (XnR)$                                  | $Rn(R \Rightarrow Y) \vDash Y$                                  | $Rn(XuY) = (RnX)u(RnY)$                                  | $R \Rightarrow (XnY) = (R \Rightarrow X) n (R \Rightarrow Y)$ | $Rn(R \Rightarrow (XnR)) = XnR$                             | $R \Rightarrow (Rn(R \Rightarrow Y)) = R \Rightarrow Y$         |
| $\mathcal{D} \dashv \cdot T$                           | $(A \rightarrow B) \rightarrow \text{Cor } A$                         | $R \mapsto (\mathcal{D}R)T$       | $R \vDash (\mathcal{D}R)T$                                      | $\mathcal{D}(AT) \vDash A$                                      | $\mathcal{D}(RuS) = \mathcal{D}Ru \mathcal{D}S$          | $(AnB)T = ATnBT$                                              | $\mathcal{D}((\mathcal{D}R)T) = \mathcal{D}R$               | $(\mathcal{D}(AT))T = AT$                                       |
| $\mathcal{R} \dashv T \cdot$                           | $(A \rightarrow B) \rightarrow \text{Cor } B$                         | $R \mapsto T(\mathcal{R}R)$       | $R \vDash T(\mathcal{R}R)$                                      | $\mathcal{R}(TA) \vDash A$                                      | $\mathcal{R}(RuS) = \mathcal{R}Ru \mathcal{R}S$          | $T(AnB) = TAnTB$                                              | $\mathcal{R}(T(\mathcal{R}R)) = \mathcal{R}R$               | $T(\mathcal{R}(TA)) = TA$                                       |
| $\cdot f \dashv \cdot f^\circ$                         | $(A \rightarrow B) \rightarrow (A \rightarrow C)$                     | $ff^\circ$                        | $1 \vDash ff^\circ$                                             | $f^\circ f \vDash 1$                                            | $(RuS)f = RfuSf$                                         | $(RnS)f^\circ = Rf^\circ nSf^\circ$                           | $ff^\circ f = f$                                            | $f^\circ ff^\circ = f^\circ$                                    |
| $f^\circ \cdot \dashv f \cdot$                         | $(A \rightarrow C) \rightarrow (B \rightarrow C)$                     | $ff^\circ$                        | $1 \vDash ff^\circ$                                             | $f^\circ f \vDash 1$                                            | $f^\circ(XuY) = f^\circ Xu f^\circ Y$                    | $f(XnY) = fXn fY$                                             | $ff^\circ f = f$                                            | $f^\circ ff^\circ = f^\circ$                                    |
| $i \dashv E$                                           | $\text{Map} \dashv \text{Rel}$                                        | $E$                               | $\frac{1}{\exists} : A \rightarrow EA$                          | $\exists : EB \rightarrow B$                                    | —                                                        | —                                                             | $\frac{\exists}{\exists} = 1$                               | $\frac{R \exists}{\exists} = R$                                 |
| $\cdot \exists \dashv \frac{f \exists}{\exists}$       | $\text{Map}(A, EB) \rightarrow (A \rightarrow B)$                     | $1$                               | $\frac{f \exists}{\exists} = f$                                 | $\frac{R \exists}{\exists} = R$                                 | —                                                        | —                                                             | $\frac{\frac{f \exists}{\exists}}{\exists} = f \exists$     | $\frac{\frac{R \exists}{\exists}}{\exists} = \frac{R}{\exists}$ |
| $\frac{\exists}{\exists} \dashv \cdot \exists$         | $(A \rightarrow B) \rightarrow \text{Map}(A, EB)$                     | $1$                               | $\frac{R \exists}{\exists} = R$                                 | $\frac{f \exists}{\exists} = f$                                 | —                                                        | —                                                             | $\frac{R \exists}{\exists} = \frac{R}{\exists}$             | $\frac{f \exists}{\exists} = f \exists$                         |

## §1.2. Composing adjunctions

(1.2a)

$$\begin{array}{c}
 \boxed{\begin{array}{c} X \\ f(R/T) \end{array}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} f^\circ X \\ R/T \end{array}} \xLeftrightarrow{\cdot T \dashv /T} \boxed{\begin{array}{c} (f^\circ X)T \\ R \end{array}} \\
 \text{f a map} \\
 \\
 \xLeftrightarrow{\text{associativity}} \boxed{\begin{array}{c} f^\circ(XT) \\ R \end{array}} \xLeftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} XT \\ fR \end{array}} \xLeftrightarrow{\cdot T \dashv /T} \boxed{\begin{array}{c} X \\ (fR)/T \end{array}} \\
 \\
 \xRightarrow{\text{indirect equality}} \boxed{\begin{array}{c} f(R/T) \\ (fR)/T \end{array}}
 \end{array}$$

|                       | $\cdot T$                                   | $T \cdot$                                         | $\cdot g$                                         | $g \cdot$                                        | $Tn$                                                      | $\circ$                                                                                                               | $\circ \triangleleft$                                                                                                 | $\Delta$                                                                |
|-----------------------|---------------------------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| $\cdot S$             | $R/(ST) = (R/T)/S$                          | $T \setminus (R/S) = (T \setminus R)/S$           | $R/(Sg) = (Rg^\circ)/S$                           | $g(R/S) = (gR)/S$                                | —                                                         | $(Y/S)^\circ = S^\circ \setminus (Y^\circ)$                                                                           | $(RS)^\wedge = R^\wedge (S \otimes 1)$                                                                                | $(T_1 n T_2)/S = T_1/S n T_2/S$                                         |
| $S \cdot$             | $S \setminus (R/T) = (S \setminus R)/T$     | $(TS) \setminus R = S \setminus (T \setminus R)$  | $S \setminus (Rg^\circ) = (S \setminus R)g^\circ$ | $(g^\circ S) \setminus R = S \setminus (gR)$     | —                                                         | $(S \setminus Y)^\circ = Y^\circ/S^\circ$                                                                             | $((S \otimes 1)R)^\wedge = SR^\wedge$                                                                                 | $S \setminus (T_1 n T_2) = S \setminus T_1 n S \setminus T_2$           |
| $\cdot f$             | $R/(fT) = (R/T)f^\circ$                     | $T \setminus (Rf^\circ) = (T \setminus R)f^\circ$ | $(fg)^\circ = g^\circ f^\circ$                    | —                                                | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                        | —                                                                                                                     | $(T_1 n T_2)f^\circ = T_1 f^\circ n T_2 f^\circ$                        |
| $f \cdot$             | $f(R/T) = (fR)/T$                           | $(Tf^\circ) \setminus R = f(T \setminus R)$       | —                                                 | $(fg)^\circ = g^\circ f^\circ$                   | —                                                         | $(fY)^\circ = Y^\circ f^\circ$                                                                                        | —                                                                                                                     | $f(T_1 n T_2) = fT_1 n fT_2$                                            |
| $Rn$                  | —                                           | —                                                 | —                                                 | —                                                | $(RnT) \Rightarrow Y = R \Rightarrow (T \Rightarrow Y)$   | $(R \Rightarrow Y)^\circ = R^\circ \Rightarrow (Y^\circ)$                                                             | —                                                                                                                     | $R \Rightarrow (T_1 n T_2) = (R \Rightarrow T_1) n (R \Rightarrow T_2)$ |
| $\circ$               | $(Y/T)^\circ = T^\circ \setminus (Y^\circ)$ | $(T \setminus Y)^\circ = Y^\circ/T^\circ$         | $(gY)^\circ = Y^\circ g^\circ$                    | $(gY)^\circ = Y^\circ g^\circ$                   | $(T \Rightarrow Y)^\circ = T^\circ \Rightarrow (Y^\circ)$ | $Y^\circ \Rightarrow Y$                                                                                               | $(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $\circ$ section's display | $(T_1 n T_2)^\circ = T_1^\circ n T_2^\circ$                             |
| $\circ \triangleleft$ | $(RT)^\wedge = R^\wedge (T \otimes 1)$      | $((T \otimes 1)R)^\wedge = TR^\wedge$             | —                                                 | —                                                | —                                                         | $(R^\wedge)^\circ = (R^\circ \otimes 1)(1 \otimes \triangleright \circ)$<br>both bends: the $\circ$ section's display | —                                                                                                                     | —                                                                       |
| $u$                   | $(X_1 u X_2)T = X_1 T u X_2 T$              | $T(X_1 u X_2) = TX_1 u TX_2$                      | $(X_1 u X_2)g = X_1 g u X_2 g$                    | $g^\circ(X_1 u X_2) = g^\circ X_1 u g^\circ X_2$ | $Tn(X_1 u X_2) = (TnX_1)u(TnX_2)$                         | $(X_1 u X_2)^\circ = X_1^\circ u X_2^\circ$                                                                           | —                                                                                                                     | —                                                                       |
| $\perp$               | $\perp T = \perp$                           | $T \perp = \perp$                                 | $\perp g = \perp$                                 | $g^\circ \perp = \perp$                          | $Tn \perp = \perp$                                        | $\perp^\circ = \perp$                                                                                                 | —                                                                                                                     | —                                                                       |

### §1.3. Adjoint triples

Beyond  $u \dashv \Delta \dashv n$ , which §1.1 already carries as two rows, the table holds one adjoint triple with content:  $\exists_f \dashv f^* \dashv \forall_f$  along a map  $f$ , once on each side of the composite.  $f : A \rightarrow B$  throughout this subsection.

$$\begin{aligned} \cdot f \dashv \cdot f^\circ \dashv /f^\circ \\ f^\circ \dashv f \dashv f \setminus \end{aligned} \quad (1.3a)$$

The first two links of each chain are the map shunting rules, and the third is §1.1's division row read at a chosen divisor:  $\cdot S \dashv /S$  at  $S := f^\circ$ , and  $S \cdot \dashv S \setminus$  at  $S := f$ .

The third link is not the first over again. For a map  $f$ ,  $/f$  collapses to  $\cdot f^\circ$ ; being a map is exactly what that collapse spends  $R/(fS) = (R/S)f^\circ$ . —  $f^\circ$  is not a map, so  $/f^\circ$  does not collapse in turn, and the three operators stay distinct. On  $\text{Rel}(\mathbb{I}, A) = \text{EA}$  they are the image triple:

| operator        | acts by                                   | name                      |
|-----------------|-------------------------------------------|---------------------------|
| $\cdot f$       | $S \mapsto f[S]$                          | direct image, $\exists_f$ |
| $\cdot f^\circ$ | $T \mapsto f^{-1}[T]$                     | inverse image, $f^*$      |
| $/f^\circ$      | $S \mapsto \{b : f^{-1}(b) \subseteq S\}$ | $\forall_f$               |

§1.1's  $\mathcal{D} \dashv \cdot T$  is this same chain along the projection  $A \otimes B \rightarrow A$ , read through  $\text{Rel}(A, B) = E(A \otimes B)$ :  $\mathcal{D}R = \{(a, a) : \exists b. aRb\}$ ,  $AT$  is the relation that ignores  $b$  altogether, and the third link is  $R \mapsto 1 n R/T = \{(a, a) : \forall b. aRb\}$ .

$$\mathcal{D} \dashv \cdot T \dashv 1 n \cdot /T \quad (1.3c)$$

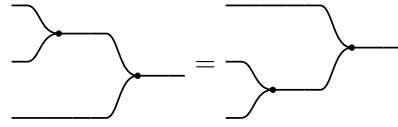
Three is where it stops, and one map breaks both ends. Take  $A = \{a_1, a_2\}$  and  $B = \{b\}$ :  $f[\{a_1\}] n f[\{a_2\}] = \{b\}$  while  $f[\{a_1\} n \{a_2\}] = \emptyset$ , so  $\cdot f$  does not preserve meets and has no left adjoint, and the same two subsets give  $\forall_f(\{a_1\} u \{a_2\}) = \{b\}$  against  $\forall_f\{a_1\} u \forall_f\{a_2\} = \emptyset$ , so  $/f^\circ$  does not preserve joins and has no right adjoint. A monic  $f$  restores the binary case and no more —  $\tau f$  still reaches only  $\text{im } f$ , and  $\forall_f \perp = B \setminus \text{im } f$  is still not  $\perp$ . The chain extends only when  $f^\circ$  is a map as well, and then  $/f^\circ = \cdot f^\circ = \cdot f$  and it repeats forever. For an  $S$  with neither  $S$  nor  $S^\circ$  a map there is no triple at all:  $\cdot S \dashv /S$  is two links and stops.

The rows that chain forever say nothing by it:  $\circ$  is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and  $\circ \dashv \circ \dashv \circ$  has no end, and §1.1's other row of that kind,  $\circ \triangleleft \dashv \triangleright \circ$ , goes the same way.

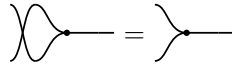
## §2. Relations

$\mathbf{Rel}$  is a poset-enriched category with  $(\otimes, \circ)$  where  $\otimes$  is commutative cartesian product, and converse  $\circ^{-1}$ , and

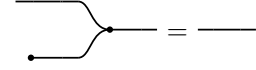
$\mathbf{C} : (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$  a commutative monoid with  $\mathbf{C}^\circ : \mathbf{C}$ . We write  $\mathbf{C}^\circ$  as  $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$



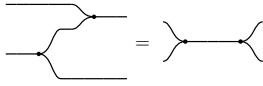
$\triangleright$  associative



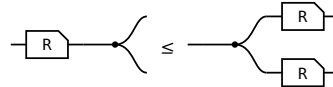
$\triangleright$  commutative



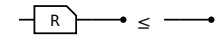
$\circ$  is its unit



Frobenius, one half — the other is its  $\circ$



$R \triangleleft \leq \triangleleft (R \otimes R)$

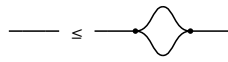


$R \circ \leq \circ$

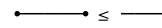
### §2.1. $\forall$ object $\mathbf{A}$ , $(\mathbf{A}, \triangleleft, \circ) \dashv (\mathbf{A}, \triangleright, \circ)$



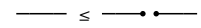
$\triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



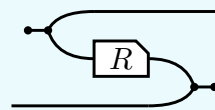
$1 \leq \circ$

(2.1a)

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In  $\mathbf{Rel}$  it reads  $\{(a, a)\} \subseteq A \times A$  — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

### §3. $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ is a 2 functor

$^\circ$  is primitive, part of the data the first section lists. It turns both of  $R$ 's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are  $R^\circ$ , not its definition:



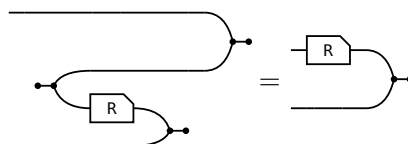
$$R^\circ = (\circ \leftarrow \triangleleft \otimes 1) (1 \otimes R \otimes 1) (1 \otimes \triangleright \rightarrow \circ)$$

where  $\circ \leftarrow \triangleleft : \mathbb{I} \rightarrow a \otimes a$  opens a pair of wires out of nothing and  $\triangleright \rightarrow \circ : b \otimes b \rightarrow \mathbb{I}$  closes one, so the input of  $R^\circ$  is where the output of  $R$  was. Taking that formula as the definition would now be circular:  $\triangleleft$  and  $\rightarrow$  are  $\triangleright^\circ$  and  $\circ \leftarrow$ . The laws of  $^\circ$  are that it is a contravariant 2-functor  $^\circ : \mathcal{C}^{\text{op}} \rightarrow \mathcal{C}$ :

$$(i) \ 1^\circ = 1 \quad (ii) \ (RS)^\circ = S^\circ R^\circ \quad (iii) \ (R \otimes S)^\circ = R^\circ \otimes S^\circ \quad (iv) \ R \leq S \text{ implies } R^\circ \leq S^\circ$$

#### §3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:

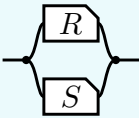


$R^\circ$  is DEFINED as the bending of  $(R \otimes 1) \triangleright \rightarrow \circ$ , so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the  $\circ \leftarrow \triangleleft$  bends the  $a$  strand down and the  $\triangleright \rightarrow \circ$  brings it back up, while the  $b$  strand rides through untouched. Nothing else is spent below.

§4.  $\cap$  is a commutative idempotent monoid on every hom-set

(4a)

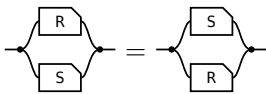
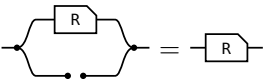
The **meet** of  $R, S : a \rightarrow b$ , the paper's **convolution**, is  $R \cap S := \langle (R \otimes S) \rangle$  — copy the input, run  $R$  and  $S$  on the two copies, merge the results — so what comes out is what both of them do.



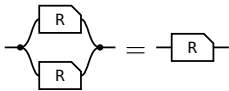
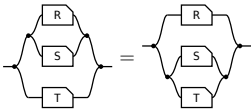
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow  $\top = \text{---} \bullet \text{---} \bullet \text{---}$  (the paper's Lemma 4.11).

(4b)



**unit:** one half of  $\top$  per end — the merge's unit law absorbs the  $\bullet$ , the copy's counit law the  $\bullet$  **commutative:**  $\sigma$  crosses  $R \otimes S$  by naturality and is absorbed by cocommutativity and commutativity



**associative:** coassociativity and associativity;  $\otimes$  re-brackets for nothing, **idempotent:** the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So  $\leq$  is the order this monoid induces.  $R \cap S \leq R$  comes from the unit, and idempotency turns anything under both  $S$  and  $T$  into something under  $S \cap T$ , since  $R = R \cap R \leq S \cap T$ .

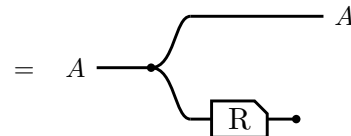
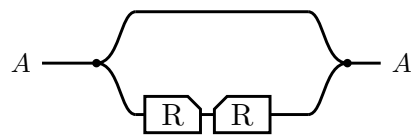
And one law relating  $\cap$  to composition, which is **not** an equation:

(4c)

| semi-distributivity, and what supplies it                                                                                                                         | picture |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $R \cap (S \cap T) \leq R \cap S \cap R \cap T$ — the lax copy law. Equality exactly when $R$ is single valued: the Maps section's $F(R \cap S) = F R \cap F S$ . |         |

## §5. Domain and range

The **domain**  $\text{Dom}(R) \triangleq 1 \cap RR^\circ$  and the **range**  $\text{Ran } R \triangleq \text{Dom}(R^\circ)$ .



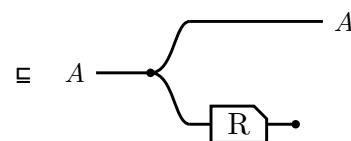
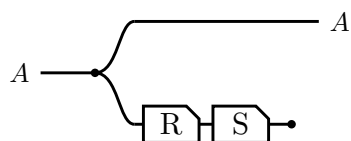
$\triangleright$  lands the return leg back on the value  $\triangleleft$  handed out,  
so  $R^\circ \triangleright$  cuts to  $\circ$  — Frobenius

Running  $R$  and throwing the result away leaves only the fact that  $R$  could fire, and  $\text{Ran } R$  is the same picture with the box mirrored. In  $\text{Rel}$  both steps are  $\{(a, a) : \exists b. a R b\}$ .

|                                                                                   |
|-----------------------------------------------------------------------------------|
| $\text{Dom}(R) \triangleq 1 \cap RR^\circ$                                        |
| $\text{Dom}(R) \subseteq A \iff R \subseteq AR$ , for $A$ coreflexive             |
| $\text{Dom}(RS) \subseteq \text{Dom}(R)$                                          |
| $\text{Dom}(R \cap S) = 1 \cap SR^\circ$                                          |
| $R \text{ entire} \iff \text{Dom}(R) = 1 \iff 1 \subseteq RR^\circ$               |
| $R \text{ simple} \iff R^\circ R \subseteq 1$                                     |
| $R \text{ a map} \iff R \text{ entire and simple}$                                |
| $R, S \text{ entire} \implies RS \text{ entire}$ — likewise simple, likewise maps |
| $RS \text{ entire} \implies R \text{ entire}$                                     |

### §5.1. Sliding the discard

$\text{Dom}(RS) \subseteq \text{Dom}(R)$ , and a single glyph for  $\text{Dom}$  would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.



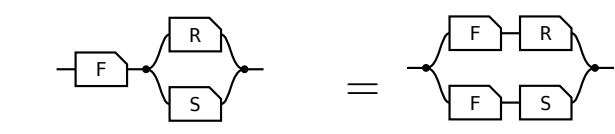
$S^\circ \circ \circ$ , the lax axiom for  $\circ$  in the first section  
— the discard slides back past  $S$

Equality is  $S$  entire, which is the same picture read as  $\text{Dom}(R) = 1 \iff R \text{ entire}$ .

## §6. Maps

|                                                                                               |                                                          |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>surjective</b><br>$\mathbb{1} \subseteq R^\circ R$ , that is, $R^\circ$ entire.            | <b>single valued</b><br>$R^\circ R \subseteq \mathbb{1}$ |
| <b>entire</b><br>$\mathbb{1} \subseteq RR^\circ$ . With <b>single valued</b> , a <b>map</b> . | <b>injective</b><br>$RR^\circ \subseteq \mathbb{1}$      |

(6a)

|                                                         |                                                                                                                                                             |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>law</b><br><br>$F(RnS) = FRnFS$<br>$F$ single valued | <b>picture</b><br><br><br>one person who admires both      a at A, b at B |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|

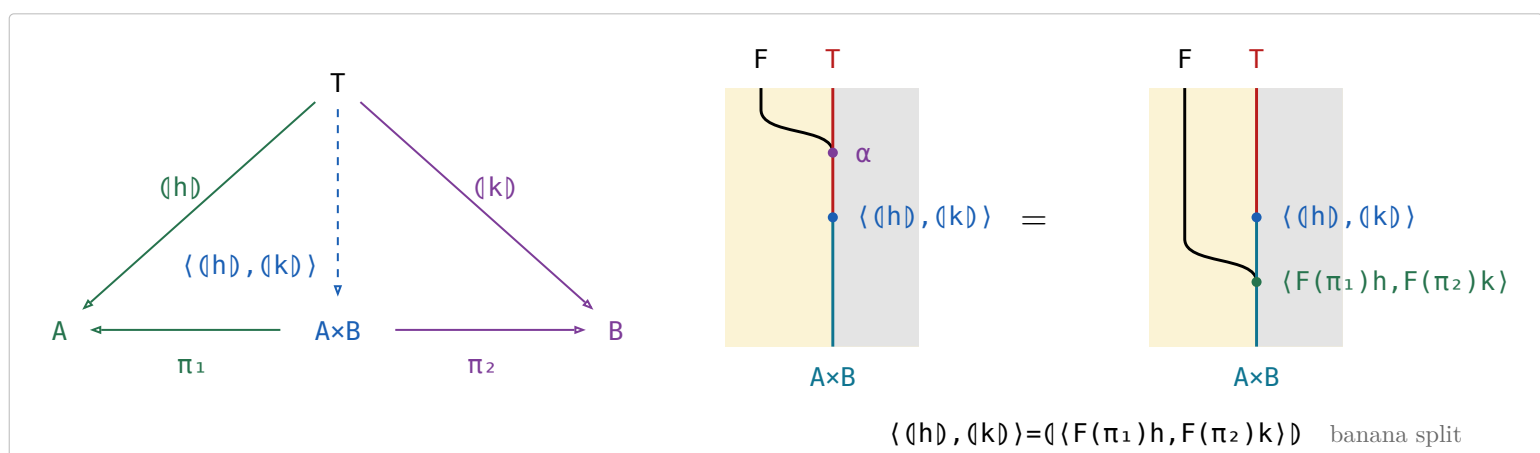
(6b)

§7. Reduce in  $\mathcal{S}et$ 

| definition                                                                                                                                                                         | type                                                        | note                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $\text{listr } A ::= \text{nil} \mid \text{cons } (A, \text{listr } A)$                                                                                                            | $\text{Set} \rightarrow \text{Set}$                         | The cons-lists over $A$ , the datatype every row below folds (B&dM p. 55).                                   |
| $\text{sum} \triangleq ([\text{zero}, \text{plus}])$                                                                                                                               | $\text{listr Nat} \rightarrow \text{Nat}$                   | $\text{plus}(a, b) = a + b$ .                                                                                |
| $\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$                                                                                                                     | $\text{listr } A \rightarrow \text{Nat}$                    | $\pi_2$ drops the head and keeps the count of the tail, $\text{succ}$ adds one for the head.                 |
| $\text{average} \triangleq (\text{sum}, \text{length}) \text{ div}$                                                                                                                | $\text{listr Nat} \rightarrow \text{Real}$                  | $\text{div}(m, n) = m/n$ , with $\text{div}(0, 0) = 0$ so <b>average</b> is total. Traverses the list twice. |
| banana-split law<br>$\langle \langle h \rangle, \langle k \rangle \rangle = \langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$                                                  | $T \rightarrow A \times B$                                  | Any fork of folds is a single fold, hence one traversal — $F$ the base functor.                              |
| what it reduces to<br>$\alpha \langle \langle h \rangle, \langle k \rangle \rangle = F(\langle \langle h \rangle, \langle k \rangle \rangle) \langle F(\pi_1)h, F(\pi_2)k \rangle$ | $FT \rightarrow A \times B$                                 | All that (11.6.1a) leaves to check: the fork satisfies the defining equation.                                |
| the instance<br>$\langle \text{sum}, \text{length} \rangle = ([\text{zeros}, \text{pluss}])$                                                                                       | $\text{listr Nat} \rightarrow \text{Nat} \times \text{Nat}$ | $\text{pluss}(a, (b, n)) = (a + b, n + 1)$ , so <b>average</b> runs in one pass.                             |
| $\text{preds } n = [n, n-1, \dots, 1]$                                                                                                                                             | $\text{Nat} \rightarrow [\text{Nat}]$                       | B&dM Ex 3.6 (p. 57): apply Ex 3.4 to write <b>preds</b> as $\langle k \rangle \pi_1$ .                       |

(7a)

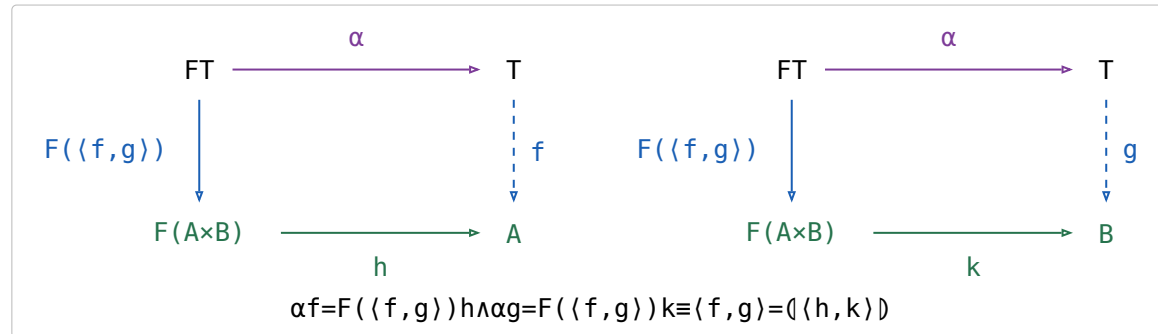
(7b)





## §7.1. Fokkinga's mutual recursion theorem

- $F\text{-Alg}(\mathcal{A})$  — the algebras and their homomorphisms — has binary products whenever  $\mathcal{A}$  does, and the forgetful  $U : F\text{-Alg}(\mathcal{A}) \rightarrow \mathcal{A}$  creates them: the product of  $h : FA \rightarrow A$  and  $k : FB \rightarrow B$  is carried by  $A \times B$ , with structure  $\langle F(\pi_1)h, F(\pi_2)k \rangle : F(A \times B) \rightarrow A \times B$ .
- $\pi_1$  and  $\pi_2$  are homomorphisms out of it, and  $\langle p, q \rangle : X \rightarrow A \times B$  is a homomorphism **iff**  $p$  and  $q$  both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (7a) IS that product's universal property read at the initial algebra:  $\langle h \rangle$  and  $\langle k \rangle$  are the unique homomorphisms to  $(A, h)$  and  $(B, k)$ , so their fork is the unique homomorphism into the product, which is  $\langle \langle F(\pi_1)h, F(\pi_2)k \rangle \rangle$ .
- Fokkinga's theorem is **strictly more general** and is not that product: in (7.1a) the two arrows leave  $F(A \times B)$ , not  $FA$  and  $FB$ , so each sees BOTH components. The product is the case where they factor as  $F(\pi_1)h$  and  $F(\pi_2)k$ ; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.



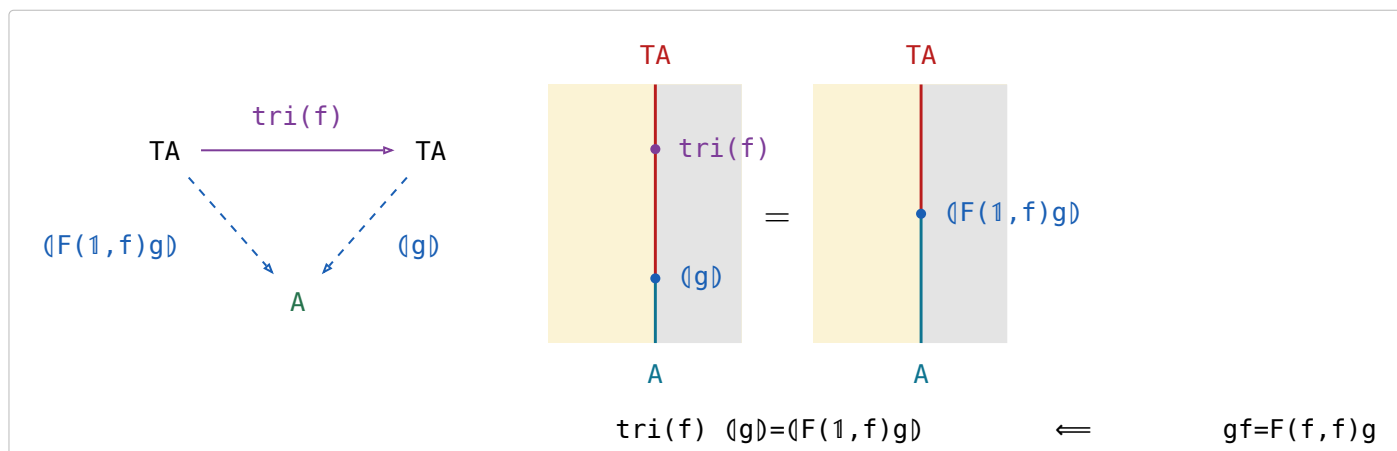
(7.1a)

## §7.2. Ruby triangles

| stage                       | definition                                                                                                                            |
|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| informally (p. 58)          | $\text{tri}(f) [a_0, a_1, \dots, a_i, \dots, a_n] = [a_0, f a_1, \dots, f^i a_i, \dots, f^n a_n]$                                     |
| for cons-lists (p. 59)      | $\text{tri}(f) = \langle [\text{nil}, (1 \times \text{listr}(f)) \text{ cons}] \rangle$                                               |
| with the base functor named | $F(A, B) = 1 + A \times B$ , $\alpha = [\text{nil}, \text{cons}]$ , so $\text{tri}(f) = \langle F(1, \text{listr}(f)) \alpha \rangle$ |
| abstractly (p. 59)          | $F$ a bifunctor with initial type $(\alpha, T)$ : $\text{tri}(f) = \langle F(1, T(f)) \alpha \rangle$                                 |

(7.2a)

For the definition to make sense  $f : A \rightarrow A$  is required, and then  $\text{tri}(f) : TA \rightarrow TA$ .



(7.2b)

(7.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to  $F(1, T(f)) \alpha \langle g \rangle = F(1, \langle g \rangle) F(1, f) g$  (p. 59).

## §7.3. Depth of a tree

|                  |                                                                                                                                                                                                                                             |        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| the datatype     | $\text{tree } A ::= \text{tip } A \mid \text{bin } (\text{tree } A, \text{tree } A)$ , base functor $F(A,B)=A+B \times B$ ,<br>initial type $([\text{tip}, \text{bin}], \text{tree})$                                                       | (7.3a) |
| the fold         | $\llbracket g, h \rrbracket$ is the unique $f$ with $f(\text{tip } a) = g\ a$ and $f(\text{bin } (x, y)) = h\ (f\ x, f\ y)$                                                                                                                 |        |
| $\text{tree}(f)$ | $\text{tree}(f) = \llbracket F(f, 1) \rrbracket [\text{tip}, \text{bin}]$ ; pointwise $\text{tree}(f)(\text{tip } a) = \text{tip } (f\ a)$ and<br>$\text{tree}(f)(\text{bin } (x, y)) = \text{bin } (\text{tree}(f)\ x, \text{tree}(f)\ y)$ |        |
| $\text{max}$     | $\text{max} = \llbracket 1, \text{bmax} \rrbracket$ , where $\text{bmax } (a, b)$ is the larger of $a$ and $b$                                                                                                                              |        |
| $\text{depths}$  | $\text{depths} = \text{tree}(\text{zero})\ \text{tri}(\text{succ})$ — replaces every tip by its depth in the tree, <b>zero</b> the constant<br>function returning 0 and <b>succ</b> the successor                                           |        |
| $\text{depth}$   | $\text{depth} = \text{depths}\ \text{max}$                                                                                                                                                                                                  |        |

## §8. / is all of

(8a)

$$x (R/S) y \iff \forall p. y S p \rightarrow x R p$$

$$x (S \setminus R) y \iff \forall p. p S x \rightarrow p R y$$

/compares images:  $S(y) \subseteq R(x)$ . \ compares preimages:  $S^\circ(x) \subseteq R^\circ(y)$ .

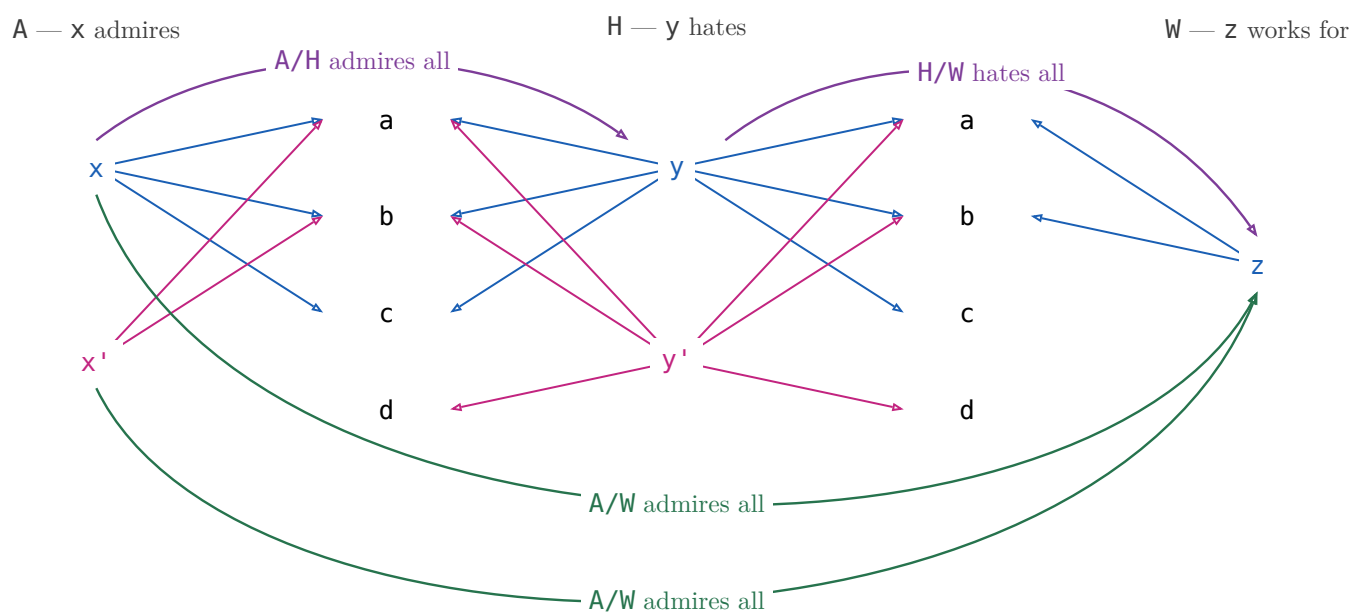
example: A admires, H hates, W works for

$x (A/H) y$  —  $x$  admires everyone  $y$  hates.

$x (H \setminus A) y$  — everyone who hates  $x$  admires  $y$ .

§8.1.  $(R/S)(S/W) \sqsubseteq R/W$ 

(8.1a)



$$x (A/H) y \text{ — } x \text{ admires everyone } y \text{ hates}$$

$$y (H/W) z \text{ — } y \text{ hates everyone } z \text{ works for}$$

$$x (A/W) z \text{ — } x \text{ admires everyone } z \text{ works for}$$

(8.1b)

$(A/H)(H/W)$  is a path:  $x \rightarrow y \rightarrow z$ , and that is all of it.  $A/W$  also holds of  $x'$ , who admires everyone  $z$  works for — but  $x'$  does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of  $(R/S)(S/W) \sqsubseteq R/W$ .

|                                                                                                                                                                                                                   |  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| $X \in R/S \iff X \subseteq R$<br>$X$ is any $X$ -to- $Y$ pairing; one that only pairs $X$ with a $Y$ such that $X$ admires everyone $Y$ hates lies inside $R/S$ , and $R/S$ is the largest such.                 |  |
| $X \in S \setminus R \iff S X \in R$<br>The mirror — divide on the left when $X$ comes first.                                                                                                                     |  |
| $(R/S) S \in R$<br>There is a $Y$ such that $X$ admires everyone $Y$ hates, and $p$ is one of the people $Y$ hates — then $X$ admires $p$ too. Strict at $S=\emptyset$ : $R/S$ is everyone, $(R/S) S=\emptyset$ . |  |
| $S (S \setminus R) \in R$<br>The mirror.                                                                                                                                                                          |  |
| <b>associate:</b> $R/(S_1 S_2) = (R/S_2)/S_1$<br>A friend's enemy is two hops: divide by the far end first.                                                                                                       |  |
| $(S_1 S_2) \setminus R = S_2 \setminus (S_1 \setminus R)$<br>The mirror.                                                                                                                                          |  |
| <b>maps:</b> $f (R/S) = (fR)/S$<br>Rename $X$ before or after dividing — the licence to write $fR/S$ .                                                                                                            |  |
| $R/(fS) = (R/S) f^\circ$<br>Rename $Y$ : a map leaves a denominator as $f^\circ$ outside the box.                                                                                                                 |  |
| $(R/S) (S/W) \in R/W$<br>Someone who admires all of a hate-set that already covers everyone $Z$ works for admires those people too.                                                                               |  |
| $1 \in R/R$<br>$R/R$ runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only $a$ and $b$ admire each other's idols too, and still stay two people.                    |  |
| $(R/R) (R/R) = R/R$<br>$R/R$ is the preorder <b>admires at least as much as</b> , and a preorder is idempotent. Freyd writes $\sqsubseteq$ ; with $1 \in R/R$ above it is an equality.                            |  |
| $(R/R)R = R$<br>Reaching $p$ through someone whose idols $X$ fully admires is reaching $p$ directly, since $X$ admires their own idols.                                                                           |  |
| $R/1 = R$<br>Dividing by $1$ : $p$ 's set is just $\{p\}$ , so admiring all of it is admiring $p$ .                                                                                                               |  |
| $R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$<br>Admiring a combined hate-set is admiring each set in full.                                                                                                               |  |
| $S \setminus (R/W) = (S \setminus R)/W$<br>Which is why $S \setminus R/W$ needs no bracket.                                                                                                                       |  |

Fifteen laws, fifteen pictures, and not one shows a generator:  $\cap$ ,  $\cup$ ,  $\circ$  and composition are what the Frobenius generators build, and  $/$  is none of those — it is posited, with nothing to unfold.

# §9. $\frac{R}{S}$

$\frac{R}{S} \triangleq (R/S) \cap (S/R)^\circ$ . In Rel  $x$  and  $y$  has the same image:  $\forall p. (x R p \iff y S p)$

(9a)

$x (A/H) y$  —  $x$  admires everyone  $y$  hates

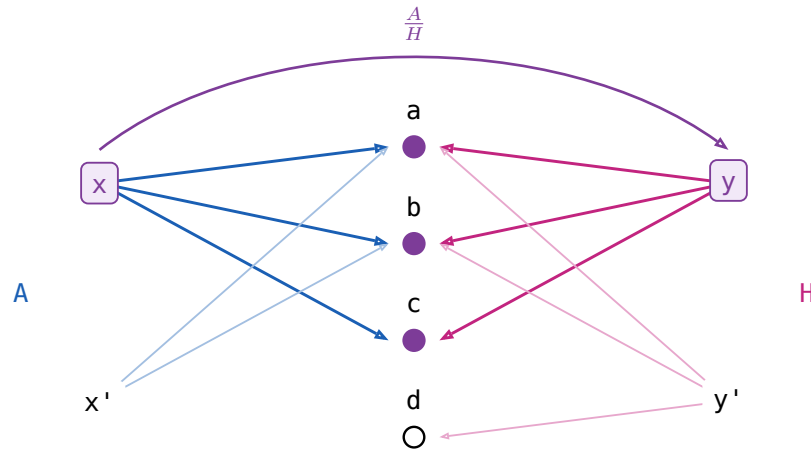
(9b)

$x ((H/A)^\circ) y$  —  $x$  admires only people  $y$  hates

$x ((A/H) \cap (H/A)^\circ) y$  —  $x$  admires only and all whom  $y$  hates

$x$  admires exactly whom  $y$  hates:  $/$  is **all of**,  $\frac{R}{S}$  is **only all of**.

(9c)



| law                                                                              | the reading                                                                                                                                        |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \subseteq \frac{R}{S} \iff XS \subseteq R \text{ and } X^\circ R \subseteq S$ | $X$ may pair $x$ with $y$ only when $x$ admires exactly whom $y$ hates. Both halves must typecheck, so the operation is <b>partial</b> .           |
| $(\frac{R}{S})^\circ = \frac{S}{R}$                                              | Matching is symmetric.                                                                                                                             |
| $\frac{R}{S} \frac{S}{W} \subseteq \frac{R}{W}$                                  | And transitive.                                                                                                                                    |
| $\frac{R}{S} S \subseteq R$                                                      | $(\exists y. x(\frac{R}{S})y \wedge ySp) \rightarrow xRp$<br>$x$ <b>only</b> admires whom $y$ hates<br>$\frac{R}{S} S = \text{Dom}(\frac{R}{S}) R$ |
| $\frac{R}{R} R = R$                                                              | $(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$<br>$x$ and $y$ admire the same people<br>$y=x$ always qualifies ( $1 \in R \circ R$ below)      |
| $1 \subseteq \frac{R}{R}$                                                        | $x(\frac{R}{R})y$ if $x$ and $y$ admires the same peoples.                                                                                         |
| $(\frac{R}{R})^2 = \frac{R}{R}$                                                  | So the relation <b>admires the same people</b> is an equivalence relation.                                                                         |
| $X \subseteq \frac{R}{R} \iff XR \subseteq R$ , for symmetric $X$                | The largest symmetric arrow that leaves $R$ alone.                                                                                                 |
| $\frac{R}{1}$ is the <b>simple part</b> of $R$                                   | The people who admire exactly one person and nobody else. It equals $R$ only when $R$ is simple, unlike $R/1=R$ .                                  |
| $\text{Dom } \frac{R}{S} = 1 \cap (R/S) (S/R)$                                   | Its domain is the <b>domain of simplicity</b> of $R$ .                                                                                             |

(9d)

## §10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows,  $\exists$  **epsilon**, subject to

$$\begin{aligned} \exists_R \square &= R\square, \quad \exists_R = \exists_R \square \\ \mathbb{1} &\subseteq (R/\exists_R)(\exists_R/R) & \exists_R \text{ is } \mathbf{thick} \\ \frac{\exists_R}{\exists_R} &= \mathbb{1} & \exists_R \text{ is } \mathbf{straight} \end{aligned}$$

$R\square$  is  $R$ 's target, an identity arrow. For  $R : A \rightarrow B$  write  $\exists : EA \rightarrow B$ , dropping the subscript.

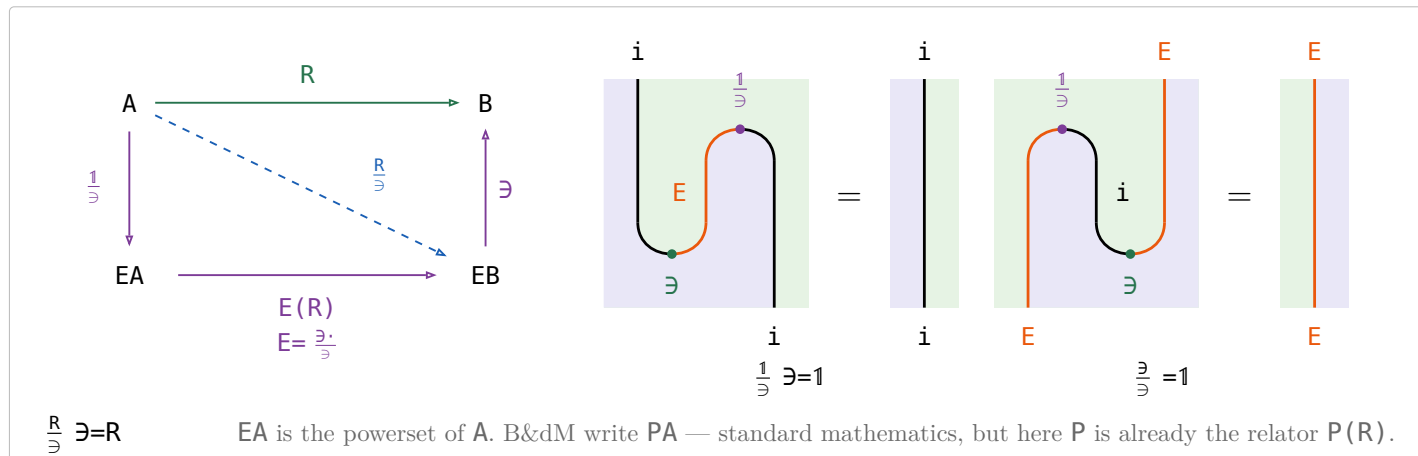
$$\epsilon \triangleq \exists^\circ : A \rightarrow EA$$

(10b)

| law                                                                                            | the reading                                                                                                                         |
|------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1 $\exists_R \square = R\square, \exists_R = \exists_R \square$                                | One $\exists$ per object, not per arrow.                                                                                            |
| 2 $\exists$ is <b>thick</b>                                                                    | <b>Comprehension:</b> every $x$ has a set of exactly the people $x$ admires.                                                        |
| 3 $\frac{R}{\exists} : A \rightarrow EB$ , for $R : A \rightarrow B$                           | convert a relation to a function. $a \frac{R}{\exists} = \{b \mid a R b\}$                                                          |
| 4 $\frac{R}{\exists}$ is a <b>map</b>                                                          |                                                                                                                                     |
| 5 $\frac{R}{\exists} \exists = R$                                                              |                                                                                                                                     |
| 6 $F \sqsubseteq \frac{F}{\exists}$ , $F$ simple                                               | A partial choice of sets is inside the total one.                                                                                   |
| 7 $\frac{1}{\exists}$ , the <b>singleton map</b> , monic                                       | The one-person set.                                                                                                                 |
| 8 $\frac{\exists}{\exists} = \mathbb{1}$                                                       | Make the set of a set, then read it back one level down.                                                                            |
| 9 <b>fusion:</b> $\frac{fR}{\exists} = f \frac{R}{\exists}$ , $f$ a map                        | Naturality of the unit, $f \frac{1}{\exists} = \frac{1}{\exists} (E(f))$ .                                                          |
| 10 $\frac{f}{\exists} = f \frac{1}{\exists}$ , $f$ a map                                       | Rename first or take singletons first — the fusion row above at $R=1$ .                                                             |
| 11 $\frac{R}{S} = \frac{R}{\exists} \left( \frac{S}{\exists} \right)^\circ$                    | $x$ and $y$ match when $R$ sends $x$ and $S$ sends $y$ to the same set.                                                             |
| 12 $E(R) \triangleq \frac{\exists R}{\exists}$ , $E \triangleq \frac{\exists \cdot}{\exists}$  | $E(R) : EA \rightarrow EB$ , image of a set of $A$                                                                                  |
| 13 $\frac{R}{\exists} = \frac{1}{\exists} E(R)$                                                | $\frac{1}{\exists} : x \mapsto \{x\}$                                                                                               |
| 14 $\frac{S}{\exists} E(R) = \frac{S}{\exists} \frac{\exists R}{\exists} = \frac{SR}{\exists}$ | absorption — the monad's composition law, $\frac{S}{\exists} \diamond \frac{R}{\exists} = \frac{SR}{\exists}$ , §10.2               |
| 15 <b>subset</b> $\triangleq \exists/\exists : EA \rightarrow EA$                              | $xs \text{ subset } ys \iff \forall a. ys \exists a \rightarrow xs \exists a$ , that is $ys \subseteq xs$ , not $xs \subseteq ys$ . |

### §10.1. $i$ -E Power Allegory defined as adjunction

(10.1a)



$i$  is the inclusion  $\text{Map}(\mathcal{A}) \rightarrow \mathcal{A}$ , doing nothing, so  $E$  is both the functor and the monad  $iE$ .

## §10.2. $\mathcal{A} \cong \text{Kleisli}(\mathbf{E})$

$$\mathbf{E}(R) \triangleq \frac{\exists R}{\exists}, \frac{1}{\exists} : A \rightarrow EA, \text{ union} = \mathbf{E}(\exists) = \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA \quad (10.2a)$$

$$f \diamond g \triangleq f \mathbf{E}(g) \text{ union} : A \rightarrow EC, \text{ for } f : A \rightarrow EB \text{ and } g : B \rightarrow EC$$

the monad is on  $\text{Map}(\mathcal{A})$ , not on the allegory:  $\mathbf{E}$  is a relator on all relations, but  $\frac{1}{\exists}$ , **union** and  $f \mathbf{E}(g) \text{ union}$  are maps, and the Kleisli construction happens where they live.

$$\begin{aligned} \boxed{\frac{S}{\exists} \diamond \frac{R}{\exists}} & \xrightarrow{\text{definition of } \diamond, \text{ union} = \mathbf{E}(\exists)} \boxed{\frac{S}{\exists} \mathbf{E}(\frac{R}{\exists}) \mathbf{E}(\exists)} \xrightarrow{\mathbf{E} \text{ a functor, } E(X)E(Y) = E(XY)} \boxed{\frac{S}{\exists} \mathbf{E}(\frac{R}{\exists} \exists)} \\ & \xrightarrow{(10b), \frac{R}{\exists} \exists = R} \boxed{\frac{S}{\exists} \mathbf{E}(R)} \xrightarrow{(10b), \text{ absorption}} \boxed{\frac{SR}{\exists}} \end{aligned} \quad (10.2b)$$

the first three steps only unfold  $\diamond$  —  $\mathbf{E}$  is a functor, so  $\mathbf{E}(\frac{R}{\exists})\text{union} = \mathbf{E}(\frac{R}{\exists} \exists) = \mathbf{E}(R)$  and the **union** is gone. Absorption, the row of (10b), is the whole law, and it is the functoriality of  $\frac{\cdot}{\exists}$ .

$1$  goes to  $\frac{1}{\exists}$ , the Kleisli identity, by definition — so  $\frac{\cdot}{\exists}$  is an isomorphism of categories  $\mathcal{A} \cong \text{Kleisli}(\mathbf{E})$ , and §10.1's  $i \dashv \mathbf{E}$  is the Kleisli adjunction.  $i$  is the identity on objects, so every object of the allegory is free.

## §11. Relator

Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from  $R$  to  $S$  IS  $R \sqsubseteq S$ . A **relator**  $F : \mathcal{C} \rightarrow \mathcal{D}$  is a 2-functor between allegories:

$$F(1) = 1 \quad F(RS) = F(R)F(S) \quad R \sqsubseteq S \implies F(R) \sqsubseteq F(S)$$

Preserving  $\circ$  is **not** asked for —  $\circ$  is an identity-on-objects involution  $\mathcal{C}^{\circ} \rightarrow \mathcal{C}$ , no part of the 2-category.

| the statement                                                                       | B&dM                      | (11b) |
|-------------------------------------------------------------------------------------|---------------------------|-------|
| For $f$ a map, $F(f)$ is a map and $F(f^{\circ}) = F(f)^{\circ}$ .                  | Lemma 5.1                 |       |
| Over a <b>tabular</b> allegory a functor is a relator $\iff$ it preserves $\circ$ . | Theorem 5.1               |       |
| $F(R^{\circ}) = F(R)^{\circ}$ for every $R$ , so $F(R)^{\circ}$ needs no bracket.   | after Theorem 5.1, p. 113 |       |
| Two relators agreeing on maps are equal.                                            | Corollary 5.1             |       |
| $F(X \cap Y) = F(X) \cap F(Y)$ for $X, Y$ coreflexive.                              | Ex 5.2                    |       |
| $F(R \cap S) \sqsubseteq F(R) \cap F(S)$ , and strictly.                            | Ex 5.2, the restriction   |       |
| $F(\text{Dom}(R)) = \text{Dom}(F(R))$ for $F$ preserving $\circ$ .                  | —                         |       |

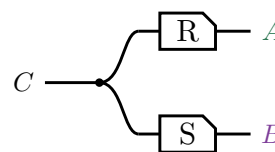
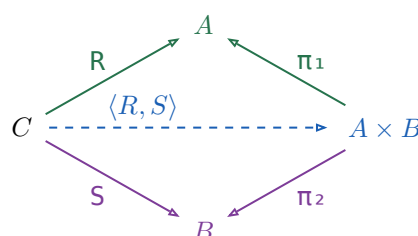
The **power relator**  $P$  —  $xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$  — is where the fourth is strict: for  $R = \{(a_1, b_1), (a_2, b_2)\}$  and  $S = \{(a_1, b_2), (a_2, b_1)\}$  the pair  $(\{a_1, a_2\}, \{b_1, b_2\})$  is in  $P(R) \cap P(S)$ , while  $R \cap S = \emptyset$ .

### §11.1. Fork $\langle R, S \rangle$

The **fork** of  $R : C \rightarrow A$  and  $S : C \rightarrow B$  is  $\langle R, S \rangle \triangleq R \pi_1^{\circ} \cap S \pi_2^{\circ}$ , where  $(\pi_1, \pi_2)$  is the tabulation of  $\tau$ .

$$\langle R, S \rangle \pi_1 = \text{Dom}(S) R$$

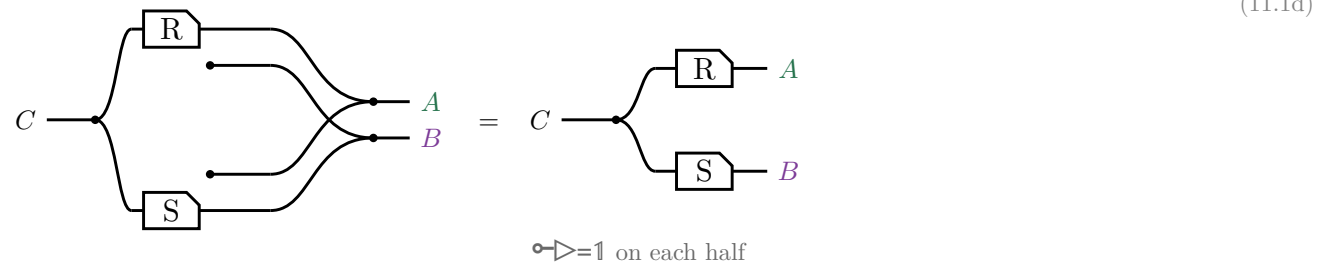
$$\langle R, S \rangle \pi_2 = \text{Dom}(R) S$$





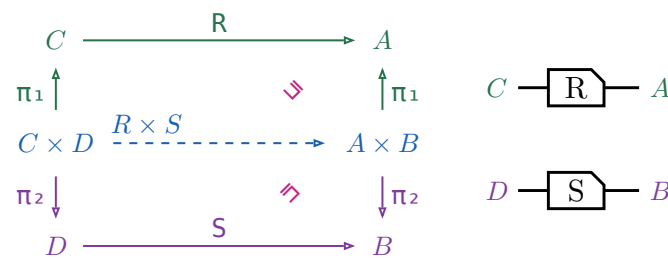
A domain is coreflexive, so  $\langle R, S \rangle \pi_1 \sqsubseteq R$ , with equality exactly when  $S$  is entire; for maps both triangles commute and  $\langle f, g \rangle$  is unique. In  $\mathbf{Rel}$ ,  $c \langle R, S \rangle (a, b)$  iff  $c R a$  and  $c S b$  — copy  $c$ , then  $R$  on one strand and  $S$  on the other, which is  $\triangleleft(R \otimes S)$  on the right.

No  $\circ$  survives the translation.  $\pi_1 = 1 \otimes \circ$  discards the second component, so  $\pi_1 \circ = 1 \otimes \circ$  **creates** one out of nothing, and  $\eta$  is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:



### §11.1.1. Relational product $R \times S$

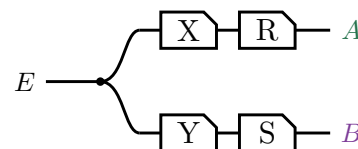
$R \times S \triangleq \langle \pi_1 R, \pi_2 S \rangle$ , a relator in each argument but no longer a categorical product.



Right-then-up is  $(R \times S) \pi_1$ , up-then-right is  $\pi_1 R$ , and  $(R \times S) \pi_1 \sqsubseteq \pi_1 R$ , equality when  $S$  is entire. In  $\mathbf{Rel}$ ,  $(c, d) (R \times S) (a, b)$  iff  $c R a$  and  $d S b$  — two strands side by side, no copy dot:  $R \times S = R \otimes S$ .

### §11.1.2. Absorption

For  $X : E \rightarrow C$  and  $Y : E \rightarrow D$ ,  $\langle X, Y \rangle (R \times S) = \langle XR, YS \rangle$ . Both sides are this picture:



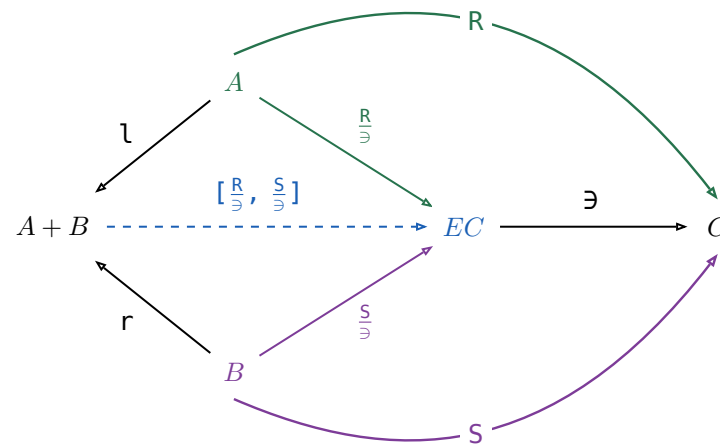
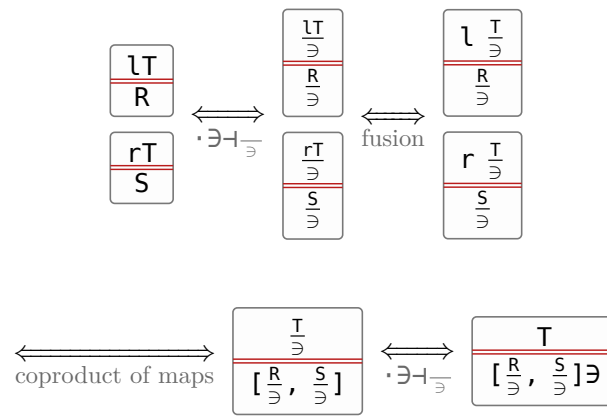
| the law                                                                                                                                                                                                                                                                                                                                                                      | picture |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 1 $R \times S = (\pi_1 R, \pi_2 S)$                                                                                                                                                                                                                                                                                                                                          |         |
| 2 $(X, Y) (R \times S) = (XR, YS)$<br>both sides are the same strokes —<br>(11.1.2a). Its $\sqsubseteq$ half is Ex 5.8, that<br>half at $S := \mathbb{1}$ and at $R := \mathbb{1}$<br>B&dM's (5.4), (5.5) — stages of<br>their proof of this row; Ex 5.6 is<br>the $(R \times S) (U \times V) = (RU) \times (SV)$ it<br>yields, at $R := \mathbb{1}$ and $V := \mathbb{1}$ . |         |
| 3 $(R, S) \pi_1 = \text{Dom}(S) R$<br>(11.1b)                                                                                                                                                                                                                                                                                                                                |         |
| 4 $(R, S) \pi_2 = \text{Dom}(R) S$<br>(11.1b)                                                                                                                                                                                                                                                                                                                                |         |
| 5 $(X, Y) (R, S)^\circ = (XR^\circ) \cap (YS^\circ)$                                                                                                                                                                                                                                                                                                                         |         |
| 6 $(R, S)^\circ (P, Q) \sqsubseteq (R^\circ P) \times (S^\circ Q)$                                                                                                                                                                                                                                                                                                           |         |
| 7 $f(R, S) = (fR, fS)$<br>$f$ a map; it fails for an arbitrary<br>arrow                                                                                                                                                                                                                                                                                                      |         |
| 8 $F(R \times S) \text{unzip}(F) = \text{unzip}(F)$<br>$(F(R) \times F(S))$<br>$\text{unzip}(F) \triangleq (F(\pi_1), F(\pi_2))$ , a<br>map                                                                                                                                                                                                                                  |         |
| 9 $g = \text{curry}(f) \iff (g \times \mathbb{1}) \text{eval} = f$<br>reading $\times$ as the relational<br>product, does $\text{Rel}$ have<br>exponentials?                                                                                                                                                                                                                 |         |

## §11.2. Coproduct $[R, S] : A + B \rightarrow C$

| the statement                                                                                                                                                                                                       | picture | (11.2a) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------|
| $[R, S] \triangleq \mathfrak{l} \circ R \mathfrak{u} r \circ S$<br>The tape is the union — a particle entering at $A+B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint. |         |         |
| $[R, S] = [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$                                                                                                                                            |         |         |
| $R + S \triangleq [R \mathfrak{l}, S \mathfrak{r}]$                                                                                                                                                                 |         |         |
| $\mathfrak{l}[R, S] = R$ , $\mathfrak{r}[R, S] = S$ , and $[R, S]$ is the only such arrow                                                                                                                           |         |         |
| $\mathfrak{l} \mathfrak{l}^\circ = \mathfrak{l} = \mathfrak{r} \mathfrak{r}^\circ$                                                                                                                                  |         |         |
| $\mathfrak{l} \mathfrak{r}^\circ = \mathfrak{l} = \mathfrak{r} \mathfrak{l}^\circ$                                                                                                                                  |         |         |
| $\mathfrak{l}^\circ \mathfrak{l} \mathfrak{u} \mathfrak{r}^\circ \mathfrak{r} = \mathfrak{l}$                                                                                                                       |         |         |
| $[U, V]^\circ [R, S] = U^\circ R \mathfrak{u} V^\circ S$                                                                                                                                                            |         |         |

### §11.2.1. $[R, S] \triangleq [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$

The universal-property row is not free:  $\mathfrak{l}, \mathfrak{r}$  were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because  $\mathfrak{S}$  sends an arrow  $A \rightarrow C$  to a map  $A \rightarrow EC$  reversibly, so the map coproduct can be applied underneath it. For any  $T : A+B \rightarrow C$ ,



Nothing here holds only up to  $\mathfrak{E}$ : every triangle commutes on the nose, which is the difference from the fork above. The border spells  $[R, S] = [\frac{R}{\mathfrak{S}}, \frac{S}{\mathfrak{S}}] \mathfrak{S}$ , and pushing  $\mathfrak{S}$  into the union that  $[\cdot, \cdot]$  on maps already is turns that back into the definition,  $(\mathfrak{l}^\circ \frac{R}{\mathfrak{S}} \mathfrak{u} \mathfrak{r}^\circ \frac{S}{\mathfrak{S}}) \mathfrak{S} = \mathfrak{l}^\circ R \mathfrak{u} r^\circ S$ .

| B&dM    | the law                                                                                                                                                                                                                           | picture |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| (5.9)   | $[R, S] = (\mathbf{l} \circ R) \cup (\mathbf{r} \circ S)$<br>(11.2a)'s first row                                                                                                                                                  |         |
| (5.10)  | $R+S = [R\mathbf{l}, S\mathbf{r}]$                                                                                                                                                                                                |         |
| (5.11)  | $[U, V] \circ [R, S] = (U \circ R) \cup (V \circ S)$<br>(11.2a)'s last row                                                                                                                                                        |         |
| Ex 5.12 | $X \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{l} \circ$ and<br>$Y \triangleq [\mathbf{1}, \mathbf{1}] = \mathbf{r} \circ$ ,<br>so $(X\mathbf{l}) \cup (Y\mathbf{r}) = [\mathbf{l}, \mathbf{r}] = \mathbf{1}$<br>which is (5.9) |         |
| Ex 5.13 | prove (5.11), and say why duality<br>does not carry it over from the<br>product law                                                                                                                                               |         |
| Ex 5.14 | $(R+S) \cap ([P, Q][U, V] \circ)$<br>$= (R \cap (PU \circ)) + (S \cap (QV \circ))$<br>[P, Q][U, V] \circ is a full 2x2 of<br>composites; R+S is diagonal, so the<br>meet cuts the two off-diagonal<br>branches                    |         |

### §11.3. The power relator $P(R)$

For  $R : A \rightarrow B$ ,

$$P(R) \triangleq ((\exists R)/\exists) \cap ((\exists R^\circ)/\exists)^\circ : EA \rightarrow EB$$

$$E(R) \triangleq \frac{\exists R}{\exists} = ((\exists R)/\exists) \cap (\exists/(\exists R))^\circ$$

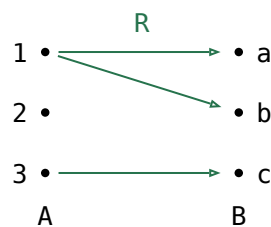
$$xs \ P(R) \ ys \iff (\forall a \in xs. \exists b \in ys. a \ R \ b) \wedge (\forall b \in ys. \exists a \in xs. a \ R \ b)$$

Every element of  $xs$  is related by  $R$  to some element of  $ys$ , and conversely.

(11.3a)

|                                     | in Rel                                                                                              | in words                                                                       | $\text{im}(xs)$              | partner                                       |
|-------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|------------------------------|-----------------------------------------------|
| $(\exists R)/\exists$               | $\forall y \in ys. \exists x \in xs. x \ R \ y$                                                     | $\forall y. \text{ some } xs \ R \ y$                                          | $ys \subseteq \text{im}(xs)$ | every $y$ has a partner                       |
| $(\exists/(\exists R))^\circ$       | $\forall y. (\exists x \in xs. x \ R \ y) \rightarrow y \in ys$                                     | every $y$ with some $xs \ R \ y$ is in $ys$                                    | $ys \supseteq \text{im}(xs)$ | $ys$ leaves out no partner                    |
| $E(R)$                              | $ys = \{y \mid \exists x \in xs. x \ R \ y\}$                                                       | $ys = \text{every } y \text{ with some } xs \ R \ y$                           | $ys = \text{im}(xs)$         | every $y$ has a partner, and none is left out |
| $((\exists R^\circ)/\exists)^\circ$ | $\forall x \in xs. \exists y \in ys. x \ R \ y$                                                     | $\forall x. x \ R \ \text{some } ys$                                           | —                            | every $x$ has a partner                       |
| $P(R)$                              | $\forall y \in ys. \exists x \in xs. x \ R \ y$ and $\forall x \in xs. \exists y \in ys. x \ R \ y$ | $\forall y. \text{ some } xs \ R \ y$ and $\forall x. x \ R \ \text{some } ys$ | —                            | every $x$ and every $y$ has a partner         |

(11.3b)



(11.3c)

| $xs$          | $(\exists R)/\exists$  | $(\exists/(\exists R))^\circ$ | $E(R)$      | $((\exists R^\circ)/\exists)^\circ$ | $P(R)$        |
|---------------|------------------------|-------------------------------|-------------|-------------------------------------|---------------|
| $\emptyset$   | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | all 8 subsets of $abc$              | $\emptyset$   |
| $\{1\}$       | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | $a, b, ab, ac, bc, abc$             | $a, b, ab$    |
| $\{2\}$       | $\emptyset$            | all 8 subsets of $abc$        | $\emptyset$ | none                                | none          |
| $\{3\}$       | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | $c, ac, bc, abc$                    | $c$           |
| $\{1, 2\}$    | $\emptyset, a, b, ab$  | $ab, abc$                     | $ab$        | none                                | none          |
| $\{1, 3\}$    | all 8 subsets of $abc$ | $abc$                         | $abc$       | $ac, bc, abc$                       | $ac, bc, abc$ |
| $\{2, 3\}$    | $\emptyset, c$         | $c, ac, bc, abc$              | $c$         | none                                | none          |
| $\{1, 2, 3\}$ | all 8 subsets of $abc$ | $abc$                         | $abc$       | none                                | none          |

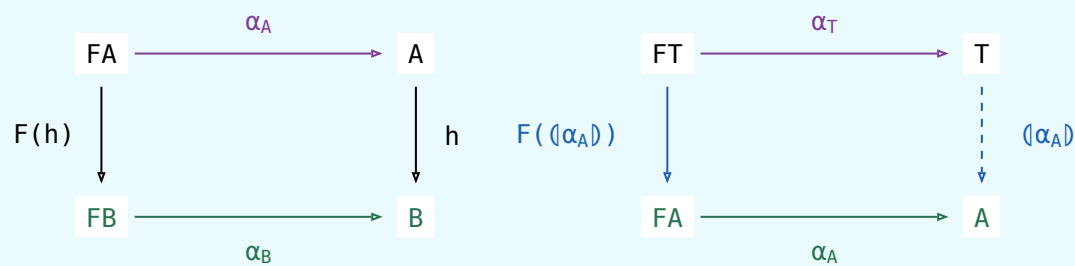
| law                                                                                          | the reading                                                                                                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $X \subseteq P(R) \iff X \subseteq \exists R \text{ and } X^\circ \subseteq \exists R^\circ$ | One containment, and the same one at $R^\circ$ — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$ , and $R \subseteq S \implies P(R) \subseteq P(S)$ .                                                                                                                        |
| $P(1) = \frac{\exists}{\exists} = 1$                                                         | The straightness axiom verbatim: extensionality is $P$ 's unit law.                                                                                                                                                                                                                                              |
| $P(f) = \frac{\exists f}{\exists}$ , for $f$ a map                                           | In Rel, $xs \ P(f) \ ys \iff ys = \{f \ a \mid a \in xs\}$ . The half at $f^\circ$ says every $a \in xs$ has its $f$ $a$ on $ys$ ; $f$ has just the one image per $a$ , so that already says $ys$ contains everything $xs$ reaches, which is the fraction's second half. For a map the two definitions coincide. |
| $P(RS) = P(R)P(S)$                                                                           | $\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(RS)$ .                                                                                                                                                      |

(11.3d)

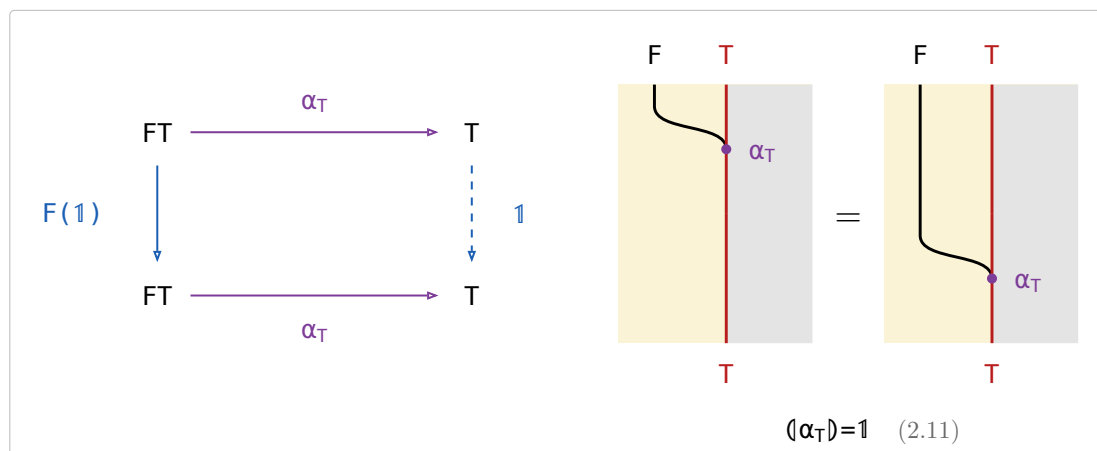
## §11.4. Initial algebra

An **F-algebra** on  $A$  is a map  $\alpha_A : FA \rightarrow A$ . An **F-homomorphism** from  $\alpha_A$  to  $\alpha_B$  is a map  $h : A \rightarrow B$  with  $\alpha_B h = F(h) \alpha_A$ . The **initial algebra**  $\alpha_T : FT \rightarrow T$  is the F-algebra with exactly one F-homomorphism  $\langle \alpha_A \rangle : T \rightarrow A$  to every F-algebra  $\alpha_A$ .

(11.4a)



### §11.4.1. Reflection



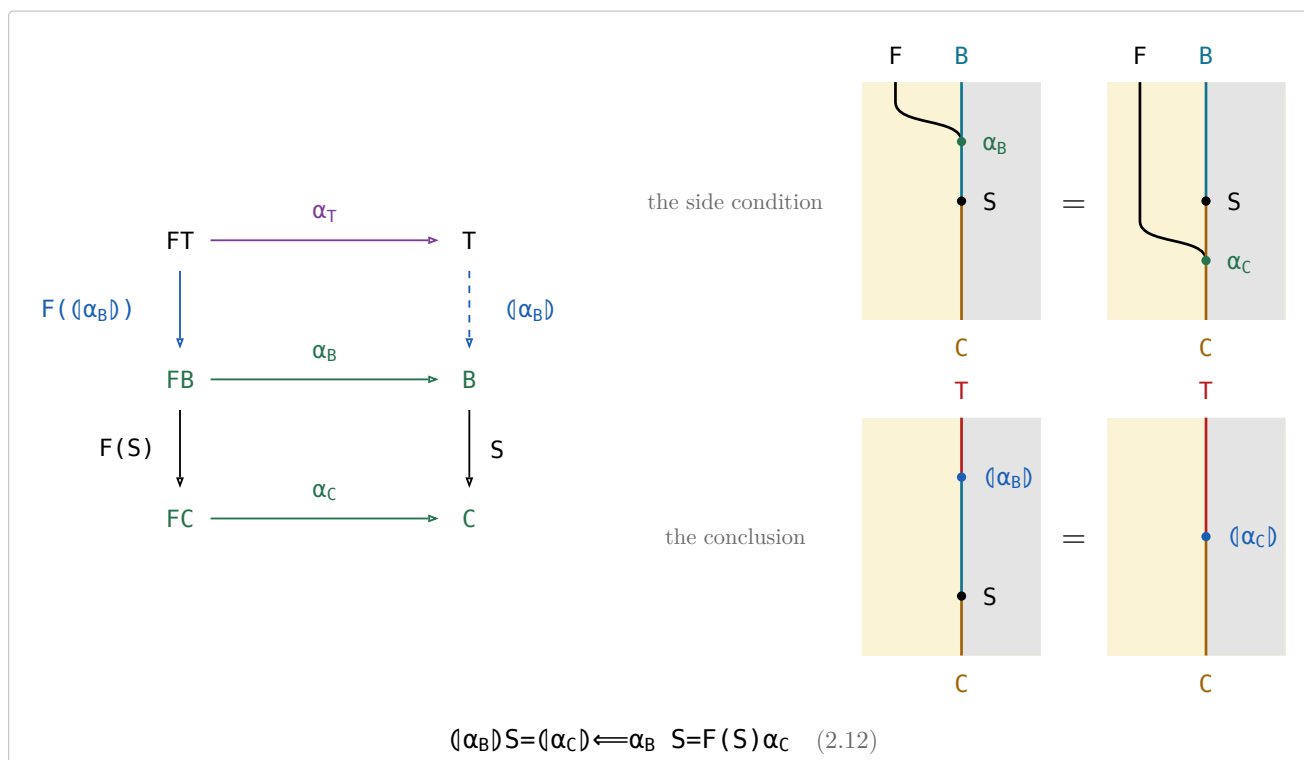
(11.4.1a)

Taking a value apart with  $\alpha_T$  and putting it straight back is doing nothing.

### §11.4.2. Fusion

Fusion rewrites  $\langle \alpha_B \rangle S$  through a second algebra  $\alpha_C : FC \rightarrow C$  along an arrow  $S : B \rightarrow C$ .

(11.4.2a)



## §11.5. Type relator

(11.5a)

Let  $F$  be a binary relator with initial type  $(\alpha, T)$ , so  $T$  is a type functor.  $F(R, S)$  is its action on a pair, and  $F(X)$  abbreviates  $F(1, X)$ , the  $F$  of the reduce section. For every object  $A$  the initial algebra is  $\alpha : F(A, TA) \rightarrow TA$ , among the maps.  $T$  acts on an arrow  $R : A \rightarrow B$  by

$$T(R) = (F(R, 1)\alpha) : TA \rightarrow TB$$

(11.5b)

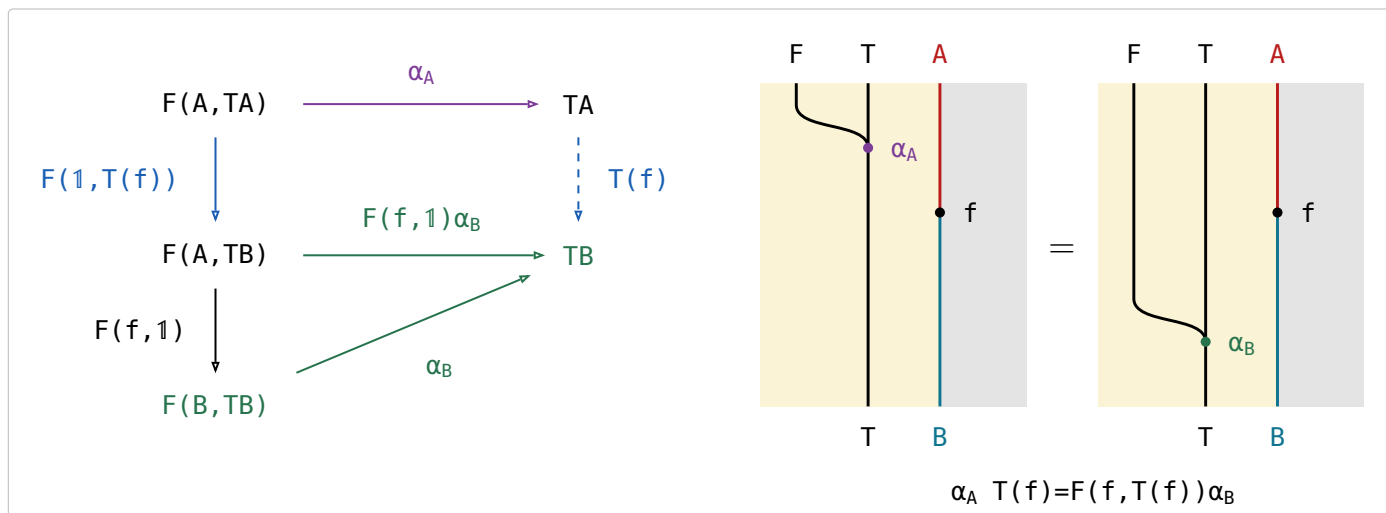
| name                   | law                               | what it says                                                                                                                                                                                                |
|------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the defining equation  | $T(R) = (F(R, 1)\alpha)$          | Rebuild the structure with $\alpha$ , applying $R$ to the parameter on the way.                                                                                                                             |
| functor                | $T(1) = 1$ and $T(R)T(S) = T(RS)$ | Acting by the identity changes nothing, and two actions in a row are one action.                                                                                                                            |
| type functor fusion    | $T(R)(Q) = (F(R, 1)Q)$            | A relator action followed by a fold is a single fold — the intermediate structure is never built. The side condition holds because $F$ is a bifunctor — $F(R, 1)F(1, (Q)) = F(R, (Q)) = F(1, (Q))F(R, 1)$ . |
| naturality of $\alpha$ | $\alpha T(R) = F(R, T(R))\alpha$  | Building and then mapping is the same as mapping the parts and then building, so $\alpha$ is natural from $G(R) = F(R, T(R))$ to $T$ .                                                                      |
| type relator           | $T(R)^\circ = T(R^\circ)$         | A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.                                                                                                        |

### §11.5.1. Type functor

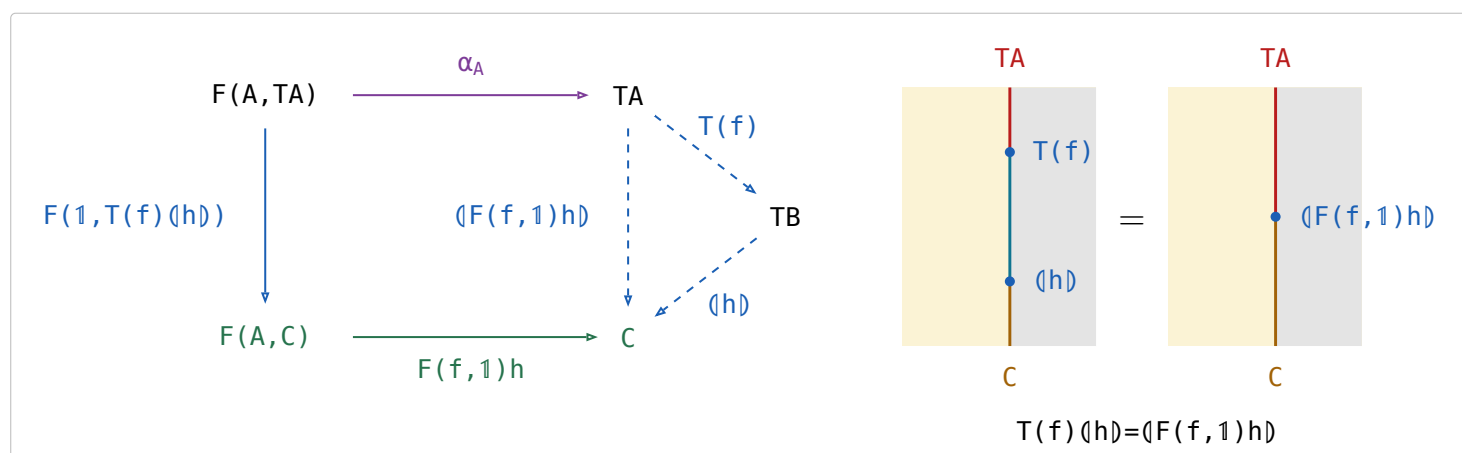
Let  $F$  be a bifunctor taking both the parameter  $A$  and the recursive position  $TA$ , with an initial algebra  $\alpha_A : F(A, TA) \rightarrow TA$  for every object  $A$ . Then  $T$  is a functor, acting on a map  $f : A \rightarrow B$  by

$$T(f) \triangleq (F(f, 1) \alpha_B) : TA \rightarrow TB$$

(11.5.1a)



(11.5.1b)



(11.5.1c)

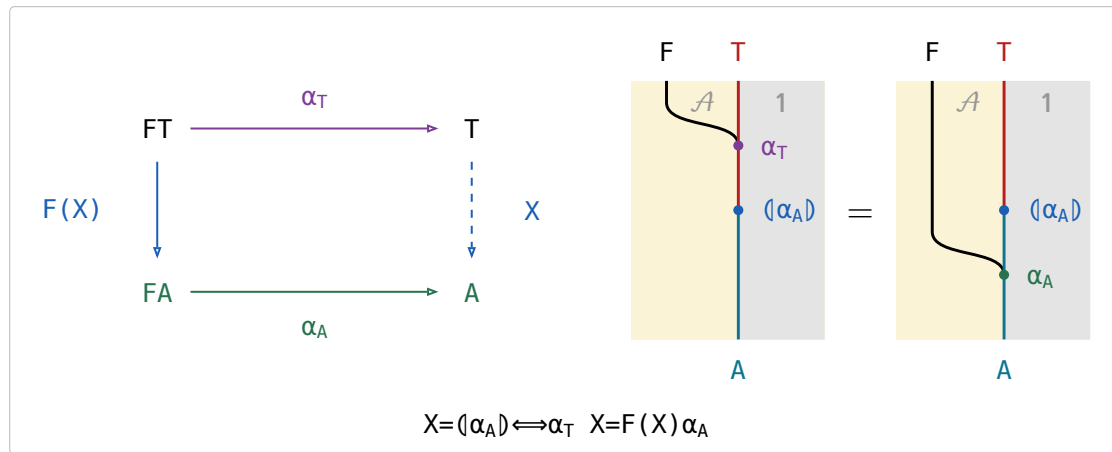


## §11.6. Reduce

let  $F$  be a relator and has **initial algebra**  $\alpha_T : FT \rightarrow T$  in the subcategory of functions.  $\alpha_T$  is also initial in the allegory:

(11.6a)

### §11.6.1. The defining equation



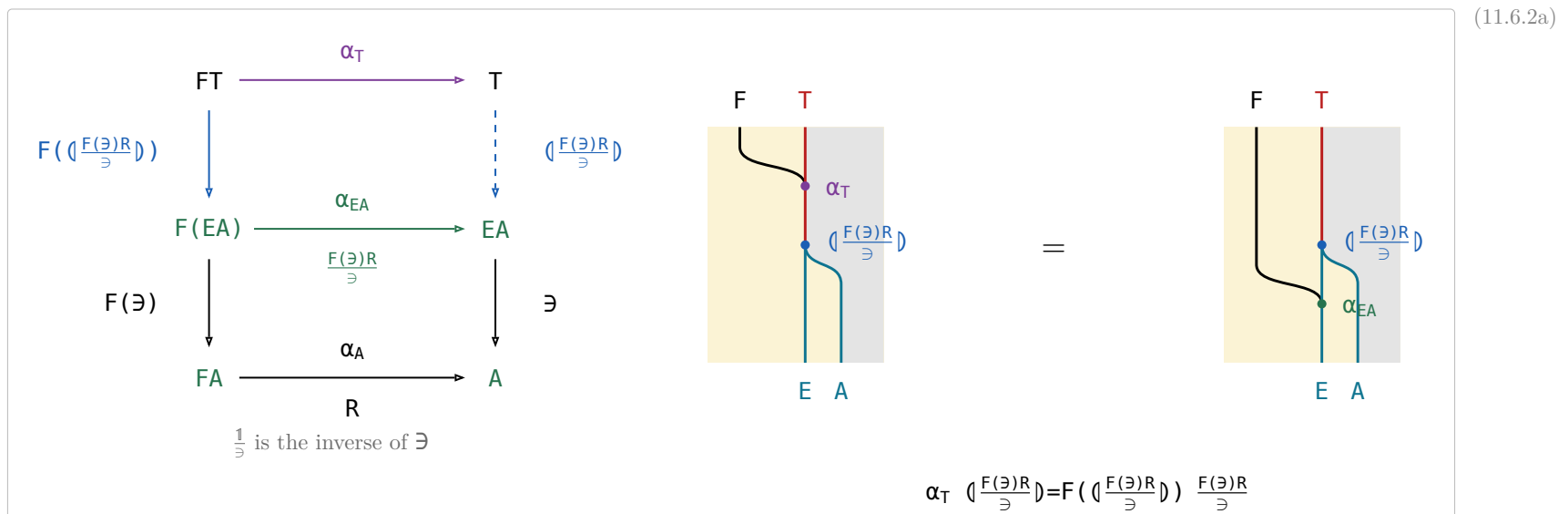
(11.6.1a)

An  $F$ -algebra is a **weakened** natural transformation. A transformation  $F \Rightarrow 1$  on  $\mathcal{A}$  would need a component  $FX \rightarrow X$  at every object and a commuting square at every arrow, but  $F$ -Algebra only need it works on  $T$  and  $A$ .

| $\llbracket [c, f] \rrbracket = \text{reduce}(c, f)$ |                                                                                                                                         |                                                                                                                                                |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| part                                                 | Nat                                                                                                                                     | [A]                                                                                                                                            |
| datatype                                             | $\text{Nat} ::= \text{zero} \mid \text{succ Nat}$                                                                                       | $[A] ::= \text{nil} \mid \text{cons } (A, [A])$                                                                                                |
| base functor $F$                                     | $F(X) = 1 + X$                                                                                                                          | $F(X) = 1 + A \times X$                                                                                                                        |
| initial algebra $\alpha$                             | $\alpha = [\text{zero}, \text{succ}]$<br>$: 1 + \text{Nat} \rightarrow \text{Nat}$                                                      | $\alpha = [\text{nil}, \text{cons}]$<br>$: 1 + A \times [A] \rightarrow [A]$                                                                   |
| the fold, pointwise                                  | $\llbracket [c, f] \rrbracket \text{ zero} = c$<br>$\llbracket [c, f] \rrbracket (\text{succ } n) = f (\llbracket [c, f] \rrbracket n)$ | $\llbracket [c, f] \rrbracket \text{ nil} = c$<br>$\llbracket [c, f] \rrbracket (\text{cons } (a, x)) = f (a, \llbracket [c, f] \rrbracket x)$ |

(11.6.1b)

### §11.6.2. $\llbracket (R) \rrbracket = (\llbracket \frac{F(\exists)R}{\exists} \rrbracket) \exists$



(11.6.2a)

$$\begin{array}{c}
 \frac{\alpha_T X}{F(X)R} \xleftrightarrow{\cdot \exists^{-1} \exists} \frac{\frac{\alpha_T X}{\exists}}{\frac{F(X)R}{\exists}} \xleftrightarrow{\cdot \exists^{-1} \exists} \frac{\frac{\alpha_T X}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}} \\
 \\
 \xleftrightarrow{\text{relator, fusion twice}} \frac{\frac{\alpha_T \frac{X}{\exists}}{F(\frac{X}{\exists})} \frac{F(\exists)R}{\exists}}{\frac{F(\frac{X}{\exists})R}{\exists}} \xleftrightarrow{\text{reduce of maps}} \frac{\frac{X}{\exists}}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket} \xleftrightarrow{\cdot \exists^{-1} \exists} \frac{X}{\llbracket \frac{F(\exists)R}{\exists} \rrbracket \exists}
 \end{array}$$

(11.6.2b)

§11.6.3.  $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$ 

(11.6.3a)

$\varphi$  a **monotonic** mapping of the hom-set  $A \rightarrow B$  into itself:  $X \sqsubseteq Y \implies \varphi(X) \sqsubseteq \varphi(Y)$ .

$(\mu X : \varphi(X))$  the least  $X : A \rightarrow B$  with  $\varphi(X) \sqsubseteq X$ .

| the law                                                                                                 | what it says                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi(Y) \sqsubseteq Y \implies (\mu X : \varphi(X)) \sqsubseteq Y$<br>Theorem 6.1, lean:AOP.A6_2.mu | to bound $(\mu X : \varphi(X))$ above, exhibit one $Y$ the body does not grow past — the half §11.6.4 and every chapter after it uses                                          |
| $\varphi((\mu X : \varphi(X))) = (\mu X : \varphi(X))$<br>Theorem 6.1, lean:AOP.A6_2.mu_fixed           | <b>Knaster–Tarski:</b> the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$ , so the least prefix point and the least fixed point are one relation |

§11.6.4.  $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$ 

(11.6.4a)

| $(S)^\circ (R) = (\mu X : S^\circ F(X) R)$<br>Theorem 6.2, lean:AOP.A6_3.hylo_eq_mu; $R : FA \rightarrow A$ , $S : FB \rightarrow B$ , $\alpha : FT \rightarrow T$ initial                                                                                                           |               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                              | Hinze–Marsden |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ (R)) \text{ --- } R \text{ --- } A$<br>$S^\circ F((S)^\circ (R)) R$<br>the body at $(S)^\circ (R)$                                                                                                                                  |               |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ) \text{ --- } F((R)) \text{ --- } R \text{ --- } A$<br>$= S^\circ F((S)^\circ) F((R)) R$<br>$F(RS) = F(R) F(S) \text{ — (11a)}$                                                                                                     |               |
| $B \text{ --- } S^\circ \text{ --- } F((S)^\circ) \text{ --- } \alpha \text{ --- } (R) \text{ --- } A$<br>$= S^\circ F((S)^\circ) \alpha (R)$<br>(11.6.1a) at $R: F((R)) R = \alpha (R)$                                                                                             |               |
| $B \text{ --- } (S)^\circ \text{ --- } \alpha^\circ \text{ --- } \alpha \text{ --- } (R) \text{ --- } A$<br>$= (S)^\circ \alpha^\circ \alpha (R)$<br>(11.6.1a) at $S$ conversed: $(S)^\circ \alpha^\circ = S^\circ F((S))^\circ$ , and $F((S))^\circ = F((S)^\circ) \text{ — (11b)}$ |               |
| $B \text{ --- } (S)^\circ \text{ --- } (R) \text{ --- } A$<br>$= (S)^\circ (R)$<br>$\alpha^\circ \alpha = 1$ , Lambek — lean:AOP.A6_2.InitialAlgebra.recip_alpha_alpha                                                                                                               |               |

| $S^\circ F(X) R \sqsubseteq X \implies (S)^\circ (R) \sqsubseteq X$<br>Theorem 6.2, lean:AOP.A6_3.hylo_le_of_prefixed                                                                                                                                                                                                                                                                                                                                       |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Hinze–Marsden |
| $B \dashv \boxed{(S)^\circ} \boxed{(R)} \dashv \sqsubseteq \dashv \boxed{X} \dashv A$ <p> <math>(S)^\circ (R) \sqsubseteq X</math><br/>                         the conclusion                     </p>                                                                                                                                                                                                                                                     |               |
| $T \dashv \boxed{(R)} \dashv \sqsubseteq \dashv \boxed{(S)^\circ \setminus X} \dashv A$ <p> <math>\Leftrightarrow (R) \sqsubseteq (S)^\circ \setminus X</math><br/>                         (1.1a)'s <math>S \cdot \neg S \setminus</math> at <math>(S)^\circ</math> </p>                                                                                                                                                                                   |               |
| $T \dashv \boxed{\alpha^\circ} \boxed{F((S)^\circ \setminus X)} \boxed{R} \dashv \sqsubseteq \dashv \boxed{(S)^\circ \setminus X} \dashv A$ <p> <math>\Leftarrow \alpha^\circ F((S)^\circ \setminus X) R \sqsubseteq (S)^\circ \setminus X</math><br/>                         (6.2) <math>(R) = (\mu X : \alpha^\circ F(X) R)</math> — (11.6.1a) and (11.6.3b);<br/>                         lean:AOP.A6_2.relCata_le_of_prefixed                     </p> |               |
| $B \dashv \boxed{(S)^\circ} \boxed{\alpha^\circ} \boxed{F((S)^\circ \setminus X)} \boxed{R} \dashv \sqsubseteq \dashv \boxed{X} \dashv A$ <p> <math>\Leftrightarrow (S)^\circ \alpha^\circ F((S)^\circ \setminus X) R \sqsubseteq X</math><br/>                         (1.1a)'s <math>S \cdot \neg S \setminus</math> at <math>(S)^\circ</math> </p>                                                                                                       |               |
| $B \dashv \boxed{S^\circ} \boxed{F((S)^\circ)} \boxed{F((S)^\circ \setminus X)} \boxed{R} \dashv \sqsubseteq \dashv \boxed{X} \dashv A$ <p> <math>\Leftrightarrow S^\circ F((S)^\circ) F((S)^\circ \setminus X) R \sqsubseteq X</math><br/> <math>(S)^\circ \alpha^\circ = S^\circ F((S)^\circ)</math> — (11.6.4a)                     </p>                                                                                                                 |               |
| $B \dashv \boxed{S^\circ} \boxed{F(X)} \boxed{R} \dashv \sqsubseteq \dashv \boxed{X} \dashv A$ <p> <math>\Leftarrow S^\circ F(X) R \sqsubseteq X</math><br/> <math>F(RS) = F(R)F(S)</math> — (11a) — and <math>(S)^\circ ((S)^\circ \setminus X) \sqsubseteq X</math> — (1.1a)                     </p>                                                                                                                                                     |               |

## §12. Combinatorial functions

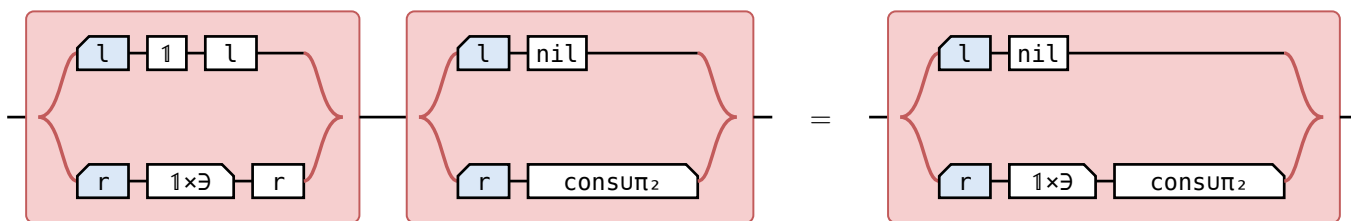
| definition                                                                                                           | type                                  | note                                                                                                                                                | (12a) |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| $[A] ::= \text{nil} \mid \text{cons } (A, [A])$                                                                      | $\mathcal{A} \rightarrow \mathcal{A}$ | B&dM's <b>listr</b> under the short name it keeps from p. 125 on.                                                                                   |       |
| $\text{list}(R) \triangleq ([\text{nil}, (R \circ 1) \text{ cons}])$                                                 | $[A] \rightarrow [B]$                 | The relator's action on $R : A \rightarrow B$ : one $R$ per element, the shape untouched.                                                           |       |
| $\text{subseq} \triangleq ([\text{nil}, \text{cons} \pi_2])$                                                         | $[A] \rightarrow [A]$                 | $xs \text{ subseq } ys$ : $ys$ is $xs$ with elements dropped — <b>cons</b> keeps the head, $\pi_2$ drops it.                                        |       |
| $\text{prefix} \triangleq ([\text{nil}, \text{nil} \cup \text{cons}])$<br>$= \text{cat}^\circ \pi_1 = \text{init}^*$ | $[A] \rightarrow [A]$                 | $ys$ is an initial segment of $xs$ ; the first <b>nil</b> is where it stops early.<br>$\text{init} \triangleq \text{snoc}^\circ \pi_1$ .            |       |
| $\text{suffix} \triangleq \text{cat}^\circ \pi_2 = \text{tail}^*$                                                    | $[A] \rightarrow [A]$                 | The dual, $\text{tail} \triangleq \text{cons}^\circ \pi_2$ ; as a reduce it needs <b>snoc</b> -lists.                                               |       |
| $\text{segment} \triangleq \text{suffix} \text{ prefix}$                                                             | $[A] \rightarrow [A]$                 | A contiguous stretch of $xs$ : a suffix, then a prefix of that.                                                                                     |       |
| $\text{partition} \triangleq \text{concat}^\circ$                                                                    | $[A] \rightarrow [[A]]$               | <b>cat</b> restricted to a non-empty first argument, so $ys$ is a list of non-empty segments of $xs$ .                                              |       |
| $\text{concat} \triangleq ([\text{nil}, \text{cat}])$                                                                | $[[A]] \rightarrow [A]$               | Joins the segments back up, which is why its converse splits a list.                                                                                |       |
| <b>inits</b>                                                                                                         | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{prefix}}{\ni}$ , listing the prefixes by increasing length.                                                                 |       |
| <b>tails</b>                                                                                                         | $[A] \rightarrow [[A]]$               | Implements $\frac{\text{suffix}}{\ni}$ by decreasing length — the opposite order.                                                                   |       |
| $\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\ni} \text{ est}(R^\circ)$                                 | $[A] \rightarrow [A]$                 | The longest subsequence of $xs$ whose every element passes $p$ . Ex 7.41; $\text{est}(R^\circ)$ is (13.1a)                                          |       |
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                                | $[A] \rightarrow [A]$                 | The preorder <b>filter</b> and <b>takewhile</b> maximise over: the longer list wins. $\geq \triangleq \leq^\circ$                                   |       |
| $\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\ni} \text{ est}(R^\circ)$                              | $[A] \rightarrow [A]$                 | The same with <b>prefix</b> for <b>subseq</b> : the longest prefix whose every element passes $p$ . Ex 7.39                                         |       |
| $\text{mss} \triangleq \frac{\text{segment sum}}{\ni} \text{ est}(\geq)$                                             | $[\text{Int}] \rightarrow \text{Int}$ | Maximum segment sum. <b>segment</b> = <b>suffix prefix</b> splits it into $\frac{\text{prefix sum}}{\ni} \text{ est}(\geq)$ on each suffix. Ex 7.40 |       |

### §12.1. $\frac{\text{subseq}}{\ni} = ([\text{nil } \frac{1}{\ni}, (\frac{1 \times \ni}{\ni} \text{ E}(\text{cons}), \pi_2) \text{ cup}])$

$$\text{cup} \triangleq \frac{\pi_1 \ni \cup \pi_2 \ni}{\ni} : EA \times EA \rightarrow EA, \text{ so } \frac{R \cup S}{\ni} = (\frac{R}{\ni}, \frac{S}{\ni}) \text{ cup.}$$

$$\frac{F(\ni) [\text{nil}, \text{cons} \pi_2]}{\ni} \xlongequal{\quad} \frac{(1 + 1 \times \ni) [\text{nil}, \text{cons} \pi_2]}{\ni} \xlongequal{\quad \text{coproduct} \quad} \frac{[\text{nil}, (1 \times \ni)(\text{cons} \pi_2)]}{\ni}$$

$$\xlongequal{\quad \text{coproduct of maps} \quad} \left[ \frac{\text{nil}}{\ni}, \frac{(1 \times \ni)(\text{cons} \pi_2)}{\ni} \right] \xlongequal{\quad \text{singleton} \quad} \left[ \text{nil } \frac{1}{\ni}, \frac{(1 \times \ni)(\text{cons} \pi_2)}{\ni} \right]$$



B&dM p. 124, “coproduct”:  $R + S \triangleq [Rl, Sr]$  and  $l[R, S] = R$ ,  $r[R, S] = S$  — (11.2a).

$$\frac{\frac{(1 \times \exists)(\text{cons} \pi_2)}{\exists}}{\text{U composes, } \pi_2 \text{ natural}} = \frac{\frac{(1 \times \exists) \text{ cons} \pi_2 \exists}{\exists}}{\text{cup}} = \left\{ \frac{(1 \times \exists) \text{ cons}}{\exists}, \frac{\pi_2 \exists}{\exists} \right\} \text{ cup} \quad (12.1d)$$

$$\frac{}{\text{absorption, fusion}} \left\{ \frac{1 \times \exists}{\exists} E(\text{cons}), \pi_2 \right\} \text{ cup}$$

$$\begin{array}{c} A \times E([A]) \xrightarrow{1 \times \exists} A \times [A] \xrightarrow{\left\{ \begin{array}{c} \text{cons} \\ \pi_2 \end{array} \right\}} [A] \\ = \\ A \times [A] \xrightarrow{\left\{ \begin{array}{c} 1 \times \exists \text{ cons} \\ 1 \times \exists \pi_2 \end{array} \right\}} A \times [A] \xrightarrow{\text{composition over U}} \\ = \\ A \times [A] \xrightarrow{\left\{ \begin{array}{c} 1 \times \exists \text{ cons} \\ \pi_2 \exists \end{array} \right\}} E([A]) \xrightarrow{\exists} [A] \end{array} \quad (12.1e)$$

composition over U π<sub>2</sub> natural

B&dM p. 124, “composition over join, naturality of  $\pi_2$ ”:  $T(X_1 \cup X_2) = TX_1 \cup TX_2$  — (1.2b);  $(1 \times \exists) \pi_2 = \pi_2 \exists$  — (11.1.1b) at  $\pi_2$ , an equality because  $1$  is entire.

$$\begin{array}{ccc} A \times E([A]) & \xrightarrow{\pi_2} & E([A]) \\ \downarrow 1 \times \exists & & \downarrow \exists \\ A \times [A] & \xrightarrow{\pi_2} & [A] \end{array}$$

$$\begin{array}{c} 1 + A \times E([A]) \xrightarrow{\left\{ \begin{array}{c} l \text{ nil } 1 \\ r \left( \begin{array}{c} 1 \times \exists \text{ cons} \\ \pi_2 \end{array} \right) \text{ cup} \end{array} \right\}} E([A]) \end{array} \quad (12.1g)$$

B&dM §5.6, p. 124, which writes  $P\text{cons}$ ;  $\text{cons}$  is a map, and there  $P(\text{cons}) = E(\text{cons})$  — (11.3d).

## §13. Optimisation Problems

### §13.1. $\text{est}(R) \triangleq \exists n(\epsilon \backslash R^\circ)$

(13.1a)

For  $R : A \rightarrow A$ ,  $\text{est}(R) \triangleq \exists n(\epsilon \backslash R^\circ) : EA \rightarrow A$ .  $\text{est}(R)$  is B&dM's  $\min R$ ,  $\text{est}(R^\circ)$  his  $\max R$

$$xs \ (\text{est}(R)) \ x \iff x \in xs \wedge (\forall y \in xs. x R y)$$

$$xs \ (\text{est}(R)) \ x \iff (x \text{ in } xs) \text{ and all } x R \backslash : xs \text{ in } q$$

$$\text{est}(R) = \exists n \text{ all } R^\circ \text{ all } R \triangleq \epsilon \backslash R, q's \text{ all; the chains below keep it written } \epsilon \backslash$$

$$E(R) \triangleq \frac{\exists R}{\exists} : EA \rightarrow EB, xs \ E(R) \ ys \iff ys = \{y \mid \exists x \in xs. x R y\} \text{ the image of } xs, (10b)$$

$$P(R) : EA \rightarrow EB, xs \ P(R) \ ys \iff (\forall x \in xs. \exists y \in ys. x R y) \wedge (\forall y \in ys. \exists x \in xs. x R y) \text{ every } x \text{ and every } y \text{ has a partner, (11.3b)}$$

| the law                                                                                                        | what it says                                                                                                  | (13.1b) |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|---------|
| $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists$ and $X^\circ \sqsubseteq R$                           | in the set, and below every element of it                                                                     |         |
| $\frac{1}{\exists} (\epsilon \backslash R) = R$                                                                | bounding a singleton is bounding its element                                                                  |         |
| $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$                                             | bound $S$ 's image without building the set                                                                   |         |
| $\text{union} \triangleq \frac{\exists \exists}{\exists} : E(EA) \rightarrow EA$                               | flattens a set of sets                                                                                        |         |
| $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$                           | bound a union by bounding each member set                                                                     |         |
| $\frac{1}{\exists} \text{est}(R) = 1 \cap R$                                                                   | a singleton's minimum is its element, where $R$ is reflexive<br>$\frac{S}{\exists} \text{est}(R)$ at $S := 1$ |         |
| $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \backslash R^\circ)$                                        | an $S$ -value that points to every $S$ -value                                                                 |         |
| $\frac{S}{\exists} \text{est}(R) = \frac{S}{\exists} \text{est}(R \cap S^\circ S)$                             | only $R$ between values $S$ gives one argument counts — context                                               |         |
| $E(S) \text{est}(R) = (\exists S) \cap ((\exists S)^\circ \backslash R^\circ)$                                 | the same for the image of a set<br>$\frac{S}{\exists} \text{est}(R)$ at $S := \exists S$                      |         |
| $P(f) \text{est}(R) = \text{est}(f R f^\circ) \ f$                                                             | shunt a function through a minimum                                                                            |         |
| $P(S) \text{est}(R) = (\exists S) \cap (\epsilon \backslash (S R^\circ))$<br>$R$ reflexive                     | fusion with the power relator<br>$\exists$ is the only proof here that tabulates                              |         |
| $P(S) \text{est}(R) \sqsubseteq (\exists S) \cap (\epsilon \backslash (S R^\circ))$                            | the half of the row above that costs nothing                                                                  |         |
| $P(\text{est}(R)) \text{est}(R) \sqsubseteq \text{union} \text{est}(R)$<br>$R$ a preorder                      | a minimum in each set, then a minimum of those                                                                |         |
| $P(\text{est}(R)) \text{est}(R) = P(\text{Dom}(\text{est}(R))) \text{ union } \text{est}(R)$<br>$R$ a preorder | the same as an equality, once empty sets are dropped                                                          |         |

#### §13.1.1. $X \sqsubseteq \text{est}(R) \iff X \sqsubseteq \exists$ and $X^\circ \sqsubseteq R$

(13.1.1a)

$$\boxed{\begin{array}{c} X \\ \text{est}(R) \end{array}} \xleftrightarrow{\Delta \dashv \cap} \boxed{\begin{array}{c} X \\ \exists \\ X \\ \epsilon \backslash R^\circ \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} X \\ \exists \\ \epsilon X \\ R^\circ \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} X \\ \exists \\ X^\circ \exists \\ R \end{array}}$$

#### §13.1.2. $\frac{1}{\exists} (\epsilon \backslash R) = R$

(13.1.2a)

$$\boxed{\begin{array}{c} X \\ \frac{1}{\exists} (\epsilon \backslash R) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} 1^\circ X \\ \frac{1}{\exists} \epsilon \backslash R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} \epsilon \frac{1}{\exists} X \\ R \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} (\frac{1}{\exists} \exists)^\circ X \\ R \end{array}} \xleftrightarrow{i \dashv E} \boxed{\begin{array}{c} X \\ R \end{array}}$$

#### §13.1.3. $\frac{S}{\exists} (\epsilon \backslash R) = S^\circ \backslash R$

$\frac{S}{\exists}$  gathers the  $S$ -image of a point into one set and  $\epsilon \backslash R$  asks that every member of that set be  $R$ -related to the target, so the set cancels and  $S^\circ \backslash R$  asks it of the  $S$ -image directly.

(13.1.3a)

$$\boxed{\begin{array}{c} X \\ \frac{S}{\exists} (\epsilon \backslash R) \end{array}} \xleftrightarrow{f^\circ \cdot \dashv f \cdot} \boxed{\begin{array}{c} \frac{S}{\exists}^\circ X \\ \frac{S}{\exists} \epsilon \backslash R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} \epsilon \frac{S}{\exists}^\circ X \\ R \end{array}} \xleftrightarrow{\circ} \boxed{\begin{array}{c} (\frac{S}{\exists} \exists)^\circ X \\ R \end{array}} \xleftrightarrow{\cdot \exists \dashv \frac{S}{\exists}} \boxed{\begin{array}{c} S^\circ X \\ R \end{array}} \xleftrightarrow{T \cdot \dashv T \backslash} \boxed{\begin{array}{c} X \\ S^\circ \backslash R \end{array}}$$

#### §13.1.4. $\text{union} (\epsilon \backslash R) = \epsilon \backslash (\epsilon \backslash R)$

$$\begin{array}{c} X \\ \hline \text{union } (\in \backslash R) \end{array} \xLeftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \text{union}^\circ X \\ \hline \in \backslash R \end{array} \xLeftrightarrow{T \cdot \neg T \backslash} \begin{array}{c} \in \text{union}^\circ X \\ \hline R \end{array} \xLeftrightarrow{\circ} \begin{array}{c} (\text{union} \exists)^\circ X \\ \hline R \end{array} \xLeftrightarrow{\cdot \exists \neg \neg} \begin{array}{c} (\exists \exists)^\circ X \\ \hline R \end{array} \xLeftrightarrow{\circ, T \cdot \neg T \backslash} \begin{array}{c} X \\ \hline \in \backslash (\in \backslash R) \end{array} \quad (13.1.4a)$$

### §13.1.5. $\frac{S}{\exists} \text{ est}(R) = S n(S^\circ \backslash R^\circ)$

$$\begin{array}{c} X \\ \hline \frac{S}{\exists} \text{ est}(R) \end{array} \xLeftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} \frac{S^\circ X}{\exists} \\ \hline \text{est}(R) \end{array} \xLeftrightarrow{\Delta \neg n} \begin{array}{c} \frac{S^\circ X}{\exists} \\ \hline \in \backslash R^\circ \end{array} \xLeftrightarrow{f^\circ \cdot \neg f^\circ} \begin{array}{c} X \\ \hline \frac{S}{\exists} (\in \backslash R^\circ) \end{array} \xLeftrightarrow{\cdot \exists \neg \neg, (13.1.3a)} \begin{array}{c} X \\ S \\ \hline X \\ S^\circ \backslash R^\circ \end{array} \xLeftrightarrow{\Delta \neg n} \begin{array}{c} X \\ \hline S n(S^\circ \backslash R^\circ) \end{array} \quad (13.1.5a)$$

### §13.1.6. $\frac{S}{\exists} \text{ est}(R) = \frac{S}{\exists} \text{ est}(R n S^\circ S)$

$$\begin{array}{c} X \\ \hline \frac{S}{\exists} \text{ est}(R n S^\circ S) \end{array} \xLeftrightarrow{(13.1.5a), \circ} \begin{array}{c} X \\ \hline S n(S^\circ \backslash (R^\circ n S^\circ S)) \end{array} \xLeftrightarrow{\Delta \neg n, T \cdot \neg T \backslash} \begin{array}{c} X \\ S \\ \hline S^\circ X \\ R^\circ n S^\circ S \end{array} \xLeftrightarrow{\Delta \neg n, S^\circ \cdot \text{monotone}} \begin{array}{c} X \\ S \\ \hline S^\circ X \\ R^\circ \end{array} \xLeftrightarrow{T \cdot \neg T \backslash, \Delta \neg n} \begin{array}{c} X \\ \hline S n(S^\circ \backslash R^\circ) \end{array} \xLeftrightarrow{(13.1.5a)} \begin{array}{c} X \\ \hline \frac{S}{\exists} \text{ est}(R) \end{array} \quad (13.1.6a)$$

### §13.1.7. $P(f) \text{ est}(R) = \text{est}(f R f^\circ) f$

$$\begin{array}{c} P(f) \text{ est}(R) \\ \hline f \text{ a map} \end{array} \xLeftrightarrow{P=E \text{ on maps, (13.1.5a)}} \begin{array}{c} (\exists f) n((\exists f)^\circ \backslash R^\circ) \end{array} \xLeftrightarrow{\circ, T \cdot \neg T \backslash, f^\circ \cdot \neg f^\circ} \begin{array}{c} (\exists f) n(\in \backslash (f R^\circ)) \end{array} \xLeftrightarrow{\text{modular law, } f \text{ simple}} \begin{array}{c} (\exists n((\in \backslash (f R^\circ)) f^\circ)) f \end{array} \xLeftrightarrow{\cdot f \neg f^\circ, \circ, \text{est}} \begin{array}{c} \text{est}(f R f^\circ) f \end{array} \quad (13.1.7a)$$

### §13.1.8. $P(S) \text{ est}(R) \in (\exists S) n(\in \backslash (S R^\circ))$

$$\begin{array}{c} P(S) \text{ est}(R) \\ \hline (\exists S) n(\in \backslash (S R^\circ)) \end{array} \xLeftrightarrow{\Delta \neg n, T \cdot \neg T \backslash} \begin{array}{c} P(S) \text{ est}(R) \\ \hline \exists S \end{array} \xLeftrightarrow{\text{UP of est}} \begin{array}{c} P(S) \exists \\ \hline \exists S \end{array} \xLeftrightarrow{\text{UP of est}} \begin{array}{c} \in P(S) \text{ est}(R) \\ \hline S R^\circ \end{array} \xLeftrightarrow{\text{UP of est}} \begin{array}{c} \in P(S) \\ \hline S \in \end{array} \quad (13.1.8a)$$

### §13.1.9. $P(\text{est}(R)) \text{ est}(R) \in \text{union est}(R)$

$$\begin{array}{c} P(\text{est}(R)) \text{ est}(R) \\ \hline \text{union est}(R) \\ \hline R \text{ a preorder} \end{array} \xLeftrightarrow{(13.1.5a)} \begin{array}{c} P(\text{est}(R)) \text{ est}(R) \\ \hline (\exists \exists) n((\exists \exists)^\circ \backslash R^\circ) \end{array} \xLeftrightarrow{\circ, \Delta \neg n, T \cdot \neg T \backslash} \begin{array}{c} P(\text{est}(R)) \text{ est}(R) \\ \hline \exists \exists \end{array} \xLeftrightarrow{\text{UP of est, } R \text{ transitive}} \begin{array}{c} P(\text{est}(R)) \exists \\ \hline \exists \text{est}(R) \end{array} \xLeftrightarrow{\text{UP of est, } R \text{ transitive}} \begin{array}{c} \in P(\text{est}(R)) \\ \hline \text{est}(R) \in \end{array} \quad (13.1.9a)$$

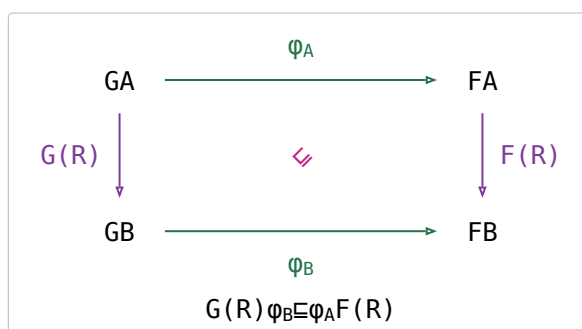
## §13.2. Lax natural transformations (LaT)

For relators  $G, F : \mathcal{C} \rightarrow \mathcal{D}$  and components  $\varphi_A : GA \rightarrow FA$ ,  $\varphi$  is **lax at**  $R : A \rightarrow B$  when  $G(R)\varphi_B \sqsubseteq \varphi_A F(R)$  — both sides  $GA \rightarrow FB$ , the left through the component at  $B$ , the right through the one at  $A$ .

(13.2a)

$\varphi : G \Rightarrow F$  is a **lax natural transformation** (LaT) when it is lax at **every**  $R$ . (5.13)

Lax at every **map** already gives LaT, and at a map the inequation is an equality  $G(f)\varphi = \varphi F(f)$ : laxness is about relations only. Theorem 5.2



(13.2b)



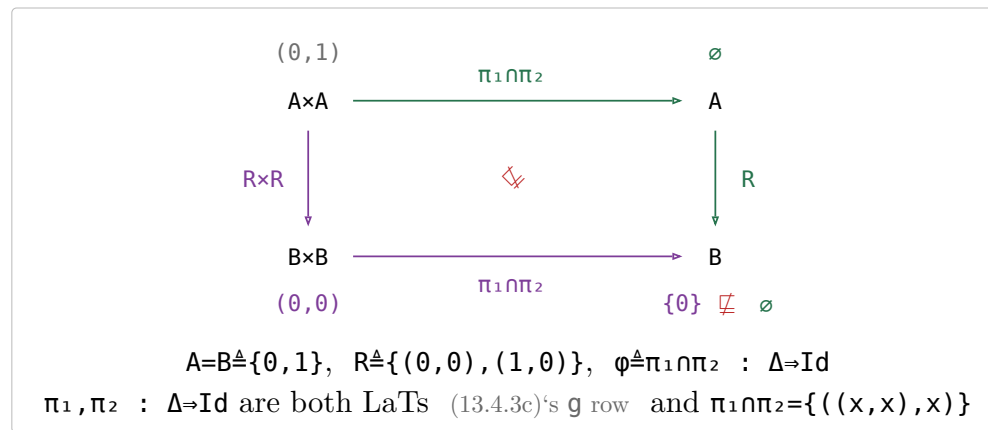
| closed under                              | commutative diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | string diagram |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| composition<br>$\psi\phi$                 | <p> <math>H(R)\psi_B \sqsubseteq \psi_A G(R)</math> and <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> give <math>H(R)</math><br/> <math>(\psi_B\phi_B) \sqsubseteq (\psi_A\phi_A) F(R)</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |
| horizontal composition<br>$\chi\circ\phi$ | <p> <math>\phi : G \Rightarrow F</math> with <math>G, F : \mathcal{C} \rightarrow \mathcal{D}</math> and <math>\chi : L \Rightarrow K</math> with <math>L, K : \mathcal{D} \rightarrow \mathcal{E}</math> give <math>\chi \circ \phi : L \circ G \Rightarrow K \circ F</math>, the family <math>A \mapsto \chi_{GA} K(\phi_A)</math><br/> <math>L(G(R)) \chi_{GB} \sqsubseteq \chi_{GA} K(G(R))</math> is <math>\chi</math> lax at <math>G(R)</math>; then<br/> <math>K(G(R)\phi_B) \sqsubseteq K(\phi_A F(R))</math> is <math>K</math> applied to <math>\phi</math>'s own inequation<br/> <math>\chi := \mathbb{1}_K</math> gives <math>K(\phi) : K \circ G \Rightarrow K \circ F</math>, <math>K</math> composed on the outside; <math>\phi := \mathbb{1}_G</math><br/>             gives <math>\chi G : L \circ G \Rightarrow K \circ G</math>, <math>G</math> composed on the inside<br/>             two candidates, <math>\chi_{GA} K(\phi_A)</math> and <math>L(\phi_A) \chi_{FA}</math>, ordered by <math>\sqsubseteq</math>; this row is<br/>             the first         </p> |                |
| union<br>$\phi\cup\psi$                   | <p> <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> and <math>G(R)\psi_B \sqsubseteq \psi_A F(R)</math> give <math>G(R)</math><br/> <math>(\phi_B \cup \psi_B) \sqsubseteq (\phi_A \cup \psi_A) F(R)</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                |
| a relator $K$<br>$K(\phi)$                | <p> <math>G(R)\phi_B \sqsubseteq \phi_A F(R)</math> gives <math>K(G(R)) K(\phi_B) \sqsubseteq K(\phi_A) K(F(R))</math> </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                |
| product<br>$\phi \times \psi$             | <p> <math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi \times \psi : G \times G' \Rightarrow F \times F'</math><br/> <math>(R \times S)(U \times V) = (RU) \times (SV)</math> and monotonicity in both slots — the<br/>             row above's <math>K</math> applied to the inequation, run on a bifunctor; the fork<br/>             is the derived case <math>\langle \phi, \psi \rangle = \triangleleft (\phi \times \psi)</math>, and its ONE cost is the lax<br/>             copy law <math>R \triangleleft \triangleleft (R \times R)</math> — (2b)         </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                |
| coproduct<br>$\phi + \psi$                | <p> <math>\phi : G \Rightarrow F</math> and <math>\psi : G' \Rightarrow F'</math> give <math>\phi + \psi : G + G' \Rightarrow F + F'</math><br/> <math>(R + S)(U + V) = (RU) + (SV)</math> and monotonicity in both slots; the co-<br/>             fork is the derived case <math>[\phi, \psi] = (\phi + \psi) \triangleright</math>, and <math>\triangleright</math> costs nothing<br/>             the step it would need is <math>(\phi_A F(R)) \cap (\psi_A F(R)) \sqsubseteq (\phi \cap \psi)_A F(R)</math>, the<br/>             wrong direction of (4c)<br/>             a counterexample: (13.2.1a)         </p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                |
| meet — <b>fails</b><br>$\phi \cap \psi$   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                |

Each row is the general slide  $R_A X \sqsubseteq X' R_B$  and  $R_B Y \sqsubseteq Y' R_C$  give  $R_A (XY) \sqsubseteq (X' Y') R_C$  at  $R_A, R_B := G(R), F(R)$  and  $X, X' := \phi_B, \phi_A$ , quantified over  $R$ ; at  $X' = X$  with  $R_A, R_B$  endorelations it is the monotonicity reading instead — the  $\forall R$  sits outside the law, and is the only difference

### §13.2.1. meet is not closed in LaT

counterexample —  $\varphi, \psi : \mathbf{G} \Rightarrow \mathbf{F}$  lax natural does NOT give  $\varphi \cap \psi$  lax natural

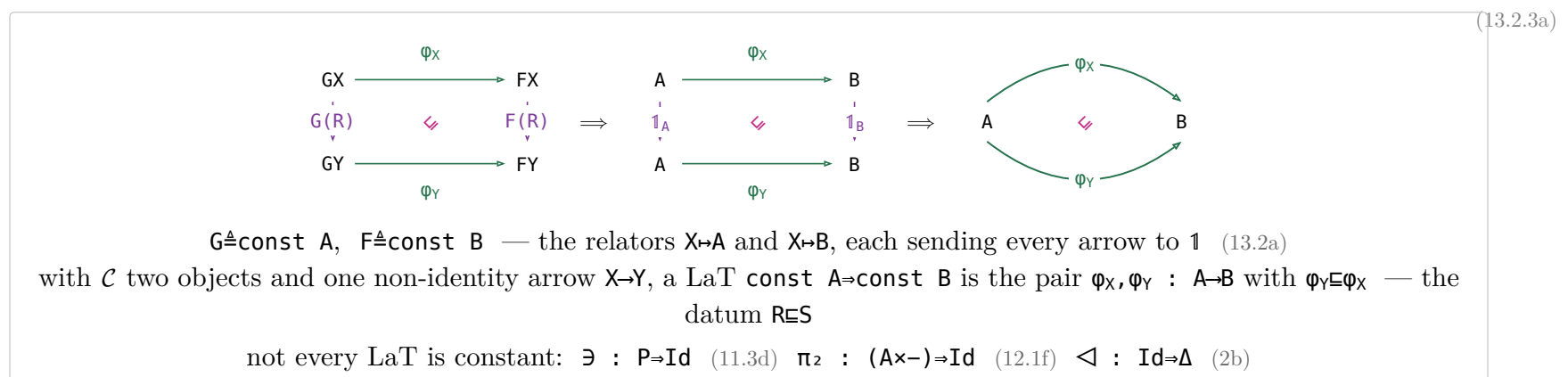
(13.2.1a)



### §13.2.2. Two stacked towers

|             | lower — inside ONE allegory      | upper — between allegories                                                                                |
|-------------|----------------------------------|-----------------------------------------------------------------------------------------------------------|
| 0-cell      | an object $A$                    | an allegory $\mathcal{C}$ — a whole lower tower                                                           |
| 1-cell      | a relation $R : A \rightarrow B$ | a relator $F : \mathcal{C} \rightarrow \mathcal{D}$                                                       |
| 2-cell      | $R \sqsubseteq S$                | a LaT $\varphi : \mathbf{G} \Rightarrow \mathbf{F}$ (13.2a)                                               |
| order       | $R \sqsubseteq S$                | $\varphi \sqsubseteq \psi$ componentwise                                                                  |
| union       | $R \cup S$                       | $\varphi \cup \psi$ , survives (13.2c)                                                                    |
| product     | $R \times S$ (11.1.1a)           | $\varphi \times \psi$ , survives; a TENSOR, not a categorical product; SYMMETRIC (11.1b) (13.2c)          |
| coproduct   | $R + S$                          | $\varphi + \psi$ , survives; a BIPRODUCT (13.2c)                                                          |
| meet        | $R \cap S$                       | $\varphi \cap \psi$ componentwise, fails (13.2.1a)                                                        |
| converse    | $R^\circ$                        | $\varphi^\circ$ , fails, oplax: $\varphi_A^\circ \mathbf{G}(R) \sqsubseteq \mathbf{F}(R) \varphi_B^\circ$ |
| zero object | $z$ with $1 \ z = 0$             | the constant relator at such a $z$ , initial and terminal                                                 |

### §13.2.3. RES is a special case of LaT

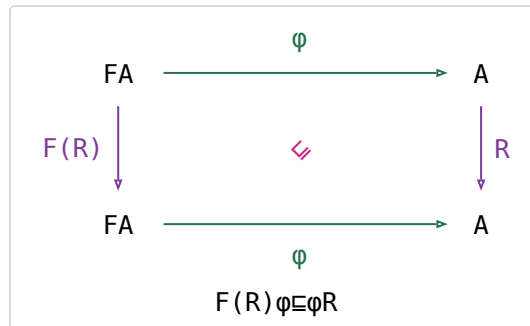


### §13.3. Monotonic algebras

An  $F$ -algebra  $\varphi : FA \rightarrow A$  is **monotonic on**  $R : A \rightarrow A$  when it is lax at  $R$  at  $G := F$ ,  $F := \text{Id}$ :  $F(R)\varphi \sqsubseteq \varphi R$ .  $R$  is an **endorelation**, so  $B=A$  and the two components  $\varphi_A, \varphi_B$  are the one arrow  $\varphi$  — an algebra, not a family.

(13.3a)

For a map  $f : FA \rightarrow A$  that is  $f \circ F(R) f \sqsubseteq R$  (1.1a)'s  $f \circ \cdot \dashv f \cdot$  at  $X := F(R) f, Y := R$ , equivalently  $F(R) \sqsubseteq f R f \circ \cdot f \dashv \cdot f \circ$  then  $f \circ \cdot \dashv f \cdot$ .  
 $(\leq x \leq) + \sqsubseteq + \leq$  — addition on  $\text{Nat}$  is monotonic on  $\leq$ , which at the point level reads  $c = a + b \wedge a \leq a' \wedge b \leq b' \implies c \leq a' + b'$ .



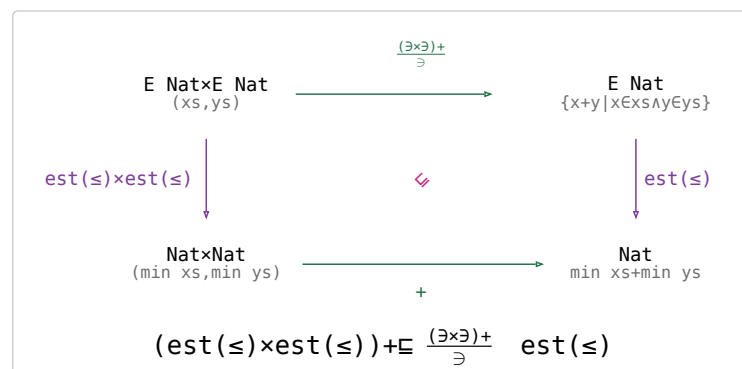
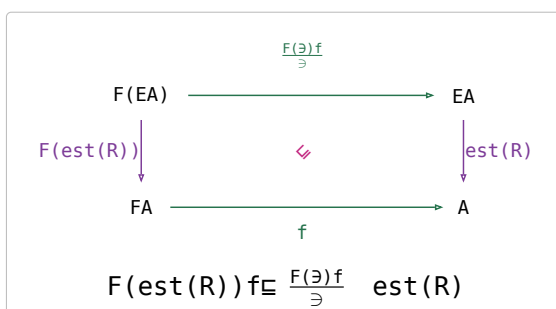
(13.3b)

#### §13.3.1. Function $f$ is monotonic on $R$ iff it distributes over $R$

$f : FA \rightarrow A$  **distributes over**  $R$  if  $F(\text{est}(R)) f \sqsubseteq \frac{F(\exists)f}{\exists} \text{est}(R)$ .

(13.3.1a)

$+$  distributes over  $\leq$ , at the point level  $\min xs + \min ys = \min\{x+y \mid x \in xs \wedge y \in ys\}$  for  $xs, ys$  non-empty and  $\min \triangleq \text{est}(\leq)$ .



(13.3.1b)

$$f \circ F(R) f \sqsubseteq R \iff F(\text{est}(R)) f \sqsubseteq \frac{F(\exists) f}{\exists} \text{est}(R)$$

 Theorem 7.1,  $f$  a map,  $R$  reflexive

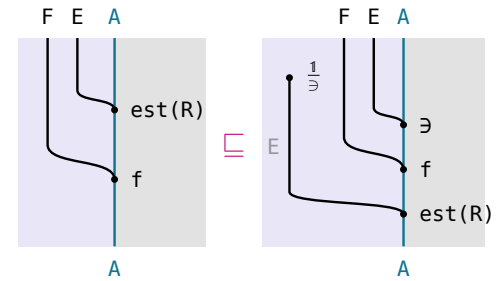
circuit

Hinze–Marsden

$$F(EA) \sqsubseteq F(\text{est}(R)) \sqsubseteq f \sqsubseteq A \quad \sqsubseteq \quad \frac{F(\exists) f}{\exists} \sqsubseteq \text{est}(R)$$

$$F(\text{est}(R)) f \sqsubseteq \frac{F(\exists) f}{\exists} \text{est}(R)$$

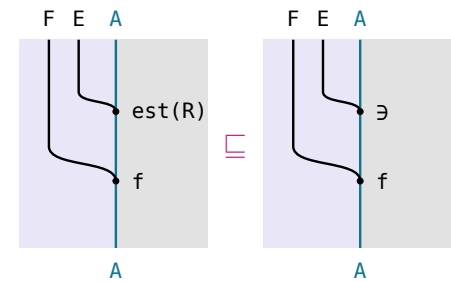
$$\frac{S}{\exists} = \frac{1}{\exists} E(S), \quad S \triangleq F(\exists) f$$

 def.  $f$  distributes over  $R$  — (13.3.1a), (10.1a)


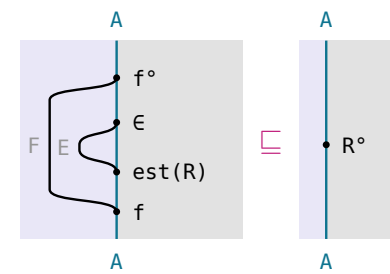
$$F(\text{est}(R)) f \sqsubseteq F(\exists) f$$

and

$$\Leftrightarrow (F(\exists) f) \circ F(\text{est}(R)) f \sqsubseteq R^\circ$$

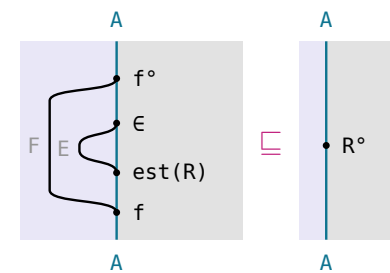
 universal property of  $\text{est}$  — (13.1b)'s  $\frac{S}{\exists} \text{est}(R) = S \cap (S^\circ \setminus R^\circ)$  at  $S := F(\exists) f$   
 $\text{est}(R) \triangleq \exists \cap (E \setminus R^\circ)$  — (13.1a)


and



$$\Leftrightarrow A \sqsubseteq \frac{F(\text{est}(R)) f}{f} \sqsubseteq \frac{F(\exists) f}{\exists} \sqsubseteq A$$

$$(F(\exists) f) \circ F(\text{est}(R)) f \sqsubseteq R^\circ$$

 $\text{est}(R) \sqsubseteq \exists$  — (13.1a); the first conjunct is free


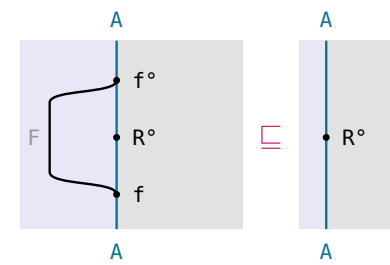
$$\Leftrightarrow A \sqsubseteq \frac{F(E \text{ est}(R)) f}{f} \sqsubseteq \frac{F(\exists) f}{\exists} \sqsubseteq A$$

$$f \circ F(E \text{ est}(R)) f \sqsubseteq R^\circ$$

 converse, relators:  $(F(\exists) f)^\circ = f^\circ F(E)$  and  $F(E) F(\text{est}(R)) = F(E \text{ est}(R))$ 

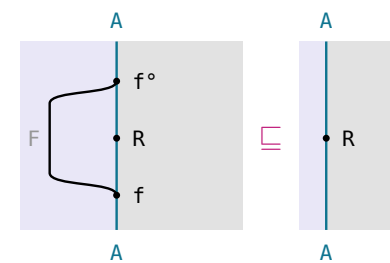
$$\Leftrightarrow A \sqsubseteq \frac{F(R^\circ) f}{f} \sqsubseteq \frac{F(\exists) f}{\exists} \sqsubseteq A$$

$$f \circ F(R^\circ) f \sqsubseteq R^\circ$$

 $E \text{ est}(R) = R^\circ$ ,  $R$  reflexive — (13.1a)


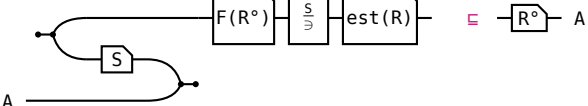
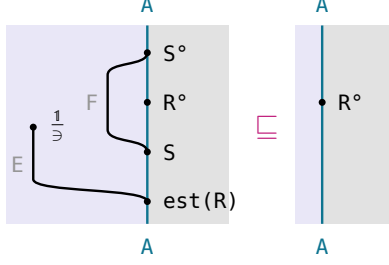
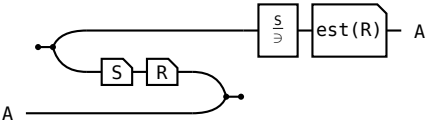
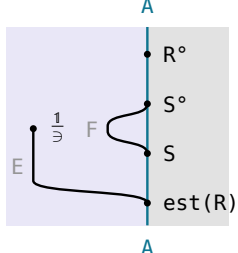
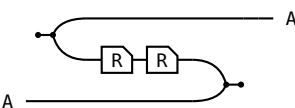
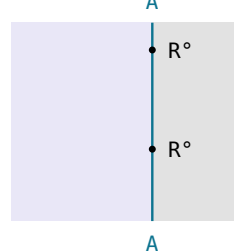
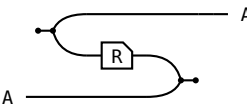
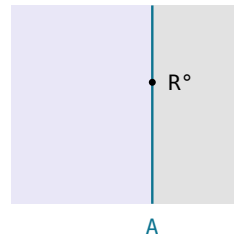
$$\Leftrightarrow A \sqsubseteq \frac{F(R) f}{f} \sqsubseteq \frac{F(\exists) f}{\exists} \sqsubseteq A$$

$$f \circ F(R) f \sqsubseteq R$$

 $f$  a map, so monotonic on  $R$  and  $R^\circ$  together


### §13.3.2. Greedy Theorem: $(\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq (\bigcup_{\subseteq} \text{est}(R))$ , given $S$ monotonic on $R$ and $R$ a preorder

(13.3.2a)

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                      | Hinze–Marsden                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| $(\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq (\bigcup_{\subseteq} \text{est}(R))$<br>the conclusion: one minimum kept at each step is below every result collected and one minimum taken at the end                                                                                                                                                                                                                                      |                                                                                       |
| $(\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq (S)$<br>and<br>$(S)^\circ (\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq R^\circ$<br>(13.1.5a) at $(S)$                                                                                                                                                                                                                                                                                    |                                                                                       |
| $\bigcup_{\subseteq} \text{est}(R) \sqsubseteq S$<br>the left conjunct; (13.1.5a), and $(\bigcup_{\subseteq})$ is monotone                                                                                                                                                                                                                                                                                                                   |                                                                                       |
|  $S^\circ F(R^\circ) (\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq R^\circ$<br>the right conjunct; $(S)^\circ (\bigcup_{\subseteq} \text{est}(R))$ is the least $X$ with $X = S^\circ F(X) (\bigcup_{\subseteq} \text{est}(R))$<br>(11.6.4a) — so Knaster–Tarski leaves this one inequation<br>$\bigcup_{\subseteq} = \frac{1}{\subseteq} E(S)$ — (10.1a) |    |
|  $R^\circ S^\circ (\bigcup_{\subseteq} \text{est}(R))$<br>$S^\circ F(R^\circ) \sqsubseteq R^\circ S^\circ$ , the converse of the hypothesis $F(R) S \sqsubseteq SR$                                                                                                                                                                                        |   |
|  $R^\circ R^\circ$<br>$S^\circ (\bigcup_{\subseteq} \text{est}(R)) \sqsubseteq R^\circ$ — (13.1.5a)                                                                                                                                                                                                                                                       |  |
|  $R^\circ$<br>$R$ transitive                                                                                                                                                                                                                                                                                                                              |  |

§13.3.3.  $\text{takewhile}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$ 

(13.3.3a)

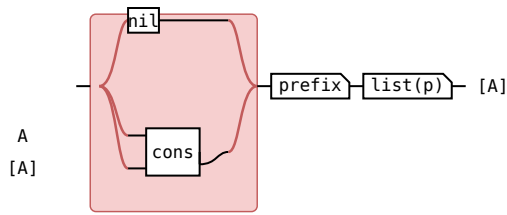
| name                     | definition                                                            | type                                                   | example                                                                                 | in words                                                                                        |
|--------------------------|-----------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| F                        | $FX = 1 + A \times X$                                                 | $\mathcal{A} \rightarrow \mathcal{A}$                  |                                                                                         | nothing, or a head and a tail                                                                   |
| nil,<br>cons             | $[A] ::= \text{nil} \mid \text{cons } (A, [A])$<br>(12a)              | $1 \rightarrow [A],$<br>$A \times [A] \rightarrow [A]$ | $\text{cons}(3, [1, 2]) = [3, 1, 2]$                                                    | the empty list; a head onto a tail                                                              |
| $\alpha$                 | $[\text{nil}, \text{cons}]$                                           | $F([A]) \rightarrow [A]$                               |                                                                                         | both constructors as one map                                                                    |
| p                        | a coreflexive                                                         | $A \rightarrow A$                                      | $p \triangleq \text{even} \rightarrow 2 \text{ p } 2$ , and<br>$3 \notin \text{Dom}(p)$ | $\{(a, a) \mid a \text{ passes the test}\}$                                                     |
| R                        | $\text{length} \leq \text{length}^\circ$ , a<br>preorder              | $[A] \rightarrow [A]$                                  | $[1] \text{ R } [1, 2]$                                                                 | $xs \text{ R } ys \iff \text{length } xs \leq \text{length } ys$                                |
| $\rightarrow \text{nil}$ | the constant nil — the<br>second nil of prefix                        | $A \times [A] \rightarrow [A]$                         | $(\rightarrow \text{nil})(3, [1, 2]) = \text{nil}$                                      | drop the pair, return nil                                                                       |
| prefix                   | $([\text{nil}, \rightarrow \text{nil} \cup \text{cons}])$<br>(12a)    | $[A] \rightarrow [A]$                                  | $[3, 1, 2] \text{ prefix } [3, 1]$                                                      | $xs \text{ prefix } ys \iff \exists zs. xs = ys \# zs$<br>— at each cons, stop or keep the head |
| S                        | $[\text{nil}, \rightarrow \text{nil} \cup (p \times 1) \text{ cons}]$ | $F([A]) \rightarrow [A]$                               | $(4, [2]) \text{ S } [4, 2]$ , and<br>$(3, [2]) \text{ S } \text{nil}$ only             | prefix's algebra with one extra p<br>— stop, or keep a head that passes p                       |
| $(g \rightarrow X, Y)$   | X where g is defined and Y<br>where it is not                         | $A \rightarrow B$ at X, Y :<br>$A \rightarrow B$       |                                                                                         | the test picks the branch                                                                       |

$\text{nil} \text{ R } \top$ ,  $\text{nil} \text{ R}^\circ = \text{nil}$  nil is the shortest list — below every list, and above only itself, so it loses every  $\text{est}(\text{R}^\circ)$

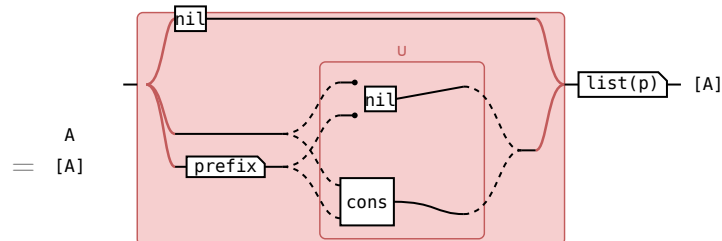
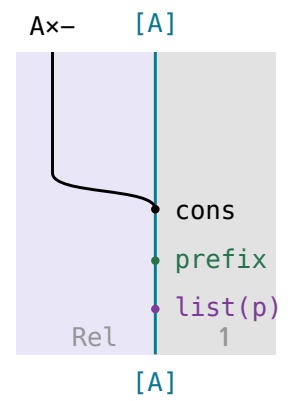
$$\alpha \text{ prefix list}(p) = F(\text{prefix list}(p))S$$

**circuit** — the fork is  $F([A]) = 1 + A \times [A]$ : nil above, the pair below

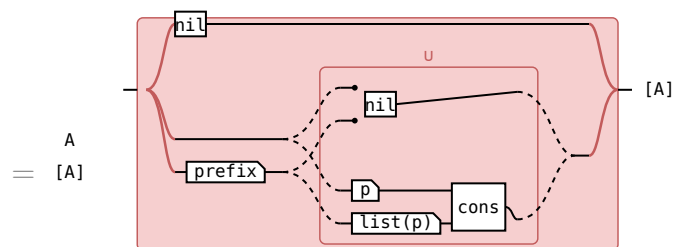
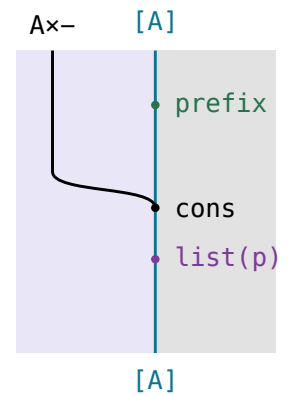
**Hinze–Marsden**



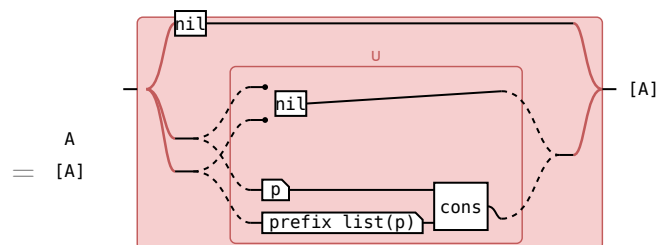
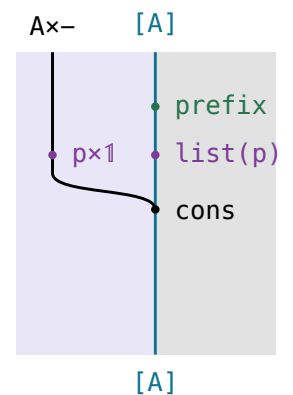
$\alpha \text{ prefix list}(p)$



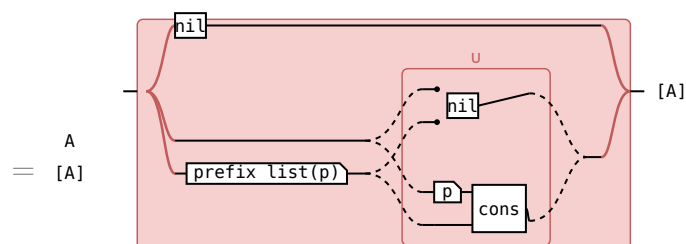
$F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup \text{cons}] \text{list}(p)$   
defining equation



$F(\text{prefix}) [\text{nil}, \multimap \text{nil} \cup (p \times \text{list}(p)) \text{cons}]$   
 $\text{list}(p)$  through cons

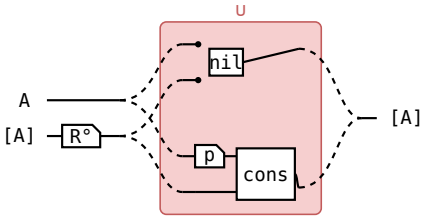
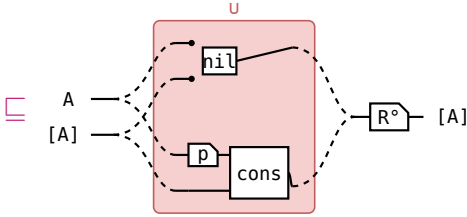


$[\text{nil}, \multimap \text{nil} \cup (p \times (\text{prefix list}(p))) \text{cons}]$   
relator, prefix entire

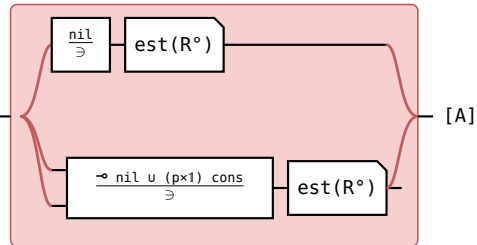
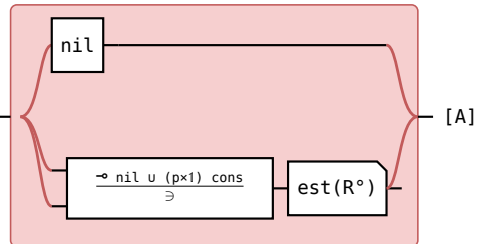
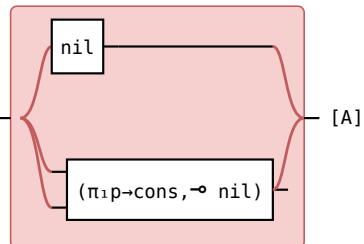


$F(\text{prefix list}(p))S$   
prefix list(p) entire

(11.6.1a) reads that off as  $\text{prefix list}(p) = \{S\}$ . (11.4.2a) cannot:  $\text{list}(p)$  is not entire,  $(1 \times \text{list}(p)) \multimap \text{nil} \sqsubseteq \multimap \text{nil}$ , and no algebra meets the side condition.

| $(1 \times R^\circ)(\neg \text{nil} \cup (p \times 1) \text{ cons}) \sqsubseteq (\neg \text{nil} \cup (p \times 1) \text{ cons}) R^\circ$ 3.3.3c) |                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula — the cons branch of $F(R^\circ) \sqsubseteq SR^\circ$                                                                                    | reason                                                                                                                                                                                    |
| $(1 \times R^\circ) \neg \text{nil} \sqsubseteq \neg \text{nil} R^\circ$                                                                          | $\text{nil} R^\circ = \text{nil}$ (13.3.3a) — the right side is $\neg \text{nil}$ , and $(1 \times R^\circ) \neg \text{nil} \sqsubseteq \neg \text{nil} R^\circ$                          |
| $(1 \times R^\circ)(p \times 1) \text{ cons} \sqsubseteq (p \times 1) \text{ cons} R^\circ$                                                       | composition (13.2c) at $(1 \times R^\circ)$<br>$(p \times 1) = (p \times R^\circ) = (p \times 1)(1 \times R^\circ)$ and $(1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons} R^\circ$ |
|                                                                  |                                                                                                                                                                                           |
|                                                                  | union (13.2c) at $\varphi := \neg \text{nil}$ , $\psi := (p \times 1) \text{ cons}$                                                                                                       |

the **nil** branch is  $\text{nil} \sqsubseteq \text{nil} R^\circ$ .  $(1 \times R^\circ) \text{ cons} \sqsubseteq \text{cons} R^\circ$  is  $\text{cons length} = (1 \times \text{length}) \pi_2 \text{ succ}$  with **SUCC** monotone — a shorter tail makes a shorter list.

| $\frac{S}{\supseteq} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \neg \text{nil})]$ 3.3.3d) |                                           |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| formula                                                                                                              | reason                                    |
| $F([A]) \xrightarrow{\frac{S}{\supseteq}} E([A]) \xrightarrow{\text{est}(R^\circ)} [A]$                              | $\frac{S}{\supseteq} \text{est}(R^\circ)$ |
| $=$                               | coproduct of maps                         |
| $=$                               | singleton, $R^\circ$ reflexive            |
| $=$                               | $\text{nil} R = \top$                     |

the set is  $\{\text{nil}\}$  where  $p$  fails on the head and  $\{\text{nil}, \text{cons}(a, xs)\}$  where it holds, and  $\text{nil}$  loses the second — (13.1a) at a two-element set.



$$\text{takewhile}(p) \triangleq \frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])$$

(13.3.3e)

 Ex 7.39: the longest prefix all of whose elements pass  $p$ 

| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                | Hinze–Marsden |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| $[A] \vdash \frac{\text{prefix list}(p)}{\supset} E([A]) \vdash \text{est}(R^\circ) \vdash [A]$ <p><math>\frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ)</math><br/>the specification — (13.1a)'s <math>\text{est}(R^\circ)</math></p>                                                                                                                                                                                 |               |
| $[A] \vdash \frac{(\frac{S}{\supset})}{\supset} E([A]) \vdash \text{est}(R^\circ) \vdash [A]$ <p><math>= \frac{(\frac{S}{\supset})}{\supset} \text{est}(R^\circ)</math><br/>(13.3.3b)</p>                                                                                                                                                                                                                                              |               |
| $[A] \vdash \frac{(\frac{S}{\supset} \text{est}(R^\circ))}{\supset} \vdash [A]$ <p><math>\sqsubseteq (\frac{S}{\supset} \text{est}(R^\circ))</math><br/>(13.3.2a) at <math>R^\circ</math>, with <math>F(R^\circ) S \subseteq SR^\circ</math> — (13.3.3c) — for its hypothesis: one longest <math>p</math>-prefix kept at each <math>\text{cons}</math>, instead of every <math>p</math>-prefix collected and one chosen at the end</p> |               |
| $[A] \vdash \frac{([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])}{\supset} \vdash [A]$ <p><math>= ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])</math><br/>(13.3.3d)</p>                                                                                                                                                                                                        |               |
| <p><math>\text{prefix}^\circ \text{ prefix} \cap R \cap R^\circ \subseteq 1</math><br/>(13.1b) at <math>\text{prefix}</math>: two prefixes of one list of equal length are equal, so <math>\frac{\text{prefix list}(p)}{\supset}</math><br/><math>\text{est}(R^\circ)</math> is simple — the longest, not a longest</p>                                                                                                                |               |
| <p><math>\frac{\text{prefix list}(p)}{\supset} \quad \text{est}(R^\circ)</math> entire<br/><math>\text{nil}</math> is always a <math>p</math>-prefix and <math>R</math> is connected on the prefixes of one list, so the longest exists</p>                                                                                                                                                                                            |               |
| <p><math>X \subseteq Y, X \text{ entire}, Y \text{ simple} \Rightarrow X=Y</math><br/><math>([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \rightarrow \text{nil})])</math> is a reduce of maps, hence entire — what turns the <math>\sqsubseteq</math> above into the heading's =</p>                                                                                                                                                |               |

#### §13.3.4. $\text{mss} = ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$

(13.3.4a)

$\text{FX} = 1 + A \times X$ ,  $A := \text{Int}$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ ,  $\text{sum} = ([\text{zero}, \text{plus}])$  and  $\text{segment} = \text{suffix prefix}$  from (7a) and (12a).

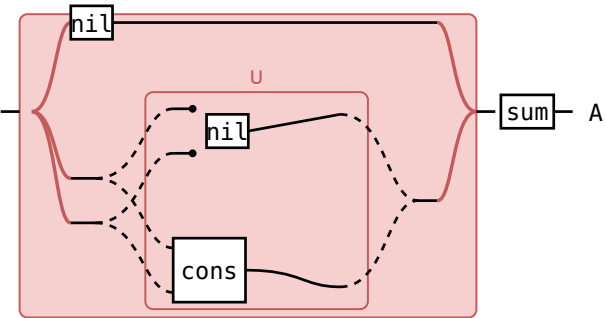
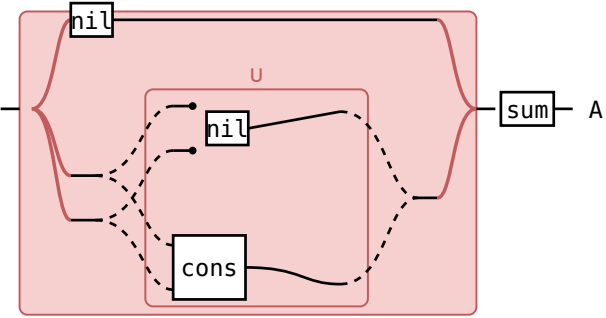
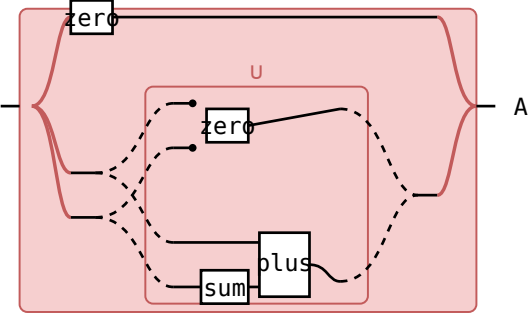
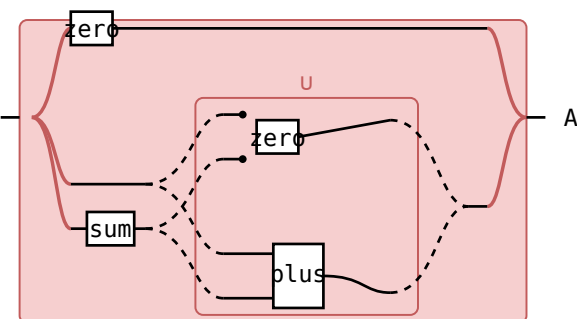
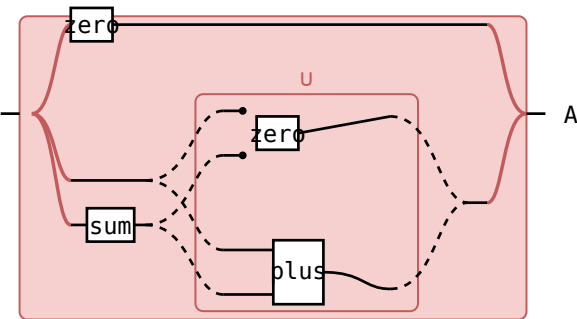
$\text{head} \triangleq \text{cons}^\circ \pi_1$ ,  $\text{wrap} \triangleq (1, \rightarrow \text{nil}) \text{ cons}$  the head of a list and the one-element list, beside (12a)'s  $\text{tail} \triangleq \text{cons}^\circ \pi_2$

$\oplus \triangleq \frac{\rightarrow \text{zero} \cup \text{plus}}{\supset} \quad \text{est}(\geq)$  B&dM's  $\text{oplus} = \max(\wedge(\text{zerouplus}))$ ; the set at  $(a, b)$  is  $\{0, a+b\}$ , so  $\oplus$  is the larger of the two

| $\frac{\text{segment sum}}{\ni} \text{est}(\geq) = \frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq)) \text{est}(\geq)$                                                                      |  |                                                                                                   |                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula — one wire from [A] to A, its type written along it                                                                                                                                                           |  |                                                                                                   | reason                                                                                                                                                                                                                                                                                                                 |
| $\cdot [A] \cdot \boxed{\frac{\text{segment sum}}{\ni}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                                                                        |  | $\frac{\text{segment sum}}{\ni} \text{est}(\geq)$                                                 |                                                                                                                                                                                                                                                                                                                        |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix (prefix sum)}}{\ni}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                                                              |  | $\frac{\text{suffix (prefix sum)}}{\ni} \text{est}(\geq)$                                         | segment=suffix<br>prefix<br>(12a), (13.3.4a)                                                                                                                                                                                                                                                                           |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\text{prefix sum})} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                                                           |  | $\frac{\text{suffix}}{\ni} E(\text{prefix sum}) \text{est}(\geq)$                                 | absorption<br>(10b) — $\text{frac}(S, \ni)$<br>$E(R) = \text{frac}(SR, \ni)$ at<br>$S := \text{suffix}, R := \text{prefix sum}$                                                                                                                                                                                        |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni})} \cdot E(EA) \cdot \boxed{\text{union}} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$        |  | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni}) \text{union est}(\geq)$               | $\frac{R}{\ni} \ni = R, \text{union} = E(\ni)$<br>(10b)'s $\text{frac}(R, \ni) \ni = R$ at<br>$R := \text{prefix sum}$ and<br>$E(R) \triangleq \text{frac}(\ni R, \ni)$ ;<br>(13.1b)'s<br>$\text{union} \triangleq \text{frac}(\ni \ni, \ni)$ ; the<br>middle equality is (11a)'s<br>$F(RS) = F(R)F(S)$ at<br>$F := E$ |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni})} \cdot E(EA) \cdot \boxed{E(\text{est}(\geq))} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$ |  | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni}) E(\text{est}(\geq)) \text{est}(\geq)$ | (13.1b), the sets non-empty                                                                                                                                                                                                                                                                                            |
| $= \cdot [A] \cdot \boxed{\frac{\text{suffix}}{\ni}} \cdot E([A]) \cdot \boxed{E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq))} \cdot EA \cdot \boxed{\text{est}(\geq)} \cdot A \cdot$                              |  | $\frac{\text{suffix}}{\ni} E(\frac{\text{prefix sum}}{\ni} \text{est}(\geq)) \text{est}(\geq)$    | relator<br>(11a) —<br>$F(RS) = F(R)F(S)$ at<br>$F := E$                                                                                                                                                                                                                                                                |

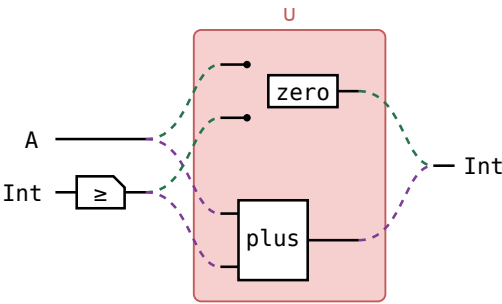
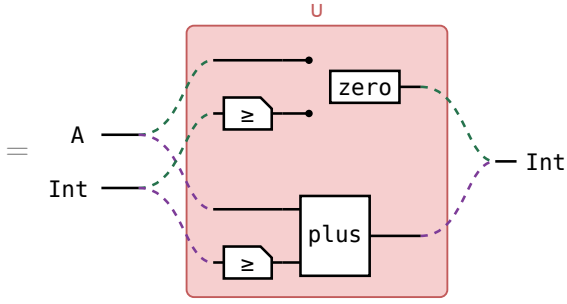
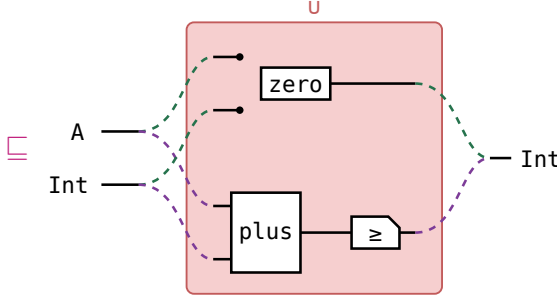
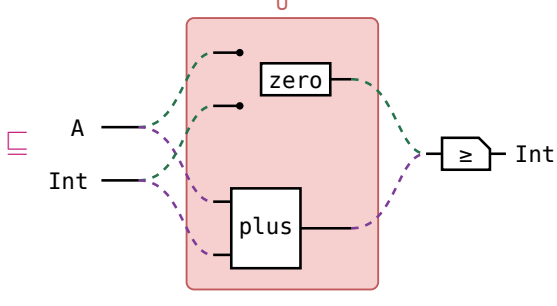
B&dM's  $\max(P(\max(\wedge(\text{sum prefix})))) \wedge \text{suffix}$ , mirrored. The **union** step is (13.1b)'s  $P(\text{est}(R)) \text{est}(R) = P(\text{Dom}(\text{est}(R))) \text{union est}(R)$  — every suffix has the empty prefix, so **Dom** is **1** here — and  $P(f) = E(f)$  at the map it is applied to ((11.3d)).

$$[\text{nil}, \multimap \text{nil } u \text{ cons}] \text{ sum} = F(\text{sum})[\text{zero}, \multimap \text{zero } u \text{ plus}]$$

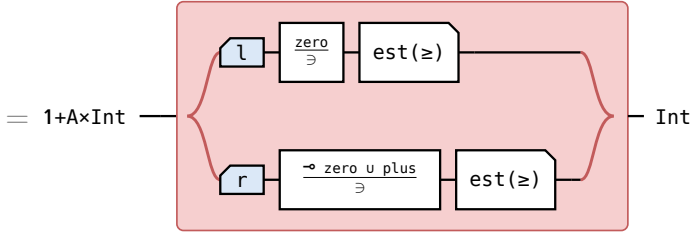
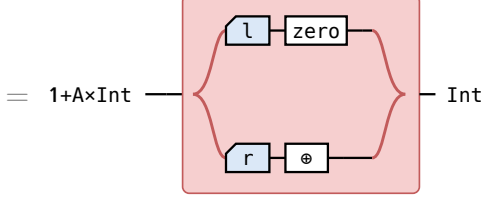
| formula — the fork is the bracket's case split $F([A]) = 1 + A \times [A]$ : nil above, the pair and its $u$ below                                                                   | reason                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
|  <p><math>[\text{nil}, \multimap \text{nil } u \text{ cons}] \text{ sum}</math></p>                 |                                                             |
|  <p><math>[\text{nil sum}, \multimap \text{nil sumucons sum}]</math></p>                            | coproduct of maps,<br>composition over $u$                  |
|  <p><math>[\text{zero}, \multimap \text{zero } u (1 \times \text{sum}) \text{ plus}]</math></p>    | sum's defining<br>equation                                  |
|  <p><math>[\text{zero}, (1 \times \text{sum}) (\multimap \text{zero } u \text{ plus})]</math></p> | $(1 \times \text{sum}) \multimap \multimap$ , sum<br>entire |
|  <p><math>F(\text{sum}) [\text{zero}, \multimap \text{zero } u \text{ plus}]</math></p>           | relator                                                     |

(11.4.2a) at  $\alpha_B := [\text{nil}, \multimap \text{nil } u \text{ cons}]$ ,  $S := \text{sum}$ : the side condition, so  $\text{prefix sum} = ([\text{zero}, \multimap \text{zero } u \text{ plus}])$ .  $\text{prefix}$  is the reduce,  $\text{sum}$  the map fused into it — the intermediate list is gone.

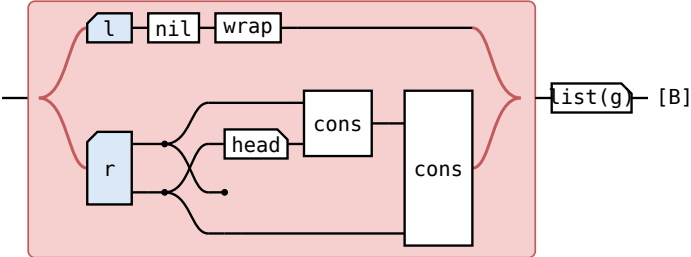
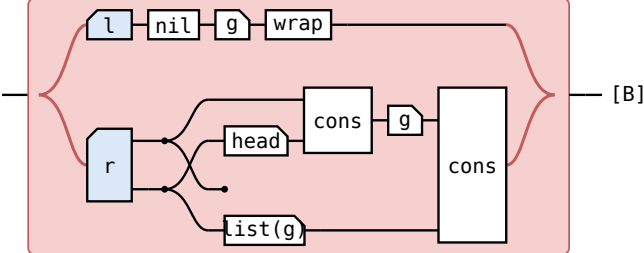
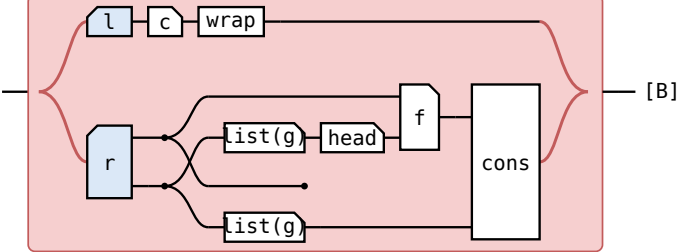
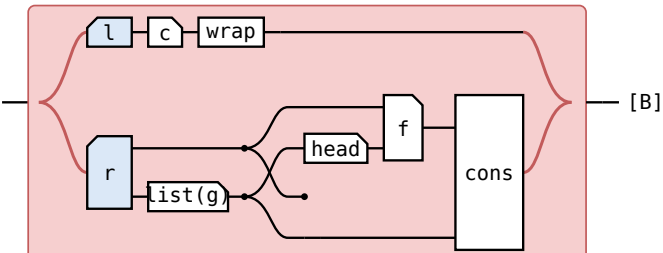
3.3.4d)

| $(1 \times \geq)(\neg \text{zero} \cup \text{plus}) \sqsubseteq (\neg \text{zero} \cup \text{plus}) \geq$<br>the <b>plus</b> branch of $F(\geq) \text{SES} \geq$ ; the <b>zero</b> branch is $\text{zero} \sqsubseteq \text{zero} \geq$ |                                                                     |                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula                                                                                                                                                                                                                                 | the head above, the running sum below; the tape is the $u$          | reason                                                                                                                                                                                                                                                                                                        |
|                                                                                                                                                        | $(1 \times \geq)(\neg \text{zero} \cup \text{plus})$                |                                                                                                                                                                                                                                                                                                               |
|                                                                                                                                                        | $(1 \times \geq) \neg \text{zero} \cup (1 \times \geq) \text{plus}$ | relator, composition over $u$                                                                                                                                                                                                                                                                                 |
|                                                                                                                                                       | $\neg \text{zero} \cup \text{plus} \geq$                            | (5.1a), $(\geq \times \geq)$<br>$\text{plus} \sqsubseteq \text{plus} \geq$<br>$(\leq \times \leq) \text{plus} \sqsubseteq \text{plus} \leq$ is<br>(13.3a), written $+$ there,<br>and <b>plus</b> is a map, so it is<br>monotonic on an order and<br>on its opposite together,<br>which carries it to $\geq$ . |
|                                                                                                                                                      | $(\neg \text{zero} \cup \text{plus}) \geq$                          | $\geq$ reflexive                                                                                                                                                                                                                                                                                              |

3.3.4e)

| $\frac{[\text{zero}, \neg \text{zero} \cup \text{plus}]}{\geq} \text{est}(\geq) = [\text{zero}, \oplus]$                                    |                                                                                                                |                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula                                                                                                                                     | the tape is the coproduct: <b>zero</b> 's branch above, <b>plus</b> 's below                                   | reason                                                                                                                                                                                               |
| $1 + A \times \text{Int} \rightarrow \frac{[\text{zero}, \neg \text{zero} \cup \text{plus}]}{\geq} \text{est}(\geq) \rightarrow \text{Int}$ | $\frac{[\text{zero}, \neg \text{zero} \cup \text{plus}]}{\geq} \text{est}(\geq)$                               |                                                                                                                                                                                                      |
|                                                          | $[\frac{\text{zero}}{\geq} \text{est}(\geq), \frac{\neg \text{zero} \cup \text{plus}}{\geq} \text{est}(\geq)]$ | coproduct of maps<br>(11.2.1a) at $T := [\text{zero}, \neg \text{zero} \cup \text{plus}]$ , then<br>$[U, V]Z = [UZ, VZ]$ —<br>(11.2a), composition over $U$                                          |
|                                                          | $[\text{zero}, \oplus]$                                                                                        | singleton, $\geq$ reflexive<br>(13.1b)'s $\frac{1}{\geq} \text{est}(R) = 1 \cap R$<br>at $R := \geq$ , <b>zero</b> a map; the<br>lower branch is $\oplus$ 's<br>definition, (13.3.4a), and no<br>law |

with (13.3.4d) the greedy theorem gives  $([\text{zero}, \oplus]) \sqsubseteq \frac{\text{prefix sum}}{\geq} \text{est}(\geq)$  — B&dM's own containment — and (13.3.4g) makes it an equality.

| $[\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}] \text{ list}(g) = F(\text{list}(g)) [c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}]$                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| formula — the 1 branch of $1 + A \times [A]$ above, the root and its list of tails below                                                                                                                                                                                                                 | reason                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|  <p style="text-align: right;">[B]</p> <p style="text-align: center;"> <math display="block">[\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}] \text{ list}(g)</math> </p>                   | $g \triangleq ([c, f])$ and<br>$\text{tails} = ([\text{nil wrap}, ((1 \times \text{head}) \text{ cons}, \pi_2) \text{ cons}])$ , since<br>$\text{tails}(\text{cons}(a, xs))$<br>is<br>$\text{cons}(\text{cons}(a, \text{head}(\text{tails } xs)), \text{tails } xs)$                                                                                                                                                                                                                                                  |
|  <p style="text-align: right;">[B]</p> <p style="text-align: center;"> <math display="block">[\text{nil } g \text{ wrap}, ((1 \times \text{head}) \text{ cons } g, \pi_2 \text{ list}(g)) \text{ cons}]</math> </p>     | coproduct of maps,<br>wrap and cons<br>natural                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|  <p style="text-align: right;">[B]</p> <p style="text-align: center;"> <math display="block">[c \text{ wrap}, ((1 \times (\text{list}(g) \text{ head})) f, (1 \times \text{list}(g)) \pi_2) \text{ cons}]</math> </p> | $g$ 's defining<br>equation, $\text{list}(g)$<br>$\text{head} = \text{head } g$<br>the $\pi_2$ slide<br>$\pi_2 \text{ list}(g) = (1 \times \text{list}(g)) \pi_2$<br>is 1 and 4 of (11.1.2b),<br>$\text{Dom}(\pi_1) = 1 \ ((5c))$ ;<br>$\text{list}(g) \text{ head} = \text{head } g$<br>$g$ is the one step the note<br>has no law for — $\text{head}$ is<br>undefined at $\text{nil}$ , and<br>the equality is a fact<br>about $\text{tails}$ , which<br>never returns an empty<br>list, not a naturality<br>square |
|  <p style="text-align: right;">[B]</p> <p style="text-align: center;"> <math display="block">F(\text{list}(g)) [c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}]</math> </p>                           | fork, relator<br>the fork slide is 7 of<br>(11.1.2b), $1 \times \text{list}(g)$ a<br>map there — and it is at<br>$c, f := \text{zero}, \oplus$ . (11.4.2a)<br>then reads off $\text{tails}$<br>$\text{list}(g) = ([c \text{ wrap}, ((1 \times \text{head}) f, \pi_2) \text{ cons}])$ , the heading's<br>fold.                                                                                                                                                                                                         |

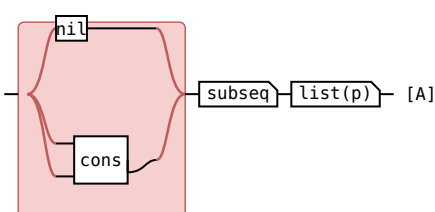
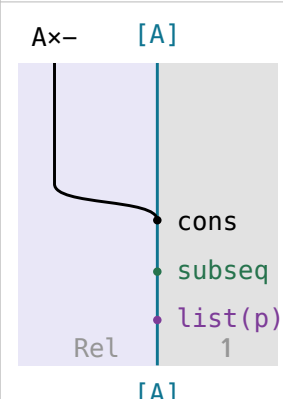
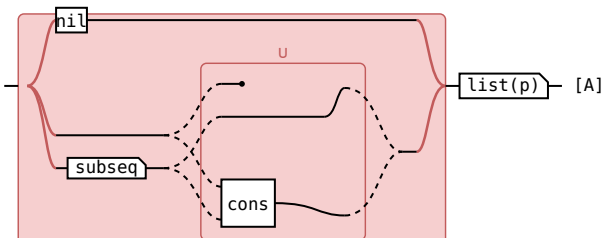
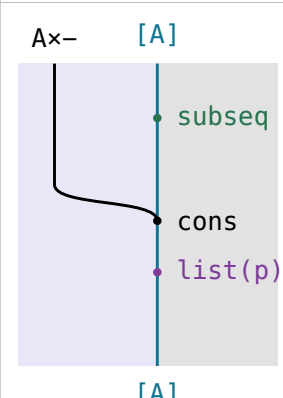
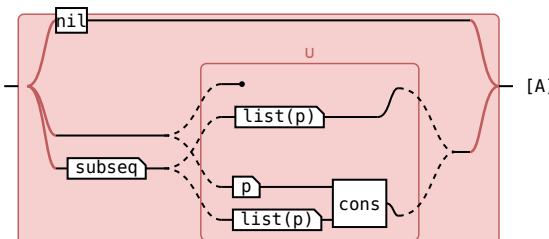
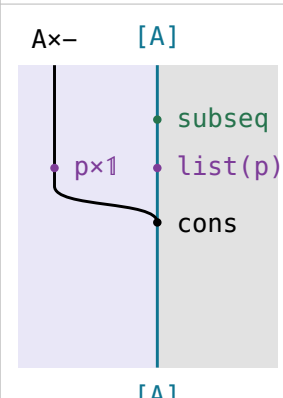
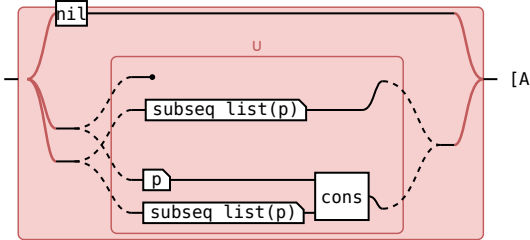
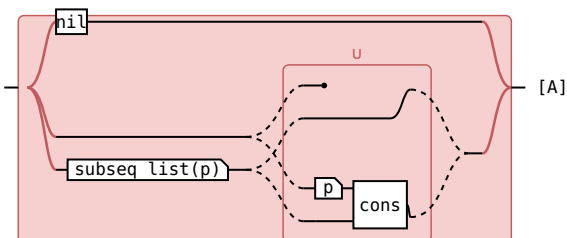
| $\text{mss} = ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$ $\text{Ex 7.40, } \oplus \stackrel{\Delta}{=} \text{zero} \cup \text{plus} \quad \text{est}(\geq) \text{ — (13.3.4a)}$                                                                                                                                                                                                                                                                                                                                                                                     |               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Hinze–Marsden |
| $[A] \rightarrow \boxed{\text{segment sum}} \rightarrow \boxed{\text{est}(\geq)} \rightarrow A$ <p><math>\text{segment sum} \quad \text{est}(\geq)</math></p> <p><math>\text{mss}</math> is the greatest of the segment sums — (13.3.4a)</p>                                                                                                                                                                                                                                                                                                                                                                          |               |
| $[A] \rightarrow \boxed{\text{suffix}} \rightarrow \boxed{E(\text{prefix sum est}(\geq))} \rightarrow \boxed{\text{est}(\geq)} \rightarrow A$ $= \text{suffix} \quad E(\text{prefix sum est}(\geq)) \text{ est}(\geq)$ <p>(13.3.4b)</p>                                                                                                                                                                                                                                                                                                                                                                               |               |
| $[A] \rightarrow \boxed{\text{suffix}} \rightarrow \boxed{E([\text{zero}, \oplus])} \rightarrow \boxed{\text{est}(\geq)} \rightarrow A$ $= \text{suffix} \quad E([\text{zero}, \oplus]) \text{ est}(\geq)$ <p>the greedy theorem (13.3.2a) at <math>R := \geq</math>, <math>S := [\text{zero}, \text{zero} \cup \text{plus}]</math> — (13.3.4d) is its condition and (13.3.4e) its <math>\frac{S}{\geq} \text{ est}(\geq)</math>; its <math>\sqsubseteq</math> is an <math>=</math> because <math>[\text{zero}, \oplus]</math> is entire and <math>\text{prefix sum est}(\geq)</math> simple (13.3.3e)'s last row</p> |               |
| $[A] \rightarrow \boxed{\text{tails}} \rightarrow \boxed{\text{list}([\text{zero}, \oplus])} \rightarrow \boxed{\text{est}(\geq)} \rightarrow A$ $= \text{tails list}([\text{zero}, \oplus]) \text{ est}(\geq)$ <p><math>\text{tails}</math> implements <math>\text{suffix}</math> and <math>\text{list}(f)</math> implements <math>E(f)</math> — (12a)</p>                                                                                                                                                                                                                                                           |               |
| $[A] \rightarrow \boxed{([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}])} \rightarrow \boxed{\text{est}(\geq)} \rightarrow A$ $= ([\text{zero wrap}, ((1 \times \text{head}) \oplus, \pi_2) \text{ cons}]) \text{ est}(\geq)$ <p>(11.4.2a) at (13.3.4f)'s side condition, <math>c, f := \text{zero}, \oplus</math></p>                                                                                                                                                                                                                                                                       |               |

one fold builds the  $n+1$  running maxima and the final  $\text{est}(\geq)$  reads them in one more pass, so  $\text{mss}$  is linear.

§13.3.5.  $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$ 

(13.3.5a)

| name                                           | definition                                                | type                           | example                                                                    | in words                                                                                                                                            |
|------------------------------------------------|-----------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $F, \alpha, \rho, R$                           | as in (13.3.3a)                                           |                                |                                                                            |                                                                                                                                                     |
| $\pi_2$                                        | $\pi_2$ where (13.3.3a) has $\rightarrow$<br>$\text{nil}$ | $A \times [A] \rightarrow [A]$ | $\pi_2(3, [1, 2]) = [1, 2]$                                                | drop the head, keep the tail                                                                                                                        |
| subseq                                         | $([\text{nil}, \pi_2 \cup \text{cons}])$ (12a)            | $[A] \rightarrow [A]$          | $[3, 1, 2] \text{ subseq } [3, 2]$                                         | $xs \text{ subseq } ys \iff ys$ is $xs$ with elements dropped — at each $\text{cons}$ , drop the head or keep it                                    |
| $S$                                            | $[\text{nil}, \pi_2 \cup (p \times 1) \text{ cons}]$      | $F([A]) \rightarrow [A]$       | $(4, [2]) S [2]$ and<br>$(4, [2]) S [4, 2]$ , but<br>$(3, [2]) S [2]$ only | subseq's algebra with one extra $p$ — drop the head, or keep a head that passes $p$                                                                 |
| $1 \sqsubseteq \pi_2 R$<br>$\text{cons}^\circ$ |                                                           |                                |                                                                            | the tail is one shorter than the $\text{cons}$ , so $\pi_2$ loses the $\text{est}(R^\circ)$ at every step — where (13.3.3a)'s loser is $\text{nil}$ |

| $\alpha \text{ subseq list}(p) = F(\text{subseq list}(p))S$                                                                                                                                                                                                                                                                   |                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <p><b>circuit</b> — the fork is <math>F([A]) = 1 + A \times [A]</math>: nil above, the pair below</p>  <p><math>\alpha \text{ subseq list}(p)</math></p>                                                                                     | <p><b>Hinze–Marsden</b></p>  |
|  <p><math>F(\text{subseq}) [\text{nil}, \pi_2 \cup \text{cons}] \text{list}(p)</math><br/>defining equation</p>                                                                                                                              |                             |
|  <p><math>F(\text{subseq}) [\text{nil}, (1 \times \text{list}(p)) \pi_2 \cup (p \times \text{list}(p)) \text{cons}] \text{list}(p)</math><br/><math>\text{list}(p)</math> through <math>\text{cons}</math>, <math>\pi_2</math> natural</p> |                            |
|  <p><math>[\text{nil}, (1 \times (\text{subseq list}(p))) \pi_2 \cup (p \times (\text{subseq list}(p))) \text{cons}]</math><br/>relator, <math>\text{subseq}</math> entire</p>                                                             |                                                                                                                 |
|  <p><math>F(\text{subseq list}(p))S</math><br/><math>\text{subseq list}(p)</math> entire</p>                                                                                                                                               |                                                                                                                 |

(11.6.1a) reads that off as  $\text{subseq list}(p) = (S)$  — (13.3.3b)'s chain with  $\pi_2$  for  $\neg \circ \text{nil}$ . Here (11.4.2a) is blocked for the same reason, and  $\pi_2$  is the branch that carries  $\text{list}(p)$  out by naturality ((12.1f) at  $\exists := \text{list}(p)$ ).



| $(1 \times R^\circ)(\pi_2 \cup (p \times 1) \text{ cons}) \sqsubseteq (\pi_2 \cup (p \times 1) \text{ cons}) R^\circ$ |                                                                                              |                                                                  | 3.3.5c) |
|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------|---------|
| formula                                                                                                               | — the cons branch of $F(R^\circ)S \sqsubseteq SR^\circ$                                      | reason                                                           |         |
|                                                                                                                       | $(1 \times R^\circ) \pi_2 = \pi_2 R^\circ$                                                   | $\pi_2$ natural (12.1f) — the right side is an EQUALITY here     |         |
|                                                                                                                       | $(1 \times R^\circ)(p \times 1) \text{ cons} \sqsubseteq (p \times 1) \text{ cons } R^\circ$ | (13.3.3c)'s cons branch, unchanged                               |         |
|                                                                                                                       | $(1 \times R^\circ)(\pi_2 \cup (p \times 1) \text{ cons})$                                   |                                                                  |         |
|                                                                                                                       | $(\pi_2 \cup (p \times 1) \text{ cons}) R^\circ$                                             | union (13.2c) at $X := \pi_2$ , $Y := (p \times 1) \text{ cons}$ |         |

$F(R^\circ)S \sqsubseteq SR^\circ$ , the nil branch again  $\text{nil} \sqsubseteq \text{nil } R^\circ$ . The first step is an equality: (13.3.3c) buys its  $\multimap \text{nil}$  branch with  $\text{nil } R^\circ = \text{nil}$ , and  $\pi_2$  needs only its naturality square.

| $\frac{S}{\sqsubseteq} \text{est}(R^\circ) = [\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$ |                                                                                          |                                            | 3.3.5d) |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------|---------|
| formula                                                                                              |                                                                                          | reason                                     |         |
| $F([A]) \sqsubseteq E([A]) \text{est}(R^\circ) [A]$                                                  | $\frac{S}{\sqsubseteq} \text{est}(R^\circ)$                                              |                                            |         |
| $=$<br>                                                                                              | $[\text{nil}, \frac{\pi_2 \cup (p \times 1) \text{ cons}}{\supset} \text{est}(R^\circ)]$ | (13.3.3d)'s first two steps                |         |
| $=$<br>                                                                                              | $[\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)]$                                 | $1 \sqsubseteq \pi_2 R \text{ cons}^\circ$ |         |

at  $(a, xs)$  the set is  $\{xs\}$  where  $p$  fails on  $a$  and  $\{xs, \text{cons}(a, xs)\}$  where it holds, and  $xs$  loses the second — (13.1a) at a two-element set. The head is dropped, not the whole tail: that is the one place  $\pi_2$  shows against (13.3.3d)'s  $\multimap \text{nil}$ .

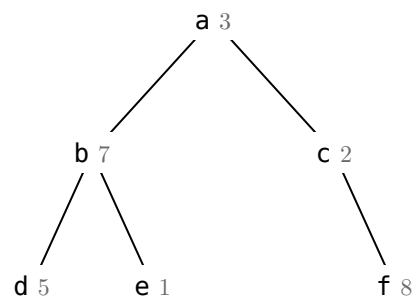
| $\text{filter}(p) = ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$ Ex 7.41, $R$ a preorder                                                                                                                                    |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit — one wire, its type written along it                                                                                                                                                                                            | Hinze–Marsden |
| $[A] \rightarrow \boxed{\frac{\text{subseq list}(p)}{\supset}} E([A]) \rightarrow \boxed{\text{est}(R^\circ)} [A]$ $\text{filter}(p) \triangleq \frac{\text{subseq list}(p)}{\supset} \text{est}(R^\circ)$ (12a)                         |               |
| $[A] \rightarrow \boxed{\frac{\langle S \rangle}{\supset}} E([A]) \rightarrow \boxed{\text{est}(R^\circ)} [A]$ $= \frac{\langle S \rangle}{\supset} \text{est}(R^\circ)$ $\text{subseq list}(p) = \langle S \rangle \text{ — (13.3.5b)}$ |               |
| $[A] \rightarrow \boxed{\langle \frac{S}{\supset} \text{est}(R^\circ) \rangle} [A]$ $\langle \frac{S}{\supset} \text{est}(R^\circ) \rangle$ (13.3.2a) at $R^\circ$ , whose hypothesis $F(R^\circ) S \subseteq SR^\circ$ is (13.3.5c)     |               |
| $[A] \rightarrow \boxed{([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])} [A]$ $= ([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$ (13.3.5d)                                                                            |               |
| $([\text{nil}, (\pi_1 p \rightarrow \text{cons}, \pi_2)])$ entire and $\text{filter}(p)$ simple<br>= an entire relation below a simple one equals it; the simplicity is (12a)'s Ex 7.41, not a law of this note                          |               |

$(\text{subseq list}(p))^\circ (\text{subseq list}(p)) \cap R \cap R^\circ \subseteq 1$  fails — two  $p$ -subsequences of one list can be of equal length and different — so §13.3.3's uniqueness argument does not transfer. What survives  $\text{est}(R^\circ)$  is the one keeping **exactly** the passing elements: a subsequence that drops a passing element is beaten by the one that keeps it.

## §13.4. Planning a company party

| definition                                                                               | type                                              | note                                                                                           | (13.4a) |
|------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------|---------|
| $\text{tree } A ::= \text{node } (A, [\text{tree } A])$                                  | $A \rightarrow A$                                 | The company hierarchy: an employee, and the list of subtrees under them.                       |         |
| $F(A, B) = A \times [B]$                                                                 | $A \times A \rightarrow A$                        | The base functor <b>tree</b> folds: an employee beside the recursive position, one layer deep. |         |
| rating                                                                                   | $A \rightarrow \text{Real}$                       | What one employee is worth as a guest.                                                         |         |
| $\text{cost} \triangleq \text{list}(\text{rating}) \text{ sum}$                          | $[A] \rightarrow \text{Real}$                     | What a guest list is worth.                                                                    |         |
| $R \triangleq \text{cost} \leq \text{cost}^\circ$                                        | $[A] \rightarrow [A]$                             | The preorder the guest list is maximised over.                                                 |         |
| $\text{choose} \triangleq \pi_1 \cup \pi_2$                                              | $[A] \times [A] \rightarrow [A]$                  | Takes one of the two parties a subtree returns.                                                |         |
| $\text{include} \triangleq (1 \times (\text{list}(\pi_2) \text{ concat})) \text{ cons}$  | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that invites the root, which puts every immediate subtree's root out. A map.         |         |
| $\text{exclude} \triangleq (1 \times (\text{list}(\text{choose}) \text{ concat})) \pi_2$ | $F(A, [A] \times [A]) \rightarrow [A]$            | The party that leaves the root out, so each subtree is free to choose. Not a map.              |         |
| $S \triangleq (\text{include}, \text{exclude})$                                          | $F(A, [A] \times [A]) \rightarrow [A] \times [A]$ | The algebra: one step returns both parties of a subtree at once.                               |         |
| $\text{party} \triangleq (S) \text{ choose}$                                             | $\text{tree } A \rightarrow [A]$                  | Every guest list the president's ruling allows.                                                |         |
| the specification<br>$\frac{\text{party}}{\exists} \text{ est}(R^\circ)$                 | $\text{tree } A \rightarrow [A]$                  | A guest list of greatest total conviviality.                                                   |         |

## §13.4.1. include and exclude



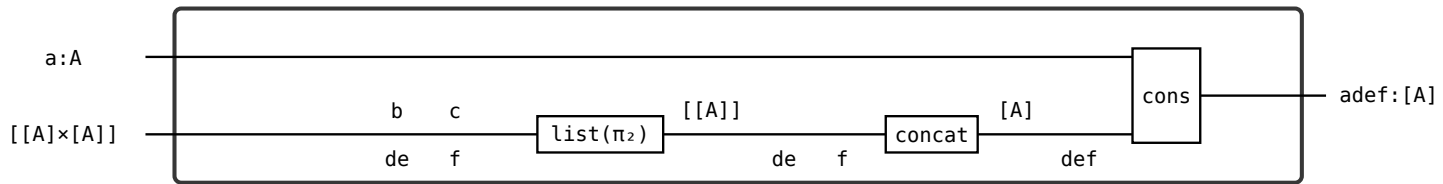
the small grey number is the employee's rating

| node    | the algebra's input<br>$A \times [[A] \times [A]]$ | include                                         | exclude                                | (13.4.1b) |
|---------|----------------------------------------------------|-------------------------------------------------|----------------------------------------|-----------|
| d, e, f | $(d, [])$                                          | $[d]$                                           | $[]$                                   |           |
| b       | $(b, ([[d], []], ([e], []]))$                      | $[b]$                                           | $[d, e], [d], [e], []$                 |           |
| c       | $(c, ([[f], []]))$                                 | $[c]$                                           | $[f], []$                              |           |
| a       | $(a, ([[b], ...], ([c], ...)))$                    | $[a] \# \text{exclude}(b) \# \text{exclude}(c)$ | $\text{choose}(b) \# \text{choose}(c)$ |           |

with those ratings,  $\text{est}(R^\circ)$  keeps  $([b]=7, [d, e]=6)$  at **b**, where **include** wins, and  $([c]=2, [f]=8)$  at **c**, where **exclude** wins, so at the root **include**  $= [a, d, e, f] = 17$  beats **exclude**  $= [b, f] = 15$ , and **choose** takes 17.

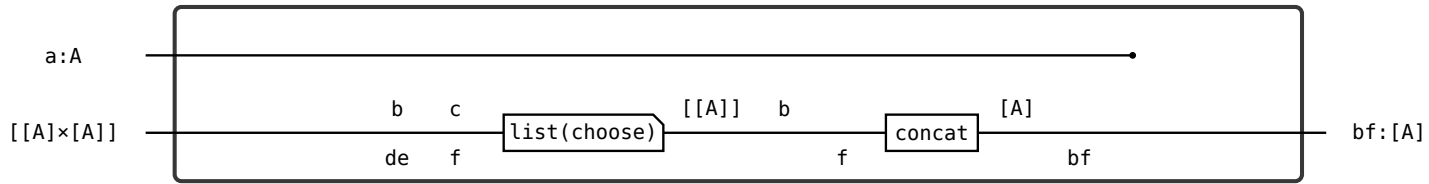
include=

(13.4.1c)



exclude=

(13.4.1d)



$\text{list}(\text{choose}) \ [([d],[]), ([e],[])]$   
 1st item  $([d],[])$  choose  $\rightarrow [d]$  or  $[]$   
 2nd item  $([e],[])$  choose  $\rightarrow [e]$  or  $[]$   
 2x2=4 combinations:  $[[d],[e]] \quad [[d],[]] \quad [[],[e]] \quad [[],[]]$   
 concat flattens:  $[d,e] \quad [d] \quad [e] \quad []$

(13.4.1e)

### §13.4.2. $\text{list}((R \times R)^\circ)$

$$\begin{aligned} & [ ([b],[d,e]) , ([c],[f]) ] \\ & \quad \downarrow \quad \quad \downarrow \\ & [ ([b],[d]) , ([c],[f]) ] \end{aligned}$$
 $\text{list}((R \times R)^\circ), \text{elementwise}$

(13.4.2a)

$([b],[d,e])$   
 $\quad \quad \quad \text{exclude: the best party in b's subtree when b does not come}$   
 $\quad \quad \quad \text{include: the best party in b's subtree when b comes}$

| element            | 1st component<br>include                      | 2nd component<br>exclude                                                                           |
|--------------------|-----------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1st, $([b],[d,e])$ | $[b] R^\circ [b]$<br>$7 \geq 7$ , reflexivity | $[d,e] R^\circ [d]$<br>$6 \geq 5$ , and $([b],[d,e])$ is the pair $\text{est}(R^\circ)$ keeps at b |
| 2nd, $([c],[f])$   | $[c] R^\circ [c]$<br>$2 \geq 2$ , reflexivity | $[f] R^\circ []$<br>$8 \geq 0$ , and $([c],[f])$ is the pair $\text{est}(R^\circ)$ keeps at c      |

(13.4.2b)

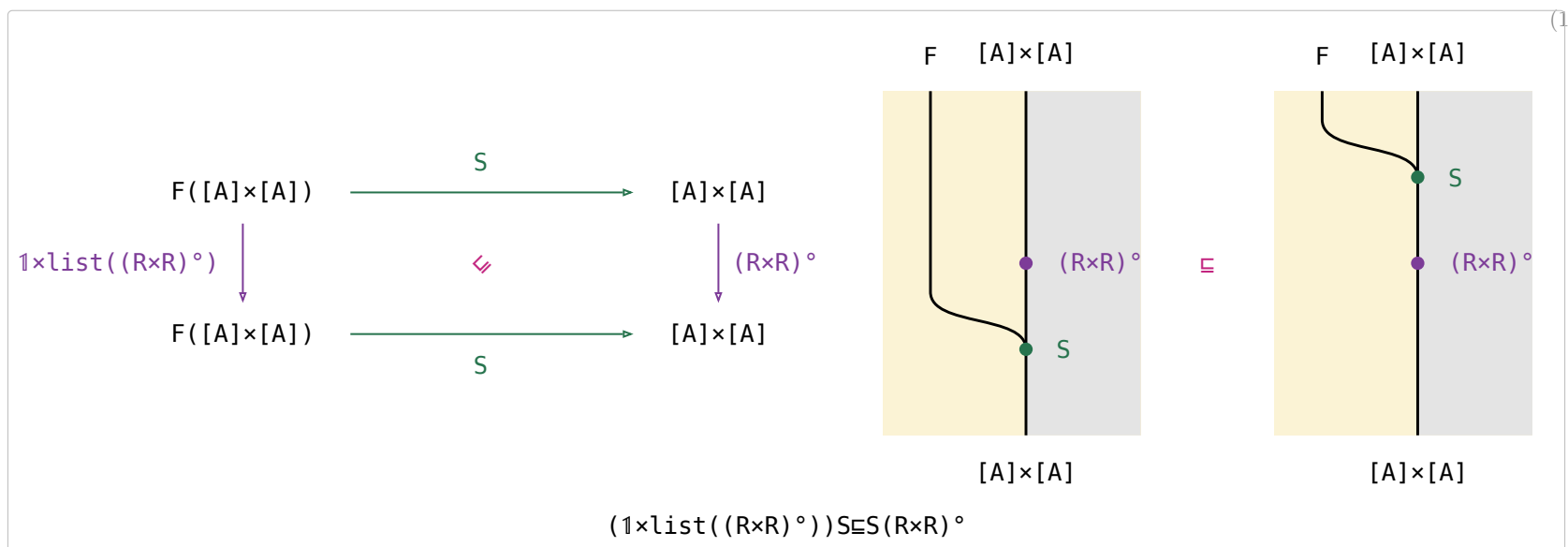
the first component of  $1 \times \text{list}((R \times R)^\circ)$  is the root employee.

$R \triangleq \text{cost} \leq \text{cost}^\circ$  is a preorder, so the components that do not move are related by reflexivity —  $(R \times R)^\circ$  demands a relation in **both** components, it cannot skip one.

$R$  compares cost only, not contents:  $[f] R^\circ []$  is legal though  $[f]$  has an element  $[]$  does not. That is why the three leaves of §13.4.3's proof all reduce to " $\text{cost}$  is a sum".

### §13.4.3. $(1 \times \text{list}((R \times R)^\circ)) S \subseteq S(R \times R)^\circ$ — $S : F([A] \times [A]) \rightarrow [A] \times [A]$ monotonic on $(R \times R)^\circ$

(13.4.3a)



(13.4.3b)

$$\begin{array}{c}
 \frac{\langle S \rangle \text{ choose}}{\exists} \\
 \text{tree } A \xrightarrow{\quad\quad\quad} E[A] \quad = \quad \text{tree } A \xrightarrow{\frac{\langle S \rangle}{\exists}} E([A] \times [A]) \xrightarrow{E(\text{choose})} E[A] \\
 \frac{\langle S \rangle \text{ choose}}{\exists} = \frac{\langle S \rangle}{\exists} E(\text{choose}) \quad (10b), \text{ absorption}
 \end{array}$$

13.4.3c)

$$\begin{array}{l}
 S \triangleq (\text{include}, \text{exclude}) \\
 (1 \times \text{list}((R \times R)^\circ)) S \sqsubseteq S(R \times R)^\circ \\
 (1 \times \text{list}((R \times R)^\circ)) \text{include} \sqsubseteq \text{include } R^\circ \\
 (1 \times \text{list}((R \times R)^\circ)) \text{exclude} \sqsubseteq \text{exclude } R^\circ
 \end{array}$$

bettering both parties of every subtree before the node's algebra runs gets no further than running it first and bettering the two parties it returns

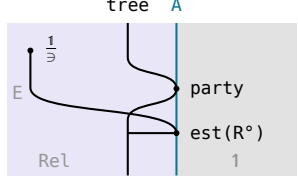
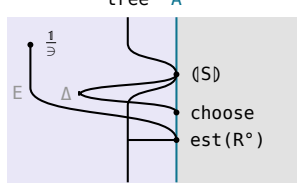
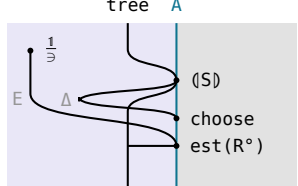
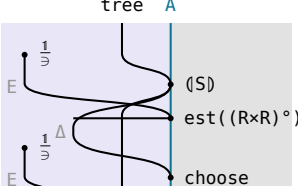
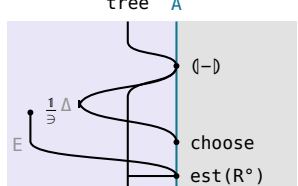
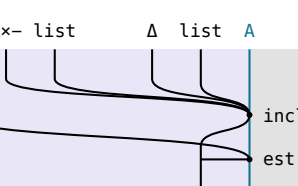
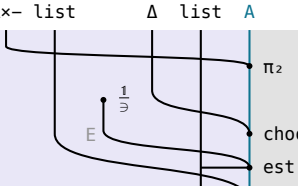
| circuit | Hinze–Marsden |
|---------|---------------|
|         |               |
|         |               |
|         |               |
|         |               |

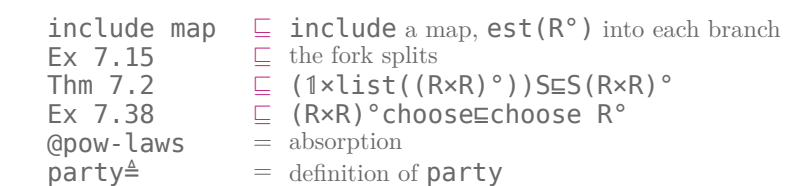
- g**  $(R \times R)^\circ g \sqsubseteq g R^\circ$   
 $g := \pi_2$  is  $(R \times R)^\circ \pi_2 = (\text{Dom}(\pi_1 R^\circ)) \pi_2 R^\circ \sqsubseteq \pi_2 R^\circ$ ,  $g := \pi_1$  its mirror  $(\text{Dom}(\pi_2 R^\circ)) \pi_1 R^\circ \sqsubseteq \pi_1 R^\circ$  — 1 and 4 of (11.1.2b), then  $\text{Dom} \sqsubseteq 1$ ;  $g := \text{choose} \triangleq \pi_1 \cup \pi_2$  is the union of the two (13.2c). **list** monotonic, (11a)  
**concat**  $\text{list}(R^\circ) \text{concat} \sqsubseteq \text{concat } R^\circ$   
 a LEAF: no law above it. **cost** is a sum, so  $\text{cost}(\text{concat } xss) = \text{sum}(\text{list}(\text{cost}) xss)$  and a cheaper part makes a cheaper whole. B&dM's exercise.  
**h**  $(1 \times R^\circ) h \sqsubseteq h R^\circ$   
 a LEAF:  $\text{cost}(\text{cons}(a, xs)) = \text{rating}(a) + \text{cost}(xs)$ , so a cheaper tail makes a cheaper list. B&dM's exercise. For  $h := \pi_2$  it is an EQUALITY  $(1 \times R^\circ) \pi_2 = (\text{Dom}(\pi_1)) \pi_2 R^\circ = \pi_2 R^\circ$ :  $\pi_1$  is a map, hence entire, so  $\text{Dom}(\pi_1) = 1$  — (5c)

|                                                |               |             |
|------------------------------------------------|---------------|-------------|
| $(1 \times (\text{list}(g) \text{ concat})) h$ | <b>g</b>      | <b>h</b>    |
| <b>include</b>                                 | $\pi_2$       | <b>cons</b> |
| <b>exclude</b>                                 | <b>choose</b> | $\pi_2$     |

|                                                                           |                                                                                                                                   |           |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------|
| $R \triangleq \{(\emptyset, \emptyset)\} : \{0, 1\} \rightarrow \{0, 1\}$ | not entire — undefined at 1                                                                                                       | (13.4.3d) |
| $(R \times R) \text{ choose at } (\emptyset, 1)$                          | nothing — $R \times R$ needs BOTH components related, and $R$ is undefined at 1                                                   |           |
| choose $R$ at $(\emptyset, 1)$                                            | $\emptyset$ — choose picks the first component $\emptyset$ , and $R \emptyset \emptyset$ holds                                    |           |
| $((\emptyset, 1), \emptyset)$                                             | in choose $R$ , not in $(R \times R) \text{ choose}$                                                                              |           |
| $(R \times R) \text{ choose} = \text{choose } R$                          | iff $R$ is entire — $\text{Dom} \sqsubseteq 1$ is the g row's only inequality step, and it is an equality iff $\text{Dom}(R) = 1$ |           |

#### §13.4.4. The derivation

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <p> <math>\frac{\text{party}}{\supset} \text{est}(R^\circ) \ni (\langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{\supset} \text{est}(R^\circ)) \text{ concat} \rangle) \frac{\text{choose}}{\supset} \text{est}(R^\circ)</math><br/>                     the best of every guest list the president allows is one pass up the tree, each subtree handing up its best party with its boss in<br/>                     and its best with the boss out, and <b>choose</b> taking the better of the two at the root                 </p> | <p>                     3.4.4a)                 </p>                                                                      |
| <p><b>circuit</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p><b>Hinze–Marsden</b> — outside the <math>\langle \rangle</math> above, inside it below; a fork drawn at one branch</p> |
| <p> <math>\text{tree } A \rightarrow \frac{\text{party}}{\supset} \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>\frac{\text{party}}{\supset} \text{est}(R^\circ)</math><br/>                     the specification — (13.4a)                 </p>                                                                                                                                                                                                                                                                                           |  <p>tree A</p> <p>list A</p>           |
| <p> <math>\text{tree } A \rightarrow \frac{\langle S \rangle \text{ choose}}{\supset} \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>= \frac{\langle S \rangle \text{ choose}}{\supset} \text{est}(R^\circ)</math><br/> <math>\text{party} \triangle \langle S \rangle \text{ choose}</math> — (13.4a)                 </p>                                                                                                                                                                                                                  |  <p>tree A</p> <p>list A</p>           |
| <p> <math>\text{tree } A \rightarrow \frac{\langle S \rangle}{\supset} E(\text{choose}) \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>= \frac{\langle S \rangle}{\supset} E(\text{choose}) \text{est}(R^\circ)</math><br/> <math>\frac{\langle S \rangle \text{ choose}}{\supset} = \frac{\langle S \rangle}{\supset} E(\text{choose})</math> — (13.4.3b)                 </p>                                                                                                                                                              |  <p>tree A</p> <p>list A</p>          |
| <p> <math>\text{tree } A \rightarrow \frac{\langle S \rangle}{\supset} \text{est}((R \times R)^\circ) \frac{\text{choose}}{\supset} \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>\frac{\langle S \rangle}{\supset} \text{est}((R \times R)^\circ) \frac{\text{choose}}{\supset} \text{est}(R^\circ)</math><br/> <math>(R \times R)^\circ \text{ choose} \sqsubseteq \text{choose } R^\circ</math> — Ex 7.38, (13.4.3c)'s g row                 </p>                                                                                        |  <p>tree A</p> <p>list A</p>         |
| <p> <math>\text{tree } A \rightarrow \frac{\langle S \rangle}{\supset} \text{est}((R \times R)^\circ) \frac{\text{choose}}{\supset} \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>\frac{\langle S \rangle}{\supset} \text{est}((R \times R)^\circ) \frac{\text{choose}}{\supset} \text{est}(R^\circ)</math><br/> <math>(1 \times \text{list}((R \times R)^\circ)) S \sqsubseteq S(R \times R)^\circ</math> — Theorem 7.2 at <math>(R \times R)^\circ</math>, (13.4.3a)                 </p>                                                 |  <p>tree A</p> <p>list A</p>         |
| <p> <math>\text{tree } A \rightarrow \frac{\langle \langle \text{include} \frac{\text{choose}}{\supset} \text{est}(R^\circ), \text{exclude} \frac{\text{choose}}{\supset} \text{est}(R^\circ) \rangle \rangle}{\supset} \text{est}(R^\circ) \rightarrow [A]</math><br/> <math>\langle \frac{\text{include}}{\supset} \text{est}(R^\circ), \frac{\text{exclude}}{\supset} \text{est}(R^\circ) \rangle \sqsubseteq \frac{\text{choose}}{\supset} \text{est}((R \times R)^\circ)</math> — Ex 7.15                 </p>                                 |  <p>tree A</p> <p>list A</p>         |
| <p> <math>\text{tree } A \rightarrow \frac{\langle \langle \text{include}, \pi_2 \text{ list}(\frac{\text{choose}}{\supset} \text{est}(R^\circ)) \text{ concat} \rangle \rangle}{\supset} \text{est}(R^\circ)</math><br/>                     include a map, <math>\text{est}(R^\circ)</math> into each branch                 </p>                                                                                                                                                                                                                 |  <p>tree A</p> <p>list A</p>         |





## §13.5. Shortest paths on a cylinder

| definition                                                                                                  | type                                                     | note                                                                                         | (13.5a) |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|---------|
| $F(A, X) = A + A \times X$                                                                                  | $\mathcal{A} \times \mathcal{A} \rightarrow \mathcal{A}$ | The base functor: one square of the new column, alone or in front of the path so far.        |         |
| $L = \text{list}^+$                                                                                         | $\mathcal{A} \rightarrow \mathcal{A}$                    | A path is a non-empty list of squares, one per column crossed.                               |         |
| $\alpha$ the initial algebra                                                                                | $F(A, L\mathcal{A}) \rightarrow L\mathcal{A}$            | $\alpha = [\text{wrap}, \text{cons}]$ : a path is started by one square, or extended by one. |         |
| $N$<br>the n-tuple relator                                                                                  | $\mathcal{A} \rightarrow \mathcal{A}$                    | One component per row of a column, so the fold carries n answers at once.                    |         |
| $R \triangleq \text{sum} \leq \text{sum}^\circ$                                                             | $L \text{ Nat} \rightarrow L \text{ Nat}$                | The cost of a path, which the cheapest minimises.                                            |         |
| setify                                                                                                      | $NA \rightarrow EA$                                      | Forgets which row a component came from, so the n answers become one set.                    |         |
| moves                                                                                                       | $NA \rightarrow E(NA)$                                   | The three columns a path may step to: rotated up, unrotated, rotated down.                   |         |
| trans                                                                                                       | $E(NA) \rightarrow N(EA)$                                | Transposes a set of tuples, so every row collects its own paths.                             |         |
| zip                                                                                                         | $F(NA, NB) \rightarrow NF(A, B)$                         | Commutates $N$ with the base functor, pairing each new square with its own row's paths.      |         |
| $\text{cp} \triangleq \frac{F(1, \exists)}{\exists}$                                                        | $F(A, EB) \rightarrow E(F(A, B))$                        | Turns a square paired with a set of paths into the set of extensions of those paths.         |         |
| $\text{generate} \triangleq F(1, \text{moves trans } N(\text{union})) \text{ zip } N(\text{cp } P(\alpha))$ | $F(NA, N(E(LA))) \rightarrow N(E(LA))$                   | One fold step: extends every path of every row by the new column.                            |         |
| $\text{paths} \triangleq (\text{generate}) \text{ setify union}$                                            | $L \text{ N Nat} \rightarrow E(L \text{ Nat})$           | Every path across the cylinder.                                                              |         |
| the specification<br>$\text{paths est}(R)$                                                                  | $L \text{ N Nat} \rightarrow L \text{ Nat}$              | A cheapest path from the entry side to the exit side.                                        |         |

§13.5.1.  $N(E(L \ A))$ 

$$\begin{array}{rcl} 1 & 5 & \\ 2 & 6 & \\ 3 & 7 & [(1,2,3,4), (5,6,7,8)] : L \text{ N Nat} \\ 4 & 8 & \end{array} \quad (13.5.1a)$$

$$(\text{generate})[(5,6,7,8)] = (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\}) \quad (13.5.1b)$$

$$\begin{aligned} \text{generate}((1,2,3,4), (\{[5]\}, \{[6]\}, \{[7]\}, \{[8]\})) = \\ ( \{ [1,5], [1,6], [1,8] \}, \\ \{ [2,5], [2,6], [2,7] \}, \\ \{ [3,6], [3,7], [3,8] \}, \\ \{ [4,5], [4,7], [4,8] \} ) \end{aligned}$$

|     |                                                        |           |
|-----|--------------------------------------------------------|-----------|
| $N$ | the 4-tuple, one component per row of the entry column | (13.5.1c) |
| $E$ | the set of paths that can start in that row            |           |
| $L$ | a path is the non-empty list of the squares it crosses |           |

row 1's set has no  $[1,7]$  — from row 1 only rows 4, 1 and 2 are reachable — and it does have  $[1,8]$ , because the cylinder glues the bottom row to the top. That is **moves trans  $N(\text{union})$**  of (13.5a): component  $k$  collects the paths of rows  $k-1$ ,  $k$ ,  $k+1$ .

## §13.5.2. The derivation

(13.5.2a)

| <b>paths est(R) <math>\exists(Q)</math> setify est(R)</b><br>B&dM §7.4, p. 182; $Q$ is (13.5.2c)'s algebra                                                                                                                                                                                           |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                              | Hinze–Marsden |
| $\text{L N Nat} \text{---} \boxed{\text{paths}} \text{---} \boxed{\text{est(R)}} \text{---} \text{L Nat}$ $\text{paths est(R)}$                                                                                                                                                                      |               |
| $\text{L N Nat} \text{---} \boxed{(\text{generate})} \text{---} \boxed{\text{setify}} \text{---} \boxed{\text{union}} \text{---} \boxed{\text{est(R)}} \text{---} \text{L Nat}$ $= (\text{generate}) \text{ setify union est(R)}$ (13.5a) at paths                                                   |               |
| $\text{L N Nat} \text{---} \boxed{(\text{generate})} \text{---} \boxed{\text{setify}} \text{---} \boxed{P(\text{est(R)})} \text{---} \boxed{\text{est(R)}} \text{---} \text{L Nat}$ $\sqsubseteq (\text{generate}) \text{ setify } P(\text{est(R)}) \text{ est(R)}$ (13.1b) at union, $R$ a preorder |               |
| $\text{L N Nat} \text{---} \boxed{(\text{generate})} \text{---} \boxed{N(\text{est(R)})} \text{---} \boxed{\text{setify}} \text{---} \boxed{\text{est(R)}} \text{---} \text{L Nat}$ $\sqsubseteq (\text{generate})N(\text{est(R)}) \text{ setify est(R)}$ setify lax natural                         |               |
| $\text{L N Nat} \text{---} \boxed{(Q)} \text{---} \boxed{\text{setify}} \text{---} \boxed{\text{est(R)}} \text{---} \text{L Nat}$ $\sqsubseteq (Q) \text{ setify est(R)}$ (11.4.2a) at (13.5.2b)                                                                                                     |               |

(13.5.2b)

| <b>generate <math>N(\text{est}(R)) \sqsubseteq F(1, N(\text{est}(R)))Q</math></b><br>the fusion condition of (13.5.2a)'s last step                                            |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                       | Hinze–Marsden |
| $F(NA, N(E(LA))) \rightarrow \text{generate} \rightarrow N(\text{est}(R)) \rightarrow N(LA)$<br>generate $N(\text{est}(R))$                                                   |               |
| $F(NA, N(E(LA))) \rightarrow F(1, N(\text{est}(R))) \rightarrow Q \rightarrow N(LA)$<br>$\sqsubseteq$ $F(1, N(\text{est}(R)))Q$<br>(7.13), then zip, trans, moves lax natural |               |

(7.13) is  $F(1, \text{est}(R)) \alpha \sqsubseteq_{\text{cp}} P(\alpha) \text{ est}(R)$ , (13.3.1c) at the map  $\alpha$  with  $\frac{F(1, \exists)\alpha}{\exists} = \text{cp } P(\alpha)$ : extending every path in a set and then taking a minimum is beaten by extending one minimum. It is the crux here, not the greedy theorem.

(13.5.2c)

| <b><math>Q = [N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]</math></b><br>the algebra (13.5.2a)'s last step folds                                                                                           |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit — the fork is $F(NA, N(LA)) = NA + NA \times N(LA)$                                                                                                                                                                                                | Hinze–Marsden |
| $F(NA, N(LA)) \rightarrow Q \rightarrow N(LA)$<br>$Q$                                                                                                                                                                                                      |               |
| $F(NA, N(LA)) \rightarrow F(1, \text{moves trans } N(\text{est}(R))) \rightarrow \text{zip} \rightarrow N(\alpha) \rightarrow N(LA)$<br>$= F(1, \text{moves trans } N(\text{est}(R))) \text{ zip } N(\alpha)$<br>the fusion condition read as a definition |               |
| $NA$<br>$NA$<br>$N(LA)$<br><br>$[N(\text{wrap}), (1 \times \text{moves trans } N(\text{est}(R))) \text{ zip}' N(\text{cons})]$<br>$\text{zip} = 1 + \text{zip}', \alpha = [\text{wrap}, \text{cons}]$                                                      |               |

## §13.6. The security van problem

| definition                                                                                                                   | type                                                                            | note                                                                                            | (13.6a) |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|---------|
| $R \triangleq \text{length} \leq \text{length}^\circ$                                                                        | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The order the schedule is minimised over: fewer van visits is better.                           |         |
| $\text{ceiling} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\geq)$                                                   | $[\text{Int}] \rightarrow \text{Int}$                                           | The highest the bank's balance reaches over a stretch of transactions.                          |         |
| $\text{floor} \triangleq \frac{\text{prefix sum}}{\ni} \text{est}(\leq)$                                                     | $[\text{Int}] \rightarrow \text{Int}$                                           | The lowest it reaches, so $\text{ceiling} - \text{floor}$ is the cash the stretch has to carry. |         |
| <b>secure</b><br>the coreflexive on $x$ with<br>$\text{bmax}(\text{ceiling } x, \text{ceiling } x - \text{floor } x) \leq N$ | $[\text{Int}] \rightarrow [\text{Int}]$                                         | The stretches one van visit can serve: some starting reserve keeps the cash between 0 and N.    |         |
| <b>ok</b><br>the coreflexive on $(a, xs)$ with $xs$ non-empty and $[a] \# \text{head } xs$ secure                            | $\text{Int} \times [[\text{Int}]] \rightarrow \text{Int} \times [[\text{Int}]]$ | The test the final program runs, in place of <b>old</b> 's <b>secure</b> .                      |         |
| $\text{new} \triangleq (\text{wrap} \times 1) \text{ cons}$                                                                  | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | Calls the van: the transaction opens a segment of its own.                                      |         |
| $\text{glue} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } (\text{cons} \times 1) \text{ cons}$                    | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | Puts the transaction on the front of the first segment.                                         |         |
| $\text{old} \triangleq (1 \times \text{cons}^\circ) \text{ assocl } ((\text{cons } \text{secure}) \times 1) \text{ cons}$    | $\text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$                   | <b>glue</b> restricted to a first segment that stays secure.                                    |         |
| $\text{partition} = ([\text{nil}, \text{newuglue}])$                                                                         | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | Every splitting of the transactions into consecutive segments.                                  |         |
| $S \triangleq [\text{nil}, \text{newuold}]$                                                                                  | $1 + \text{Int} \times [[\text{Int}]] \rightarrow [[\text{Int}]]$               | The algebra whose fold is every splitting of the transactions into secure segments.             |         |
| $\text{partition list}(\text{secure}) = (S)$                                                                                 | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | Fusion: keeping only secure segments is what turns <b>glue</b> into <b>old</b> .                |         |
| $H \triangleq (\text{head prefix}^\circ \text{ head}^\circ) \cup (\text{nil}^\circ \text{ nil})$                             | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | One schedule's first segment is a prefix of the other's, or both are empty.                     |         |
| $R; H \triangleq R \cap (R^\circ \Rightarrow H)$                                                                             | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The refinement of $R$ that makes both halves of $S$ monotonic.                                  |         |
| $ R  \triangleq R \cap R^\circ$                                                                                              | $[[\text{Int}]] \rightarrow [[\text{Int}]]$                                     | The strict part $R$ splits into: $R; H =  R  \cup (R \cap H)$ .                                 |         |
| the specification<br>$\frac{\text{partition list}(\text{secure})}{\ni} \text{est}(R)$                                        | $[\text{Int}] \rightarrow [[\text{Int}]]$                                       | A schedule with the fewest secure segments.                                                     |         |

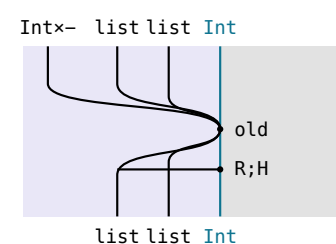
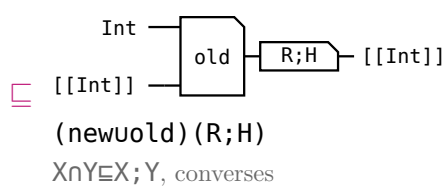
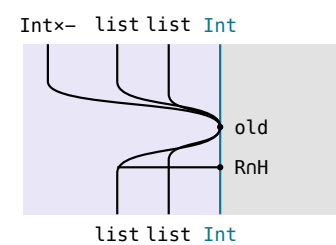
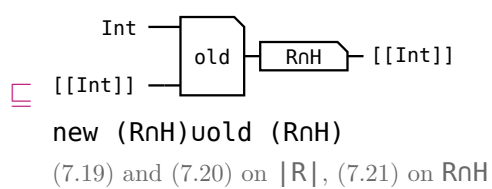
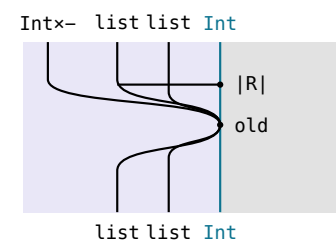
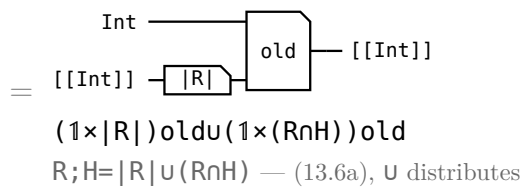
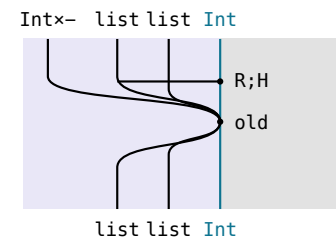
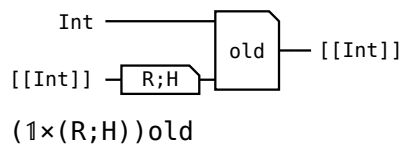
| $\frac{\text{partition list}(\text{secure})}{\supset} \text{est}(R) \sqsubseteq ([\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})])$ <p>the fewest secure segments the transactions can be cut into are one pass along them, the next transaction glued onto the open segment wherever that segment stays secure and the van called where it does not</p>                                                                                                                    |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Hinze–Marsden |
| $[\text{Int}] \rightarrow \boxed{\frac{\text{partition list}(\text{secure})}{\supset}} \text{est}(R) \rightarrow [[\text{Int}]]$ <p><math>\frac{\text{partition list}(\text{secure})}{\supset} \text{est}(R)</math><br/>the specification — (13.6a)</p>                                                                                                                                                                                                                                  |               |
| $[\text{Int}] \rightarrow \boxed{\frac{\langle S \rangle}{\supset}} \text{est}(R) \rightarrow [[\text{Int}]]$ <p><math>= \frac{\langle S \rangle}{\supset} \text{est}(R)</math><br/><math>\text{partition list}(\text{secure}) = \langle S \rangle</math> — (13.6a), (11.4.2a) at <math>\text{secure prefix} \sqsubseteq \text{prefix secure}</math></p>                                                                                                                                 |               |
| $[\text{Int}] \rightarrow \boxed{\frac{\langle S \rangle}{\supset}} \text{est}(R; H) \rightarrow [[\text{Int}]]$ <p><math>\sqsubseteq \frac{\langle S \rangle}{\supset} \text{est}(R; H)</math><br/><math>R; H \sqsubseteq R</math> — (13.6a); (7.15) <math>(1 \times R) \text{old} \sqsubseteq (\text{new} \text{old}) R</math> is FALSE, the shorter partition need not stay secure, where (7.14) <math>(1 \times R) \text{new} \sqsubseteq (\text{new} \text{old}) R</math> holds</p> |               |
| $[\text{Int}] \rightarrow \boxed{\frac{\langle S \rangle}{\supset} \text{est}(R; H)} \rightarrow [[\text{Int}]]$ <p><math>\sqsubseteq \frac{\langle S \rangle}{\supset} \text{est}(R; H)</math><br/>(13.3.2a) at <math>R; H</math>, its hypothesis <math>F(R; H) S \sqsubseteq S(R; H)</math> the old half (7.17) — (13.6c) — and the new half (7.16), which rests on (7.18) <math>(1 \times T) \text{new} \sqsubseteq \text{new } H</math></p>                                          |               |
| $[\text{Int}] \rightarrow \boxed{\langle [\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})] \rangle} \rightarrow [[\text{Int}]]$ <p><math>\sqsubseteq \langle [\text{nil}, (\text{ok} \rightarrow \text{glue}, \text{new})] \rangle</math><br/><math>\text{old} \sqsubseteq \text{new } (R; H) \circ</math>: old returns the shorter result wherever it returns one, and ok is where it does</p>                                                                              |               |

$$(1 \times (R;H)) \text{old} \sqsubseteq (\text{newuold})(R;H)$$

gluing the transaction onto a better schedule for the rest gets no further than gluing it on, or calling the van, and bettering the whole schedule after

**circuit** — the old branch of each union

**Hinze–Marsden**



## §14. Thinning Algorithms

### §14.1. Thinning

For  $Q : A \rightarrow A$ ,  $\text{thin } Q \triangleq (\exists/\exists) \cap (\epsilon \setminus (Q^\circ \epsilon)) : EA \rightarrow EA$  (8.1).

$ys \text{ (thin } Q) \text{ } xs \iff xs \subseteq ys \wedge (\forall a \in ys. \exists b \in xs. b \ Q \ a)$

(14.1a)

| the law                                                                                                                                                 | what it says                                                                                                              | (14.1b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------|
| $X \subseteq \frac{S}{\exists} \text{ thin } Q \iff X \subseteq S \text{ and } S^\circ X \subseteq Q^\circ \epsilon$                                    | the universal property: everything kept is an $S$ -value, and every $S$ -value has a $Q$ -lower bound among the kept ones |         |
| $Q \subseteq R \implies \text{thin } Q \subseteq \text{thin } R$                                                                                        | the fewer pairs $Q$ relates, the fewer subsets count as thinnings                                                         |         |
| $1 \in \text{thin } Q$ , and $\text{thin } Q$ is a preorder if $Q$ is                                                                                   | keeping everything is always a legal thinning                                                                             |         |
| $\text{est}(R) = \text{thin } Q \text{ est}(R) \quad Q \subseteq R, \text{ both preorders}$                                                             | <b>thin-introduction</b> : thinning first cannot lose an $R$ -minimum                                                     |         |
| $\text{thin } Q \supseteq \text{est}(Q) \quad \frac{1}{\exists} \quad (8.2)$                                                                            | <b>thin-elimination</b> : keeping one element is a thinning, but its domain is the sets $\text{est}(Q)$ is defined on     |         |
| $\frac{S}{\exists} \text{ thin } Q \supseteq \frac{S}{\exists} \text{ est}(R) \quad \frac{1}{\exists}$<br>(8.3), $R \cap (S^\circ S) \subseteq Q$       | the usable variant: $R$ need only refine $Q$ between values $S$ gives one argument                                        |         |
| $\text{union thin } Q \supseteq P(\text{thin } Q) \text{ union} \quad (8.4)$                                                                            | thinning each member set is a thinning of the union                                                                       |         |
| $Q(\frac{F(\exists,1)S}{\exists} \text{ thin } Q) \subseteq \frac{QS}{\exists} \text{ thin } Q$<br>Theorem 8.1, $S$ monotonic on $Q$                    | the <b>thinning theorem</b> : thinning at every step beats thinning only at the end                                       |         |
| $Q(\frac{F(\exists,1)S}{\exists} \text{ thin } Q) \text{ est}(R) \subseteq \frac{QS}{\exists} \text{ est}(R)$<br>Corollary 8.1, $Q \subseteq R$ as well | the same against the optimisation problem itself, by thin-introduction                                                    |         |

### §14.2. Paths in a layered network

$F(A, X) = A + A \times X$ ,  $L = \text{list}^+$  with initial algebra  $\alpha \triangleq [\text{wrap}, \text{cons}] : F(A, LA) \rightarrow LA$ .

$\text{wrapz} \triangleq (\text{wrap}, \text{zero})$ ,  $\text{consw } (a, (xs, n)) = (\text{cons } (a, xs), \text{wt } (a, \text{head } xs) + n)$ .

$\text{cost} \triangleq ([\text{wrapz}, \text{consw}]) \pi_2$ ,  $Q([\text{wrapz}, \text{consw}]) = (1, \text{cost})$ ,  $R \triangleq \text{cost} \leq \text{cost}^\circ$ .

$Q \triangleq R \cap (\text{head } \text{head}^\circ)$ ,  $S \triangleq F(1, \exists) \alpha$ ,  $\frac{F(\exists, 1)}{\exists} = 1 + \text{cpl}$ ,  $\frac{F(1, \exists)}{\exists} = 1 + \text{cpr}$ ,  $\text{step} \triangleq \text{cpr } P(\text{cons}) \text{ est}(R)$ .

(14.2a)

| the law                                                                                                                                                                                                                       | what it says                                                                                                             | (14.2b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R) = \frac{L(\exists)}{\exists} \text{ est}(R)$                                                                                                                          | the specification: a least-cost path across the network                                                                  |         |
| $F(\exists, Q) \alpha \subseteq F(\exists, 1) \alpha Q$                                                                                                                                                                       | $S$ is monotonic on $Q$ ; on $R$ it is not, since the next edge can cost arbitrarily much                                |         |
| $Q(\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q) \text{ est}(R) \subseteq \frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R)$<br>Corollary 8.1                                                              | thinning applies: one partial path per starting vertex is enough                                                         |         |
| $S \text{ head} \in [1, \pi_2]$<br>$S \text{ head}$ simple, so $R \cap (S^\circ S) \subseteq Q$                                                                                                                               | the side condition of (8.3): between two paths $S$ builds from one argument, equal cost and equal head already means $Q$ |         |
| $\frac{F(\exists, \exists) \alpha}{\exists} \text{ thin } Q$<br>$\supseteq \frac{F(\exists, 1)}{\exists} P(\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R))$<br>(8.4), (8.3), $P(\frac{1}{\exists}) \text{ union} = 1$ | thin eliminated: split the algebra at $F(\exists, 1)F(1, \exists)$ , thin each part, one minimum per part                |         |
| $\frac{F(1, \exists)}{\exists} P(\alpha) \text{ est}(R) = [\text{wrap}, \text{step}]$                                                                                                                                         | that inner part read off the two branches of $\alpha$                                                                    |         |
| $\frac{Q(F(\exists, 1) \alpha)}{\exists} \text{ est}(R) \supseteq ([P(\text{wrap}), \text{cpl } P(\text{step})]) \text{ est}(R)$                                                                                              | the program: one fold, a best path per vertex of the current layer                                                       |         |

## §14.3. Implementing thin

(14.3a)

setify :  $[A] \rightarrow EA$ ,  $\text{cup} : EA \times EA \rightarrow EA$ ,  $\text{cp}(F) \triangleq \frac{F(\exists)}{\exists}$ ,  $\text{listcp}(F) : F([A]) \rightarrow [FA]$ ,  $\text{sort } P \triangleq \text{setify}^\circ \text{ ordered } P$  for  $P$  a connected preorder.

$\text{thinlist } Q$  is any  $\text{thinlist } Q \subseteq \text{subseq}$  with  $\text{thinlist } Q \text{ setify} \subseteq \text{setify thin } Q$ ; one is  $\langle [\text{nil}, \text{bump } Q] \rangle$ ,  $\text{bump } Q(a, []) = [a]$ ,  $\text{bump } Q(a, [b] \# xs) = (b \text{ } Q \text{ } a \rightarrow [a] \# xs, a \text{ } Q \text{ } b \rightarrow [b] \# xs, [a] \# [b] \# xs)$ .

**Binary thinning** data:  $S = (f_1 p_1) \cup (f_2 p_2)$  with  $p_1, p_2$  coreflexive;  $Q$  a preorder with  $Q \subseteq R$  and both  $f_1 p_1, f_2 p_2$  monotonic on  $Q$ ;  $P$  a connected preorder with both  $f_1, f_2$  monotonic on  $P$ ;  $g_i \triangleq \text{list}(f_i) \text{ filter}(p_i)$ .

| the law                                                                                                                                                                                                     | what it says                                                                                                     | (14.3b) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------|
| $\text{thinlist } Q \text{ xs} = [\text{minlist } Q \text{ xs}]$<br>(8.5), $Q$ connected, $XS$ non-empty                                                                                                    | what thinning should come to when it can: one element                                                            |         |
| $\text{sort } P \text{ thinlist } Q \subseteq \text{thin } Q \text{ sort } P$ (8.6)                                                                                                                         | thinning a sorted list is a thinning of the set — this is what $\text{thinlist } Q \subseteq \text{subseq}$ buys |         |
| $\text{sort } P \text{ minlist } Q \subseteq \text{est}(Q)$ (8.7)                                                                                                                                           | a minimum of the sorted list is a minimum of the set                                                             |         |
| $\text{sort}(f P f^\circ) \text{ list}(f) \subseteq P(f) \text{ sort } P$ (8.8)                                                                                                                             | shunt a function through a sort                                                                                  |         |
| $\text{sort } P \text{ filter}(p) \subseteq E(p) \text{ sort } P$ (8.9), $p$ coreflexive                                                                                                                    | filtering a sorted list sorts the restricted set                                                                 |         |
| $(\text{sort } P \times \text{sort } P) \text{ merge } P \subseteq \text{cup sort } P$ (8.10)                                                                                                               | merging two sorted lists sorts their union                                                                       |         |
| $F(\text{sort } P) \text{ listcp}(F) \subseteq \text{cp}(F) \text{ sort}(FP)$<br>(8.11), $F$ linear                                                                                                         | $\text{listcp}(F)$ is the list implementation of the cartesian product $\text{cp}(F)$                            |         |
| $F(\text{sort } P) \text{ listcp}(F) \text{ list}(f) \text{ filter}(p) \subseteq \frac{F(\exists) f p}{\exists} \text{ sort } P$<br>Lemma 8.1, $f$ monotonic on $P$ , $p$ coreflexive                       | one sorted list built from sorted arguments, instead of a set built and then sorted                              |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } P \text{ thinlist } Q \rangle \text{ minlist } R$<br>$\subseteq \frac{\langle \text{est} \rangle}{\exists} \text{ est}(R)$<br>Theorem 8.2 | the <b>binary thinning theorem</b> : a fold on sorted lists of partial solutions, thinned at every step          |         |

## §14.4. The knapsack problem

(14.4a)

$\text{vol}, \text{wt} : \text{Item} \rightarrow \text{Real}$ ,  $\text{value} \triangleq \text{list}(\text{vol}) \text{ sum}$ ,  $\text{weight} \triangleq \text{list}(\text{wt}) \text{ sum}$ .

$\text{subseq} = \langle [\text{nil}, \text{cons}] \cup [\text{nil}, \pi_2] \rangle$ , within  $w$  the coreflexive on  $xs$  with  $\text{weight } xs \leq w$ .

$R \triangleq \text{value} \geq \text{value}^\circ$ ,  $Q \triangleq R \cap (\text{weight} \leq \text{weight}^\circ)$ ,  $P \triangleq R$ .

$FA = 1 + \text{Item} \times A$ ,  $\text{listcp}(F) = \text{wrap} + \text{cpr}$ ,  $g_1 \triangleq \text{list}([\text{nil}, \text{cons}]) \text{ filter}(\text{within } w) = [\text{list}(\text{nil}), h_1]$ ,  
 $g_2 \triangleq \text{list}([\text{nil}, \pi_2]) = [\text{list}(\text{nil}), h_2]$ .

$h_1 \triangleq \text{list}(\text{cons}) \text{ filter}(\text{within } w)$ ,  $h_2 \triangleq \text{list}(\pi_2)$ .

| the law                                                                                                                                                             | what it says                                                                                              | (14.4b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{subseq (within } w)}{\exists} \text{ est}(R)$                                                                                                          | the specification: a selection of greatest total value that stays within the capacity $w$                 |         |
| $\text{subseq (within } w) = \langle ([\text{nil}, \text{cons}] (\text{within } w)) \cup [\text{nil}, \pi_2] \rangle$<br>fusion, weights non-negative               | the selections that fit are themselves a fold, and the fold's algebra already has binary thinning's shape |         |
| $(1 \times R) (\text{cons (within } w)) \subseteq \text{cons (within } w) R$<br>FALSE                                                                               | not monotonic on $R$ : a selection of greater value need not still fit once one more item goes in         |         |
| $(1 \times Q) (\text{cons (within } w)) \subseteq \text{cons (within } w) Q$<br>$(1 \times Q) \pi_2 \subseteq \pi_2 Q$                                              | both halves are monotonic on $Q$ once ties in value are broken by weight                                  |         |
| $\langle \text{listcp}(F) \langle g_1, g_2 \rangle \text{ merge } R \text{ thinlist } Q \rangle \text{ minlist } R$<br>Theorem 8.2 at $P \triangleq R$ , $F$ linear | binary thinning, sorting in descending order of value                                                     |         |
| $\text{knapsack } w = \langle [\text{nil}, \text{cpr} \langle h_1, h_2 \rangle \text{ merge } R \text{ thinlist } Q] \rangle \text{ minlist } R$                    | the program; $\text{minlist } R$ becomes <b>head</b> , since packings come out in descending value        |         |



## §14.5. The paragraph problem

(14.5a)

```

Line=list+ Word, Para=list+ Line, FA=Word+Word×A, listcp(F)=wrap+cpr.

new (a,xs)=[a]*xs, glue (a,xs)=[a]*head xs*tail xs, partition≐([wrap wrap,newuglue]) : list+
Word→Para.

width≐([length,(length×1) plus succ]), fits w the coreflexive on a line x with width x≤w, ok w the coreflexive
on [x]*xs with width x≤w.

white w x=w-width x, collect≐list(sqr) sum, waste w≐init list(white w) collect.

R≐(waste w)≤(waste w)°, Q≐Rn(head head°), P≐T.

g1≐list([wrap wrap,new]), g2≐list([wrap wrap,glue]) filter(ok w), start≐wrap wrap wrap, h1≐list(new),
h2≐list(glue) filter(ok w).

```

| the law                                                                                                                                                                                            | what it says                                                                                                             | (14.5b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{partition list}^+(\text{fits } w)}{\ni} \text{ est}(R)$                                                                                                                               | the specification: least waste among the paragraphs whose every line fits                                                |         |
| $\text{partition list}^+(\text{fits } w) = ([\text{wrap wrap}, \text{newu}(\text{glue } (ok \ w))] \downarrow)$<br>fusion, every word fits on a line by itself                                     | the fitting paragraphs are themselves a fold                                                                             |         |
| $[\text{wrap wrap}, \text{newu}(\text{glue } (ok \ w))] = [\text{wrap wrap}, \text{new}] \cup ([\text{wrap wrap}, \text{glue}] (ok \ w))$                                                          | rewritten into binary thinning's shape $(f_1 p_1) \cup (f_2 p_2)$                                                        |         |
| $(1 \times R) \text{ glue} \sqsubseteq \text{glue } R \quad \text{FALSE}$                                                                                                                          | glue is not monotonic on R: the waste of a paragraph depends on its whole first line, so no greedy algorithm solves this |         |
| $(1 \times Q) \text{ new} \sqsubseteq \text{new } Q$<br>$(1 \times Q) (\text{glue } (ok \ w)) \sqsubseteq \text{glue } (ok \ w) Q$<br>cons monotonic on $\text{collect} \leq \text{collect}^\circ$ | both halves are monotonic on Q once ties in waste are broken by the first line                                           |         |
| merge $T = \text{cat}$ ; $P \triangleq \text{head prefix head}^\circ$ also serves                                                                                                                  | T needs no sorting at all, and prefix is a linear order on first lines of paragraphs of one input                        |         |
| $(\text{listcp}(F) \ (g_1, g_2) \ \text{cat thinlist } Q) \ \text{minlist } R$<br>Theorem 8.2 at $P \triangleq T$                                                                                  | binary thinning, one candidate kept per first line                                                                       |         |
| paragraph $w = ([\text{start}, \text{cpr } (h_1, h_2) \ \text{cat thinlist } Q]) \ \text{minlist } R$                                                                                              | the program, $g_1$ and $g_2$ split along the coproduct                                                                   |         |

## §14.6. Bitonic tours

(14.6a)

```

FA=(City×City)+(City×A), the base functor of cons-lists of length at least two; listcp(F)=wrap+cpr; tc :
City×City→Real, neither positive nor symmetric.

start (a,b)=[a,b],[a,b], dropl (a,([b]*xs,ys))=[a]*xs,[a]*ys, dropr (a,(xs,[b]*ys))=[a]*xs,
[a]*ys, tour≐([start,dropludropr]).

cost (xs,ys)=outcost xs+incost ys, outcost [a0,...,an]=tc (a0,a1)+...+tc (an-1,an), incost [a0,...,an]=tc
(a1,a0)+...+tc (an,an-1).

next≐tail head, next2≐next×next, R≐cost≤cost°, Q≐Rn(next2 next2°), P≐T, g1≐list([start,dropl]),
g2≐list([start,dropr]).

```

| the law                                                                                                                                                    | what it says                                                                                                                  | (14.6b) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{tour}}{\ni} \text{est}(R)$                                                                                                                    | the specification: a least-cost bitonic tour, outward journey and return kept as a pair of lists                              |         |
| $(1 \times R) \text{ dropl} \sqsubseteq \text{dropl } R \quad \text{FALSE}$<br>$(1 \times R) \text{ dropr} \sqsubseteq \text{dropr } R \quad \text{FALSE}$ | neither drop is monotonic on <b>R</b> : the two edges it adds and removes depend on <b>head</b> and <b>next</b> of both lists |         |
| $(1 \times Q) \text{ dropl} \sqsubseteq \text{dropl } Q$<br>$(1 \times Q) \text{ dropr} \sqsubseteq \text{dropr } Q$                                       | both are, once ties in cost are broken by the two second cities — the heads already agree among tours of one input            |         |
| $(\text{listcp}(F) \langle g_1, g_2 \rangle \text{ cat thinlist } Q) \text{ minlist } R$<br>Theorem 8.2 at $P \triangleleft T$ , merge $T = \text{cat}$    | binary thinning, one candidate per pair of second cities                                                                      |         |
| $\text{mintour} = ((\text{start wrap, cpr } (\text{list}(\text{dropl}), \text{list}(\text{dropr}))$<br>$\text{cat thinlist } Q)) \text{ minlist } R$       | the program: quadratic, because each step adds just two tours to the list kept                                                |         |

## §15. Dynamic Programming

### §15.1. Theory

$h : FB \rightarrow B$  a map,  $T : FA \rightarrow A$  an  $F$ -algebra,  $R : B \rightarrow B$ .

(15.1a)

$H \triangleq (T)^\circ \circ (h) : A \rightarrow B$ ,  $M \triangleq \frac{H}{\supseteq} \text{est}(R)$  the problem to be solved,  $(\mu X : G(X))$  as in (11.6.3a).

| the law                                                                                                                                                                                                                                                                   | what it says                                                                                       | (15.1b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|---------|
| $(\mu X : \frac{T^\circ}{\supseteq} P(F(X)h) \text{ est}(R)) \sqsubseteq M$<br>Theorem 9.1, $h$ monotonic on $R$                                                                                                                                                          | <b>dynamic programming:</b> decompose in every way, solve each part, keep one optimum per part     |         |
| $\frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq M \quad (9.1)$                                                                                                                                                                                             | all Knaster–Tarski leaves to prove                                                                 |         |
| $\frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq H \quad (9.2)$<br>$H^\circ \frac{T^\circ}{\supseteq} P(F(M)h) \text{ est}(R) \sqsubseteq R \quad (9.3)$                                                                                                    | (9.1) split by the universal property of <b>est</b>                                                |         |
| $P(X) \text{ est}(R) \sqsubseteq (\exists X) n(\epsilon \setminus (XR^\circ))$<br>(9.4) = (7.10), (13.1b)                                                                                                                                                                 | the only fact about <b>est</b> either proof uses                                                   |         |
| $(\mu X : \frac{T^\circ}{\supseteq} \text{thin } Q P(F(X)h) \text{ est}(R)) \sqsubseteq M$<br>Theorem 9.2, $Q$ a preorder with $QF(H)h \sqsubseteq F(H)hR$ ; $\text{thin } Q$ as in (14.1b)                                                                               | the same with a thinning step: decompositions that can never win are dropped                       |         |
| the fixed point is unique, and entire<br>Theorem 6.3, $T^\circ$ followed by $F$ 's membership relation inductive; $\frac{T^\circ}{\supseteq}$ finite and non-empty, $R$ connected                                                                                         | when the recursion can be refined to a recursive function                                          |         |
| $\frac{[V_1, V_2]^\circ}{\supseteq} \text{thin}(Q_1 + Q_2) P([U_1, U_2]) \text{ est}(R)$<br>$= (\text{Ran } V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supseteq} \text{thin}(Q_i) P(U_i) \text{ est}(R)$<br>Proposition 9.1, $V_2 V_1^\circ = \perp$ | $FA$ is usually a coproduct, and disjoint ranges split the fixed point into one branch per summand |         |
| $F(R)h \sqsubseteq hR$<br>$QF(H)h \sqsubseteq F(H)hR$                                                                                                                                                                                                                     | the two conditions to be checked, in the order the three propositions below discharge them         |         |
| $F(R)h \sqsubseteq hR$<br>Proposition 9.2, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\text{cost})k$ , $F(\leq)k \sqsubseteq k \leq$                                                                                                         | monotonicity when the cost is itself a fold with a step $k$ monotonic on $\leq$                    |         |
| $F(Rn(H^\circ H))h \sqsubseteq hR$<br>Proposition 9.3, $R \triangleq \text{cost} \leq \text{cost}^\circ$ , $h \text{ cost} = F(\langle \text{cost}, H^\circ \rangle)k$ , $F(\leq \times \mathbb{1})k \sqsubseteq k \leq$ , $H^\circ$ simple                               | monotonicity <b>in context</b> : $k$ may also read the input the part was built from               |         |
| $Q \triangleq F(U, V)$<br>Proposition 9.4, $U, V$ preorders, $F(U, R)h \sqsubseteq hR$ , $VH \sqsubseteq HR$                                                                                                                                                              | both conditions of Theorem 9.2 at once, split along the two arguments of a bifunctor               |         |

### §15.2. The string edit problem

$Op ::= \text{cpy Char} \mid \text{del Char} \mid \text{ins Char}$ ,  $F(A, B) = 1 + (A \times B)$ ,  $\alpha \triangleq [\text{nil}, \text{cons}]$ .

(15.2a)

$\text{edit} \triangleq ([\text{base}, \text{step}]) : [Op] \rightarrow [\text{Char}] \times [\text{Char}]$ ,  $\text{base}$  returning  $([], [])$ ,  $\text{step}(\text{cpy } a, (xs, ys)) = ([a] \# xs, [a] \# ys)$ ,  $\text{step}(\text{del } a, (xs, ys)) = ([a] \# xs, ys)$ ,  $\text{step}(\text{ins } a, (xs, ys)) = (xs, [a] \# ys)$ .

$\text{length} \triangleq ([\text{zero}, \pi_2 \text{ succ}])$ ,  $R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U \triangleq T$ ,  $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$ ,  $Q \triangleq 1 + (U \times V)$ ,  $\text{empty}$  the coreflexive on  $(xs, ys)$  with both lists empty.

$\text{unstep}$  implements  $\frac{\text{step}^\circ}{\supseteq} \text{thin}(U \times V)$ :  $\text{unstep}([a] \# xs, []) = [(\text{del } a, (xs, []))]$ ,  $\text{unstep}([], [b] \# ys) = [(\text{ins } b, ([], ys))]$ ,  $\text{unstep}([a] \# xs, [b] \# ys) = (a \rightarrow b \rightarrow [(\text{cpy } a, (xs, ys)), [(\text{del } a, (xs, [b] \# ys)), (\text{ins } b, ([a] \# xs, ys))])$ .

| the law                                                                                                                                                                                                                                                                                                                               | what it says                                                                                                                                          | (15.2b) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{edit}^\circ}{\supset} \text{est}(R)$                                                                                                                                                                                                                                                                                     | the specification: a shortest edit sequence from which both strings can be reconstituted                                                              |         |
| $F(R) \alpha \sqsubseteq \alpha R$<br>Proposition 9.2 at <code>length</code> , <code>succ</code> monotonic on $\leq$                                                                                                                                                                                                                  | <code>cons</code> is monotonic on $R$ , so Theorem 9.1 already applies                                                                                |         |
| $QF(1, \text{edit}^\circ) \alpha \sqsubseteq F(1, \text{edit}^\circ) \alpha R$                                                                                                                                                                                                                                                        | the thinning condition sought, over $F(0p, [\text{Char}] \times [\text{Char}])$                                                                       |         |
| $F(T, R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                                             | $U \triangleq T$ costs nothing: any two operations may be compared                                                                                    |         |
| $\text{edit}(1 \times \text{suffix}) \sqsubseteq R^\circ \text{edit}$<br>$\text{edit}(\text{suffix} \times 1) \sqsubseteq R^\circ \text{edit}$                                                                                                                                                                                        | the other half of Proposition 9.4 for $V \triangleq \text{suffix}^\circ \times \text{suffix}^\circ$                                                   |         |
| $\text{edit}(1 \times \text{tail}) \sqsubseteq R^\circ \text{edit}$<br>$\text{edit}(\text{tail} \times 1) \sqsubseteq R^\circ \text{edit}$<br>$\text{suffix} = \text{tail}^*$ , and $BA \sqsubseteq CB \implies BA^* \sqsubseteq C^*B$                                                                                                | one step is enough: drop the operation that produced the head, or weaken its <code>cpy</code> to a <code>del</code> — never lengthening the sequence  |         |
| $(\mu X : \frac{[\text{base}, \text{step}]^\circ}{\supset} \text{thin } Q P([\text{nil}, (1 \times X) \text{ cons}]) \text{ est}(R)) \sqsubseteq \frac{\text{edit}^\circ}{\supset} \text{est}(R)$<br>Theorem 9.2                                                                                                                      | a copy, when available, beats a delete or an insert                                                                                                   |         |
| $X = (\text{empty} \rightarrow \text{nil}, \frac{\text{step}^\circ}{\supset} \text{thin}(U \times V) P((1 \times X) \text{ cons}) \text{ est}(R))$<br>Proposition 9.1                                                                                                                                                                 | <code>base</code> and <code>step</code> have disjoint ranges                                                                                          |         |
| $\text{mle} = (\text{empty} \rightarrow \text{nil}, \text{unstep list}((1 \times \text{mle}) \text{ cons}) \text{ minlist } R)$                                                                                                                                                                                                       | the program: at most two decompositions survive, but the same subproblem is solved many times, so the running time is exponential                     |         |
| $\text{column } xs \text{ } ys = [\text{mle } (u, ys) \mid u \leftarrow \text{tails } xs]$<br>$\text{column } xs = ([\text{fstcol } xs, \text{nextcol } xs]), \text{fstcol} = \text{list}(\text{del})$<br>$\text{tails}$                                                                                                              | the tabulation: $\text{mle } (xs, ys)$ needs $\text{mle } (u, v)$ for every tail $u$ of $xs$ and $v$ of $ys$ , so the columns are built right to left |         |
| $\text{column } xs \text{ } ([b] \# ys) = \text{nextcol } xs \text{ } (b, \text{column } xs \text{ } ys)$<br>$\text{nextcol } xs \text{ } (b, us)$<br>$= ([\text{base } (b, \text{last } us), \text{step } b]) \# us$<br>$us = \text{zip } (xs, \text{zip } (\text{init } us, \text{tail } us))$                                      | each column is a fold built bottom to top, over $xs$ zipped with the adjacent pairs of the column to its right                                        |         |
| $\text{base } (b, u) = [[\text{ins } b] \# u]$<br>$\text{step } b \text{ } ((a, (u, v)), ws) =$<br>$(a = b \rightarrow [[\text{cpy } a] \# v] \# ws,$<br>$[b \text{ min}(R) ([\text{del } a] \# w, [\text{ins } b] \# u)] \# ws)$<br>$w = \text{head } ws$ ; these <code>base</code> , <code>step</code> are not <code>edit</code> 's | an entry depends on the one below it (a delete), the one to its right (an insert), and the one below that (a copy) — quadratic in the two lengths     |         |

### §15.3. Optimal bracketing

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <p><code>tree A ::= tip A   bin (tree A, tree A), <math>FX = A + X^2</math>, so <math>F(R) = 1 + R^2</math>; <math>h \triangleq [\text{tip}, \text{bin}]</math>, <math>\text{flatten} \triangleq ([\text{wrap}, \text{cat}]) : \text{tree } A \rightarrow \text{list}^+ A</math>, <math>H = \text{flatten}^\circ</math>.</code></p> <p><math>(\text{cost}, \text{size}) \triangleq ([\text{opt}, \text{opb}])</math>, <math>\text{opt} \triangleq (\text{zero}, \text{st})</math>, <math>\text{opb } ((cx, sx), (cy, sy)) = (\text{cb } (sx, sy) + cx + cy, \text{sb } (sx, sy))</math>.</p> <p><code>sb</code> associative, so <math>\text{size} = \text{flatten } sz</math> for a map <math>sz</math>; <math>R \triangleq \text{cost} \leq \text{cost}^\circ</math>, <math>g \triangleq [\text{zero}, (1 \times sz)^2 \text{ opb } \pi_1]</math>, single the coreflexive on singleton lists.</p> <p><math>\text{splits} \triangleq (\text{inits}^+, \text{tails}^+) \text{ zip}</math>, an implementation of <math>\frac{\text{cat}^\circ}{\supset}</math>; <math>\text{array} \triangleq \text{inits list}(\text{row})</math>, <math>\text{row} \triangleq \text{tails list}(\text{mct})</math>, <math>\text{col} \triangleq \text{inits list}(\text{mct})</math>.</p> <p><math>\text{mix} \triangleq \text{zip list}(\text{bin}) \text{ minlist } R</math>, <math>\text{next} \triangleq (\pi_1, \text{mix}) \text{ snoc}</math>, <math>\text{process} \triangleq ((\text{tip wrap}) \times 1) \text{ loop}(\text{next})</math>.</p> | (15.3a) |
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| the law                                                                                                                                                                                                                                                                                                                                                                                             | what it says                                                                                                                           | (15.3b) |
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| $\frac{\text{flatten}^\circ}{\ni} \text{est}(R)$                                                                                                                                                                                                                                                                                                                                                    | the specification: a least-cost bracketing of $a_1 \oplus \dots \oplus a_n$ , the tree whose flattening is the given list              |         |
| $[\text{tip}, \text{bin}] \text{ cost} = (1 + (\text{cost}, \text{flatten})^2) \text{ g} \quad (9.5)$                                                                                                                                                                                                                                                                                               | the cost of a node reads only the cost and the flattening of its two subtrees                                                          |         |
| $(1 + (\leq \times 1)^2) \text{ g} \in \text{g} \leq \quad (9.6)$                                                                                                                                                                                                                                                                                                                                   | $\text{g}$ is monotonic on $\leq$ in its two cost arguments                                                                            |         |
| $F(Rn(\text{flatten } \text{flatten}^\circ))h \in hR$<br>Proposition 9.3, $H^\circ = \text{flatten}$ a map                                                                                                                                                                                                                                                                                          | monotonicity in context: only trees with the same flattening are compared                                                              |         |
| $(\mu X : \frac{[\text{wrap}, \text{cat}]^\circ}{\ni} P([\text{tip}, (X \times X) \text{ bin}] \text{ est}(R)) \in \frac{\text{flatten}^\circ}{\ni} \text{est}(R) \quad \text{Theorem 9.1}$                                                                                                                                                                                                         | split the list in every way, bracket both halves, join                                                                                 |         |
| $X = (\text{single} \rightarrow \text{wrap}^\circ \text{ tip}, \frac{\text{cat}^\circ}{\ni} P((X \times X) \text{ bin}) \text{ est}(R))$<br>Proposition 9.1                                                                                                                                                                                                                                         | <code>wrap</code> and <code>cat</code> have disjoint ranges                                                                            |         |
| $\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, \text{splits } \text{list}((\text{mct} \times \text{mct}) \text{ bin}) \text{ minlist } R)$                                                                                                                                                                                                                                       | the program; exponential, since the segments of one list overlap                                                                       |         |
| $\text{mct} = (\text{single} \rightarrow \text{head } \text{tip}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ mix}) \quad (9.7)$                                                                                                                                                                                                                                                      | the tabulation: <code>mct xs</code> is needed for every non-empty segment <code>xs</code> , so the values are held as an array of rows |         |
| $\text{col} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{init } \text{col}, \text{tail } \text{row}) \text{ next}) \quad (9.8)$<br>$\text{cons } \text{col} = (1 \times \text{array}) \text{ process} \quad (9.9)$<br>$\text{row} = (\text{single} \rightarrow \text{head } \text{tip } \text{wrap}, (\text{mct}, \text{tail } \text{row}) \text{ cons}) \quad (9.10)$ | a column extends the column to its left, a row the row below it; (9.9) is (9.8) rewritten as a loop                                    |         |
| $\text{array} = ([\text{fstcol}, \text{addcol}]), \text{fstcol} \triangleq \text{tip } \text{wrap } \text{wrap}$<br>$\text{addcol} \triangleq (\pi_1 \text{ tip } \text{wrap}, \text{step}) \text{ cons}$<br>$\text{step} \triangleq (\text{process } \text{tail}, \pi_2) \text{ zip } \text{list}(\text{cons})$                                                                                    | the program: one fold building the array column by column, cubic in the length of the input                                            |         |

## §15.4. Data compression

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
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| <pre> [A] ::= nil   snoc ([A], A), list+ A ::= wrap A   snoc (list+ A, A), String = [Char], Code ::= sym Char   ptr (String, String+), F(Code, String) = 1 + (String × Code), α ≜ [nil, snoc].  decode ≜ ([nil, extend]) : [Code] → String, extend (xs, sym a) = xs # [a], extend (xs, ptr (ys, zs)) = xs # zs when ys # zs is a proper prefix of xs # zs; H = decode°.  size ≜ ([zero, distr [1 × c, 1 × p] plus]) with c, p the constant costs of a symbol and a pointer; R ≜ size ≤ size°, Q ≜ F(T + T, prefix°) = 1 + (prefix° × (T + T)), the two T on symbols and on pointers.  lrt ws = est(prefix° × (T + T)) {(xs, (ys, zs))   xs # zs = ws, ys # zs a proper prefix of ws}, the longest repeated tail; reduce (ws # [a]) = (zs ≠ [] → [(ws, sym a), (xs, ptr (ys, zs))], [(ws, sym a)]) with (xs, (ys, zs)) = lrt (ws # [a]). </pre> | (15.4a) |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                                                                                     | what it says                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{\text{decode}^\circ}{\ni} \text{est}(R)$                                                                                                                                                                                                             | the specification: a smallest code sequence decoding to the given string                                                             |
| $F(R) \alpha \sqsubseteq \alpha R \quad (\tau+\tau)[c,p]=[c,p]$                                                                                                                                                                                             | monotonicity is routine, the two costs being constants                                                                               |
| $F(\tau+\tau, R) \alpha \sqsubseteq \alpha R$<br>decode prefix $\in R^\circ$ decode<br>Proposition 9.4 at $Q \triangleq F(\tau+\tau, \text{prefix}^\circ)$                                                                                                  | the thinning condition split in two: pointers are ordered only by how much input they consume, symbols and pointers not at all       |
| decode init $\in R^\circ$ decode                                                                                                                                                                                                                            | enough for the second: drop the last character of the output by shortening or removing the last code element, never raising the cost |
| $(\mu X : \frac{[\text{nil}, \text{extend}]^\circ}{\ni} \text{thin } Q \text{ } P([\text{nil}, (X \times 1) \text{ snoc}]) \text{ est}(R)) \sqsubseteq \frac{\text{decode}^\circ}{\ni} \text{est}(R)$<br>Theorem 9.2                                        | between a symbol and a pointer nothing can be decided in advance; between two pointers the longer match wins                         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau+\tau)) \text{ } P((X \times 1) \text{ snoc}) \text{ est}(R))$<br>Proposition 9.1                                                     | <code>nil</code> and <code>extend</code> have disjoint ranges                                                                        |
| $\frac{\text{extend}^\circ}{\ni} (\text{ws} \# [a]) = \{(\text{ws}, \text{sym } a)\} \cup \{(x\text{s}, \text{ptr } (y\text{s}, z\text{s})) \mid x\text{s} \# z\text{s} = \text{ws} \# [a], y\text{s} \# z\text{s} \text{ a proper prefix of } \text{ws}\}$ | the decompositions of one string: take the last character as a symbol, or end with a pointer                                         |
| reduce implements $\frac{\text{extend}^\circ}{\ni} \text{thin}(\text{prefix}^\circ \times (\tau+\tau))$                                                                                                                                                     | thinning leaves at most two: the symbol, and the pointer of <code>lrt</code>                                                         |
| encode = ( <code>null</code> $\rightarrow$ <code>nil</code> , reduce list(( <code>encode</code> $\times$ 1) <code>snoc</code> ) <code>minlist</code> R)                                                                                                     | the program, again exponential; the book gives no tabulation for it                                                                  |

## §16. Greedy Algorithms

### §16.1. Theory

$h, T, R, H, M$  as in (15.1a); additionally  $Q$  a **connected** preorder on the sets  $\frac{T^\circ}{\supseteq}$  returns, so that  $\frac{T^\circ}{\supseteq} \text{est}(Q)$  is entire. (16.1a)

| the law                                                                                                                                                                                                                              | what it says                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $(\mu X : \frac{T^\circ}{\supseteq} \text{est}(Q) F(X)h) \sqsubseteq M$<br>Theorem 10.1, the hypotheses of Theorem 9.2                                                                                                               | <b>greedy</b> : one decomposition is kept at every step, so $P$ and $\frac{T^\circ}{\supseteq}$ disappear from the recursion                                                                                                                    |
| $\frac{[V_1, V_2]^\circ}{\supseteq} \text{est}(Q_1 + Q_2) [U_1, U_2]$<br>$= (\text{Ran } V_1 \rightarrow W_1, W_2)$<br>$W_i \triangleq \frac{V_i^\circ}{\supseteq} \text{est}(Q_i) U_i$<br>Proposition 10.1, $V_2 V_1^\circ = \perp$ | Proposition 9.1 with <b>est</b> for <b>thin</b>                                                                                                                                                                                                 |
| $Q \triangleq F(U, V)$ Proposition 9.4                                                                                                                                                                                               | still the way to get both conditions, but $\frac{T^\circ}{\supseteq} \text{est}(Q)$ must now be entire as well                                                                                                                                  |
| Theorem 7.2                                                                                                                                                                                                                          | that greedy theorem chooses among the <b>results</b> of one relational step of a reduce; Theorem 10.1 chooses among the <b>decompositions</b> of the input, for an arbitrary $T$ rather than an initial algebra, at the cost of $h$ being a map |

### §16.2. The detab-entab problem

$\text{detab} \triangleq ([\text{nil}, \text{expand}]) : \text{String} \rightarrow \text{String}$  over  $\text{snoc}$ -lists,  $\alpha \triangleq [\text{nil}, \text{snoc}]$ ,  $H = \text{detab}^\circ$ ;  $\text{expand } (xs, a) = (a = \text{TB} \rightarrow \text{fill } xs, xs \# [a])$ ,  $\text{fill } xs = xs \# \text{blanks } (n - (\text{col } xs) \bmod n)$ . (16.2a)

$\text{col} \triangleq ([\text{zero}, \text{count}])$ ,  $\text{count } (c, a) = (a = \text{NL} \rightarrow 0, c + 1)$ ;  $\text{TB}$  the tab,  $\text{BL}$  the blank,  $\text{NL}$  the newline, tab stops every  $n$  columns.

$R \triangleq \text{length} \leq \text{length}^\circ$ ,  $U$  the preorder with  $a U b \iff a = \text{TB} v a = b$ ,  $V \triangleq \text{prefix}^\circ n (\text{fill } \text{fill}^\circ)$ ,  $Q \triangleq 1 + (V \times U)$ .

$\text{unfill } xs$  the shortest prefix of  $xs$  with  $\text{fill } (\text{unfill } xs) = \text{fill } xs$ ;  $\text{tbc}$  the trailing blank count,  $\text{triple} \triangleq (\text{unfill } \text{entab}, \text{tbc}, \text{col})$ .

| the law                                                                                                                                                                                                                                                                                                  | what it says                                                                                                                                        | (16.2b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{detab}^\circ}{\supset} \text{est}(R)$                                                                                                                                                                                                                                                       | the specification: a shortest input <b>detab</b> expands to the given output — <b>detab entab=1</b> and nothing shorter does                        |         |
| $F(T,R) \alpha \sqsubseteq \alpha R$ left as an exercise                                                                                                                                                                                                                                                 | $U$ may be any preorder on characters; $a \sqcup b \iff a=TB \vee a=b$ puts <b>TB</b> below every character, so <b>est</b> prefers a tab to a blank |         |
| $\text{detab prefix} \in R^\circ \text{ detab}$ FALSE                                                                                                                                                                                                                                                    | at $n=8$ , <b>detab [a,b,c,d,e,TB]=[a,b,c,d,e,BL,BL,BL]</b> , whose prefix <b>[a,b,c,d,e,BL,BL]</b> is longer than any input giving it              |         |
| $\text{detab } V^\circ \in R^\circ \text{ detab } V \triangleq \text{prefix}^\circ n(\text{fill fill}^\circ)$                                                                                                                                                                                            | true once only the prefixes that do not cross a tab stop are allowed                                                                                |         |
| $\text{nil } V^\circ = \text{nil fill } V^\circ = \text{fill}$<br>$\text{snoc } V^\circ \sqsubseteq \text{snoc } U(\pi_1 V^\circ)$ left as exercises                                                                                                                                                     | the three properties of $V$ the derivation rests on; the second is what <b>fill fill</b> <sup>°</sup> was added for                                 |         |
| $\text{expand } V^\circ \sqsubseteq \text{expand } U(\pi_1 V^\circ)$                                                                                                                                                                                                                                     | the claim they add up to: shortening the output either leaves the last step alone or discards it                                                    |         |
| $\text{detab } V^\circ \sqsubseteq \text{detab } U(\text{init detab } V^\circ)$<br><i>init</i> inductive, so the greatest solution is the unique one                                                                                                                                                     | hence $\text{detab } V^\circ \sqsubseteq \text{prefix detab}$ , and $\text{prefix} \in R^\circ$ finishes Proposition 9.4                            |         |
| $(\mu X : \frac{[\text{nil}, \text{expand}]^\circ}{\supset} \text{est}(Q) (1+(X \times 1)) [\text{nil}, \text{snoc}]) \sqsubseteq \frac{\text{detab}^\circ}{\supset} \text{est}(R)$<br>Theorem 10.1                                                                                                      | greedy: one character of input is decided at each step                                                                                              |         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{expand}^\circ}{\supset} \text{est}(V \times U) (X \times 1) \text{snoc})$ Proposition 10.1                                                                                                                                                         | <b>nil</b> and <b>expand</b> have disjoint ranges                                                                                                   |         |
| $\frac{\text{expand}^\circ}{\supset} (xs \# [a]) = \{(ys, TB) \mid \text{fill } ys = xs \# [a]\} \cup \{(xs, a)\}$<br>the first set is non-empty iff $a=BL$ and $\text{col } (xs \# [a]) \bmod n = 0$                                                                                                    | the last output character came from a tab only if it is a blank landing exactly on a tab stop                                                       |         |
| $\frac{\text{expand}^\circ}{\supset} \text{est}(V \times U) (xs \# [a]) =$<br>$(a=BL \wedge \text{col } (xs \# [a]) \bmod n = 0 \rightarrow (\text{unfill } xs, TB), (xs, a))$                                                                                                                           | the greedy step: emit a tab whenever a tab is legal, consuming all the blanks back to the previous tab stop                                         |         |
| $\text{entab } xs = \text{entab } (\text{unfill } xs) \# \text{blanks } (tbc \ xs)$ (10.1)                                                                                                                                                                                                               | what makes <b>triple</b> a <b>snoc-list</b> reduce: the output splits at the last tab stop                                                          |         |
| $\text{triple} = ([\text{base}, \text{op}])$ and $\text{entab} = \text{triple assocl } \pi_1 (1 \times \text{blanks}) \text{ cat}$                                                                                                                                                                       | the program: one pass carrying the column and the count of pending blanks                                                                           |         |
| <b>base</b> returns $([], (0, 0))$<br><b>op</b> $((xs, (t, c)), a) =$<br>$(a=BL \wedge (c+1) \bmod n \neq 0 \rightarrow (xs, (t+1, c+1)),$<br>$a=BL \rightarrow (xs \# [TB], (0, c+1)),$<br>$a=NL \rightarrow (xs \# \text{blanks } t \# [NL], (0, 0)),$<br>$(xs \# \text{blanks } t \# [a], (0, c+1)))$ | hold a blank back, cash the held blanks in for a tab at a tab stop, flush them at a newline, flush them before anything else                        |         |

### §16.3. The minimum tardiness problem

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| <p><math>FX = 1 + (X \times \text{Job})</math>, <math>\alpha \triangleq [\text{nil}, \text{snoc}]</math> on schedules, <math>\beta \triangleq [\text{nil}, \text{snag}]</math> on bags, <b>snag</b> putting a job into a bag; <math>\text{bagify} \triangleq ([\beta]) : [\text{Job}] \rightarrow \text{Bag Job}</math>, <math>H = \text{bagify}^\circ</math>.</p> <p><math>\text{ct}, \text{dt}, \text{wt} : \text{Job} \rightarrow \text{Real}</math> the completion, due and weighting quantities of a job; <math>\text{penalty } (xs, j) = (\text{sum } (\text{list}(\text{ct}) \ xs) + \text{ct } j - \text{dt } j) \times \text{wt } j</math>.</p> <p><math>\text{cost} \triangleq \frac{\text{prefix}}{\supset} P(\alpha^\circ [\text{zero}, \text{penalty}]) \text{est}(\geq)</math>, <math>\text{cost } [] = 0</math>, <math>\text{cost } (xs \# [j]) = \text{bmax } (\text{cost } xs, \text{penalty } (xs, j))</math>, <math>R \triangleq \text{cost} \leq \text{cost}^\circ</math>.</p> <p><math>\text{perm} \triangleq \text{bagify bagify}^\circ = ([\text{nil}, \text{add}])</math>, <math>\text{add } (xs, j) = ys \# [j] \# zs</math> for some <math>xs = ys \# zs</math>.</p> <p><math>k \triangleq [\text{zero}, \text{assocr } (1 \times ((\text{bagify}^\circ \times 1) \text{penalty})) \text{bmax}]</math>, <math>f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{penalty}]</math>, <math>Q \triangleq f \leq f^\circ</math>.</p> | (16.3a) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|



| the law                                                                                                                                                                                                              | what it says                                                                                                                                                                                      | (16.3b) |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| $\frac{\text{bagify}^\circ}{\ni} \text{est}(R)$                                                                                                                                                                      | the specification: an ordering of the given bag of jobs with least maximum penalty                                                                                                                |         |
| $F(Rn(\text{bagify } \text{bagify}^\circ))\alpha \sqsubseteq \alpha R$ (10.2)                                                                                                                                        | monotonicity in context — only schedules of one bag are compared                                                                                                                                  |         |
| $(Qn(\beta\beta^\circ))F(\text{bagify}^\circ)\alpha \sqsubseteq F(\text{bagify}^\circ)\alpha R$ (10.3)                                                                                                               | the greedy condition, also in context                                                                                                                                                             |         |
| $\text{cost } (xs\#[j]) = \text{bmax } (\text{cost } xs, \text{penalty } (\text{perm } xs, j))$<br>$\alpha \text{ cost} = F((\text{cost}, \text{bagify}))k, F(\leq \times 1)k \sqsubseteq k \leq$<br>Proposition 9.3 | (10.2): <b>penalty</b> sums the completion times of $xs$ , and a sum does not see the order                                                                                                       |         |
| $\alpha \text{ cost} = (g, h) \text{ bmax}$ (10.4)<br>$g \triangleq [\text{zero}, \text{penalty}]$ (10.5)<br>$h \triangleq [\text{zero}, \pi_1 \text{ cost}]$ (10.6)                                                 | the cost of a schedule split into the penalty of its last job and the cost of the rest                                                                                                            |         |
| $\text{add} \sqsubseteq \pi_1 R$ (10.7)<br>$\beta \text{ bagify}^\circ = F(\text{bagify}^\circ) [\text{nil}, \text{add}]$ (10.8)                                                                                     | adding a job never lowers the cost; and <b>bagify</b> <sup>°</sup> is itself a reduce on bags. Together: $\beta \text{ bagify}^\circ \sqsubseteq F(\text{bagify}^\circ) h \leq \text{cost}^\circ$ |         |
| $F(\text{bagify})Q^\circ F(\text{bagify}^\circ) = g \geq g^\circ$ , met by<br>$Q \triangleq f \leq f^\circ, f \triangleq [\text{zero}, (\text{bagify}^\circ \times 1) \text{ penalty}]$                              | the specification $Q$ has to satisfy, and the choice that meets it: a $Q$ -minimum is a job of least penalty                                                                                      |         |
| no greedy <b>reduce</b> exists                                                                                                                                                                                       | a greedy snoc-list reduce would also solve every prefix of the input, and the best schedule of a prefix need not extend to a best schedule of the whole                                           |         |
| $X = (\text{null} \rightarrow \text{nil}, \frac{\text{snag}^\circ}{\ni} \text{est}(Q) (X \times 1) \text{ snoc})$ Theorem 10.1,<br>Proposition 10.1                                                                  | $\text{nil}$ and $\text{snag}$ have disjoint ranges                                                                                                                                               |         |
| $\text{schedule} = (\text{null} \rightarrow \text{nil}, \text{pick } (\text{schedule} \times 1) \text{ snoc})$<br>$\text{pick} \sqsubseteq \frac{\text{snag}^\circ}{\ni} \text{est}(Q)$ , a partial function         | the program: repeatedly remove a job of least penalty and put it last; quadratic in the number of jobs                                                                                            |         |

## §16.4. The TeX problem

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
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| $\text{intern} \triangleq \text{val } \text{round} : \text{Decimal} \rightarrow [0, 2^{16}), \text{val} \triangleq ([\text{zero}, \text{shift}]), \text{shift } (d, r) = (d+r)/10, \text{round } r \text{ rounds } 2^{16}r \text{ to the nearest integer: } \text{round } r = n \iff 2n-1 < 2^{17}r < 2n+1.$<br><br>$\text{interval } n = ((2n-1)/2^{17}, (2n+1)/2^{17}), r \text{ inrange } (a, b) \iff a < r < b, \text{round}^\circ = \text{interval inrange}, R \triangleq \text{length} \leq \text{length}^\circ.$<br><br>Interval the pairs $(a, b)$ with $0 < b < 1$ and $a < b$ (10.9) $[\text{arb}, \text{step}] : 1 + (\text{Digit} \times \text{Interval}) \rightarrow \text{Interval}, \text{step } (d, (a, b)) = ((d+a)/10, (d+b)/10).$<br><br>$FX = 1 + (\text{Digit} \times X), \alpha \triangleq [\text{nil}, \text{cons}], h \triangleq ([\text{arb}, \text{step}])^\circ = H^\circ, ! : \text{Digit} \times \text{Interval} \rightarrow 1, Q \triangleq (l^\circ !^\circ r) \cup 1, w \triangleq 2^{17}.$ | (16.4a) |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|

| the law                                                                                                                                                                                        | what it says                                                                                                                                                                                           |
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| $\frac{\text{intern}^\circ}{\supset} \text{est}(R) = \text{interval } \frac{\text{inrange val}^\circ}{\supset} \text{est}(R)$                                                                  | the specification: a shortest decimal whose internal representation is the given multiple of $2^{-16}$ . $\text{round}^\circ$ is not a map, but $\text{interval}$ is, so it comes out of the $\supset$ |
| $\text{val inrange}^\circ = ([\text{arb}, \text{step}])$<br>fusion; $\text{zero inrange}^\circ = \text{arb}$ , $\text{shift inrange}^\circ = (1 \times \text{inrange}^\circ)$<br>$\text{step}$ | the converse of $\text{val}$ , cut down to intervals, is a reduce on cons-lists                                                                                                                        |
| $F(R) \alpha \sqsubseteq \alpha R$                                                                                                                                                             | $\text{cons}$ is monotonic on $R$ — routine                                                                                                                                                            |
| $\alpha^\circ F(h) Q^\circ \sqsubseteq R^\circ \alpha^\circ F(h)$                                                                                                                              | the greedy condition of Theorem 10.1, converted                                                                                                                                                        |
| $\frac{\text{arb}^\circ}{\supset} (a, b) = (a < 0 \rightarrow \{*\}, \{\})$<br>$\frac{\text{step}^\circ}{\supset} (a, b) = \{(d, (10a-d, 10b-d)) \mid 0 < 10b-d < 1\}$                         | the decompositions of an interval: stop, or take one more digit                                                                                                                                        |
| $0 < 10b-d_1 < 1$ and $0 < 10b-d_2 < 1$ imply $d_1 = d_2$                                                                                                                                      | $\text{step}^\circ$ is in fact a map, $d$ the digit with $0 < 10b-d < 1$ , so $\frac{[\text{arb}, \text{step}]^\circ}{\supset}$ returns at most two elements and $Q$ need only choose between them     |
| $Q \triangleq (l^\circ !^\circ r) \cup 1$ , and $! \text{ nil} \sqsubseteq \text{cons } R^\circ$<br>from $! \text{ nil } \text{length} \sqsubseteq \text{cons } \text{length} \geq$            | stop whenever stopping is legal: the empty decimal is shorter than any other                                                                                                                           |
| $(\mu X : \frac{[\text{arb}, \text{step}]^\circ}{\supset} \text{est}(Q) F(X) \alpha) \sqsubseteq \frac{\text{intern}^\circ}{\supset} \text{est}(R)$ Theorem 10.1                               | greedy: emit the one digit the interval allows, until the interval contains zero                                                                                                                       |
| $\text{extern} = \text{interval } f$<br>$f(a, b) = (a < 0 \rightarrow [], [d] \# f(10a-d, 10b-d))$                                                                                             | the program, with $d$ the digit above                                                                                                                                                                  |
| $\text{extern } n = f(2n-1, 2n+1)$<br>$f(p, q) = (p \leq 0 \rightarrow [], [d] \# f(10p-w \ d, 10q-w \ d))$<br>$d = (10q) \text{ div } w, w = 2^{17}$                                          | the same in integer arithmetic only, as chapter 3 required of $\text{intern}$ : every interval reached is $(p/w, q/w)$                                                                                 |

## §17. Appendix

### §17.1. $P(S) \text{ est}(R) = (\exists S) \cap (\epsilon \setminus (SR^\circ))$

$(p, q)$  tabulates  $W \triangleq (\exists S) \cap (\epsilon \setminus (SR^\circ))$ , and  $y \triangleq \frac{(p \exists S) \cap (qR)}{\exists}$ , a map.

(17.1a)

(17.1b)

$$\begin{array}{c} \boxed{\begin{array}{c} W \\ P(S) \text{ est}(R) \end{array}} \xleftarrow{p^\circ q = W, 1 \sqsubseteq yy^\circ} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \\ \boxed{\begin{array}{c} y^\circ q \\ \text{est}(R) \end{array}} \end{array}$$

(17.1c)

$$\begin{array}{c} \boxed{\begin{array}{c} y^\circ q \\ \exists \end{array}} \xleftarrow{f^\circ \cdot \neg f \cdot, \cdot \exists \neg \neg} \boxed{\begin{array}{c} q \\ (p \exists S) \cap (qR) \end{array}} \xleftarrow{\Delta \neg \cap, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ q \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} 1 \\ R \end{array}} \end{array}$$

R reflexive

(17.1d)

$$\begin{array}{c} \boxed{\begin{array}{c} \epsilon y^\circ q \\ R^\circ \end{array}} \xleftarrow{\cdot \exists \neg \neg, \circ, \text{meet}} \boxed{\begin{array}{c} R^\circ q^\circ q \\ R^\circ \end{array}} \quad \text{q simple} \\ \\ \boxed{\begin{array}{c} p^\circ y \\ P(S) \end{array}} \xleftarrow{\Delta \neg \cap, \cdot T \neg / T, \circ, f^\circ \cdot \neg f \cdot} \begin{array}{c} \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \end{array} \\ \\ \boxed{\begin{array}{c} p^\circ y \exists \\ \exists S \end{array}} \xleftarrow{\cdot \exists \neg \neg, \text{meet}} \boxed{\begin{array}{c} p^\circ p \exists S \\ \exists S \end{array}} \quad \text{p simple} \\ \\ \boxed{\begin{array}{c} p \exists \\ y \exists S^\circ \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} p \exists \\ (p \exists) \cap (qRS^\circ) \end{array}} \xleftarrow{1 \sqsubseteq qq^\circ} \boxed{\begin{array}{c} q^\circ p \exists \\ RS^\circ \end{array}} \xleftarrow{\circ, T \cdot \neg T \setminus} \boxed{\begin{array}{c} W \\ \epsilon \setminus (SR^\circ) \end{array}} \quad \text{W's right half} \end{array}$$

### §17.2. $P(\text{est}(R)) \text{ est}(R) = P(\text{Dom}(\text{est}(R))) \text{ union est}(R)$

(17.2a)

$$\boxed{P(\text{est}(R)) \text{ est}(R)} \xrightarrow{\text{Dom}(\text{est}(R)) \text{ est}(R) = \text{est}(R)} \boxed{P(\text{Dom}(\text{est}(R)) \text{ est}(R)) \text{ est}(R)}$$

$$\xrightarrow{P \text{ a relator}} \boxed{P(\text{Dom}(\text{est}(R))) P(\text{est}(R)) \text{ est}(R)} \xrightarrow{(13.1.9a)} \boxed{P(\text{Dom}(\text{est}(R))) \text{ union est}(R)}$$

(17.2b)

$$\boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ P(\text{est}(R)) \text{ est}(R) \end{array}} \xleftarrow{(17.1b) \text{ at } S := \text{est}(R), \Delta \neg \cap, T \cdot \neg T \setminus} \boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \quad \boxed{\begin{array}{c} \epsilon P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}}$$

$$\boxed{\begin{array}{c} P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \exists \text{est}(R) \end{array}} \xleftarrow{P(\text{Dom}(\text{est}(R))) \sqsubseteq 1} \boxed{\begin{array}{c} \text{union est}(R) \\ \exists \text{est}(R) \end{array}}$$

$$\xleftarrow{T(\text{Un}V) \sqsubseteq T \text{Un}TV, \cdot \exists \neg \neg, (13.1.4a)} \boxed{\begin{array}{c} (\exists \exists) \cap (\epsilon \setminus (\epsilon \setminus R^\circ)) \\ \exists \text{est}(R) \end{array}} \xleftarrow{\text{modular law}} \boxed{\begin{array}{c} \exists (\exists \cap (\epsilon \setminus (\epsilon \setminus R^\circ))) \\ \exists (\exists \cap (\epsilon \setminus R^\circ)) \end{array}} \quad \text{count of } T \cdot \neg T \setminus$$

$$\boxed{\begin{array}{c} \epsilon P(\text{Dom}(\text{est}(R))) \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}} \xleftarrow{\epsilon \text{ lax natural}} \boxed{\begin{array}{c} \text{Dom}(\text{est}(R)) \epsilon \text{ union est}(R) \\ \text{est}(R) R^\circ \end{array}}$$

